

Ratio & Proportion

Key - points:-

1) What are antecedent and consequent?

In the form $x : y$; x is called the *antecedent* and y is called *consequent*.

2) $x : y$ can also be written as $\frac{x}{y}$

3) Duplicate ratio (square) :-

Duplicate ratio of $x : y$ will be $x^2 : y^2$

4) Sub-duplicate Ratio (square root):-

Sub-duplicate Ratio of $x : y$ will be $\sqrt{x} : \sqrt{y}$

5) Triplicate Ratio (cube) :-

Triplicate ratio of $x : y$ will be $x^3 : y^3$

6) Sub-triplicate Ratio (cube root)

Sub-triplicate Ratio of $x : y$ will be $\sqrt[3]{x} : \sqrt[3]{y}$

7) Inverse ratio :- $x : y$ will be $y : x$

8) Compound Ratio : If two or more ratios are given, then the antecedent of one is multiplied with antecedent of the other and respective consequents are also multiplied.

If $a : b, c : d$ and $e : f$ are three ratios, then their compound ratio will be $\frac{ace}{bdf}$.

9) 3rd Proportional:- 3rd proportion of a and b i.e. $a : b :: b : x \Rightarrow$

$$\text{Then, } x = \frac{b^2}{a}$$

10) 4th Proportional: 4th proportional of a, b and c i.e. $a : b :: c : x$

$$\text{Then, } x = \frac{b \times c}{a}$$

11) Componendo and dividendo rule:

$$\text{If } a : b :: c : d, \text{ Then } \Rightarrow \frac{a+b}{a-b} = \frac{c+d}{c-d}$$

12) Mean Proportion : Mean proportion of two numbers, a and b , is given by $\sqrt{ab}$.

Partnership

Concept 1

The ratio of profits of two or more partners will be equal to the ratio of their respective equivalent investment ratio.

Equivalent investment = (Amount Invested) $\times$ (Time for which the amount is invested).

We will take some examples to understand the concepts of partnership.

Example 1: A invests Rs. 10,000 and B invests Rs. 20,000 for a period of 1 year each. The profit, after 1 year will be divided in the ratio:

$$\frac{\text{Profit of A}}{\text{Profit of B}} = \frac{10000 \times 12}{20000 \times 12} = \frac{1}{2}$$

Example 2: A invests Rs.100 in a business. After 6 months B joins with an investment of Rs. 200. At the end of 1 year C joins with an investment of Rs.100. What will be the ratio of profit after 3 years?

	A	B	C
Amount(Rs.)	100	200	100
Time(month)	36	30	24
Equivalent Investment	3600	6000	2400
Profit Ratio	3	5	2

Example 3: In the previous case if the total profit after 3 years is 10,000. What will be the share of A?

Share of A in total profit =

$$\frac{3}{3+5+2} \times 10,000 = \frac{3}{10} \times 10000$$

$$\Rightarrow 3 \times 1000 = 3000$$

Note: In case someone invests in-between the total period of time, just take care for what time an amount is invested. Similarly there can be a case when someone opts out in that case also, the time for which the amount was invested will be taken into consideration and not the full period.

Concept 2

If $\frac{a}{b} = \frac{c}{d} = \frac{e}{f} = \frac{f}{g} = K$, where K is constant. Then, $K = \frac{a+c+e+f}{b+d+f+g}$

Example : $\frac{1+x}{2-x} = \frac{9-x}{3+x} = K$, Find the value of K ?

$$\text{Sol. :- } K = \frac{1+x+9-x}{2-x+3+x} = \frac{10}{5} = 2$$

Concept 3

If two or three ratios are given, and we have to find an individual ratio of all of them, then we take LCM of the common entities and then balancing the common entities, we get the required ratio.

Example :- $a : b = 3 : 4, b : c = 6 : 7$.
LCM of the common entities i.e $b = (4, 6) = 12$

$$a : b = 3 : 4 \Rightarrow 3 \times 3 = 9 : 12$$

$$b : c = 6 : 7 \Rightarrow 2 \times 6 = 12 : 14$$

$$\text{So, } a : b : c = 9 : 12 : 14$$

Note : In case there are more than three ratios given, then the LCM approach will be tedious in finding the individual ratio of the given entities.

Alternative method

$$\text{If } a : b = 2 : 3, \quad b : c = 4 : 5$$

$$c : d = 6 : 7, \quad d : e = 1 : 2$$

$$\begin{array}{ccccccc} a : b : c : d : e \\ \boxed{2 : 3} & \boxed{3} & \boxed{3} & \boxed{3} \\ \boxed{4} & \boxed{4 : 5} & \boxed{5} & \boxed{5} \\ \boxed{6} & \boxed{6} & \boxed{6 : 7} & \boxed{7} \\ \boxed{1} & \boxed{1} & \boxed{1} & \boxed{1 : 2} \end{array}$$

$$48 : 72 : 90 : 105 : 210$$

To understand this concept we will take some examples

Example 1 : ₹5600 is to be divided among A, B, C & D in such a way that the ratio of share of A : B is 1 : 2, B : C is 3 : 1, C : D is 2 : 3, find the share of A and B taken together?

Share of A : B = 1 : 2

Share of B : C = 3 : 1

Share of C : D = 2 : 3

A : B : C : D $\rightarrow$

$$1 \times 3 \times 2 : 2 \times 3 \times 2 : 2 \times 1 \times 2 : 2 \times 1 \times 3$$

$$= 6 : 12 : 4 : 6 \text{ or } 3 : 6 : 2 : 3$$

Sum of the share of all of them = $3 + 6 + 2 + 3 = 14$ unit

$$\text{Total share of (A + B)} = \frac{6+3}{14} \times 5600$$

$$= 9 \times 400 = ₹3600$$

Example 2 : Sum of ₹ x is divided among A, B, C such that the ratio of the shares of A and B is 6 : 7 and that of B and C is 3 : 2. If the difference of the share of A and C is ₹540. Find the value of x

Share of A and B = 6 : 7

Share of B and C = 3 : 2

Ratio of the shares of A, B and C = $6 \times 3 : 3 \times 7 : 7 \times 2 = 18 : 21 : 14$

Diff. of A and C = $18 - 14 = 4$ unit, which corresponds to ₹540

Total share of A, B and C = $18 + 21 + 14$

$$= 53 \text{ unit} = \frac{540}{4} \times 53 = ₹ 7155$$

Concept 4

If numerator and denominator of a ratio is multiplied and divided by same number, the ratio remains unchanged.

$$\text{E.g. :- } \frac{a}{b} = \frac{a \times k}{b \times k} = \frac{\frac{a}{k}}{\frac{b}{k}}$$

Ratio is independent of change in scale only and not of origin.

Concept 5

If there is 'n' number of unequal fractions

such as $\frac{a1}{b1}, \frac{a2}{b2}, \frac{a3}{b3}, \frac{a4}{b4}, \dots, \frac{an}{bn}$ then

the ratio of $(a_1+a_2+a_3+\dots+a_n)$ and $(b_1+b_2+b_3+\dots+b_n)$ lies between lowest and highest values of these fractions.

Example:- three unequal fractions are $\frac{1}{2}$

, $\frac{2}{3}, \frac{4}{7}$. So, the ratio of $(1+2+4) : (2+3+7) = 7:12$. Among these fractions,

lowest fraction is $\frac{1}{2}$ and the highest fraction is $\frac{2}{3}$ so, the required value lies between $\frac{1}{2}$ and $\frac{2}{3}$.

Concept 6:

If $aA = bB = cC$, where a, b, c are constants and A, B, C are variables. Then the ratio of $A : B : C = b \times c : c \times a : a \times b$.

Example: If $3A = 4B = 5C$, then find $A:B:C$
The ratio of $A : B : C = 4 \times 5 : 5 \times 3 : 3 \times 4 = 20 : 15 : 12$

Variety Questions

Q.1. The concentration of three wines A, B and C are 10%, 20% and 30%, respectively. They are mixed in the ratio 2 : 3 : x resulting in a 23% concentration solution. Find the value of x.
SSC CHSL Tier II 02/11/2023
(a) 7 (b) 6 (c) 8 (d) 5

Q.2. In a school of 720 students, the ratio of boys to girls is 3 : 5. Find how many new boys may be allowed to the school if 18 new girls are admitted, so that the ratio of boys to girls changes to 2 : 3.
SSC CPO 05/10/2023 (3rd Shift)
(a) 42 (b) 50 (c) 44 (d) 38

Q.3. Ram has 12 hens and 5 cows, Shyam has 18 hens and 9 cows, Preetham has 4 hens and 8 cows, and Vinay has 10 hens and 7 cows. Who has a greater ratio of hens to cows?
SSC CPO 05/10/2023 (2nd Shift)
(a) Shyam (b) Preetham
(c) Vinay (d) Ram

Q.4. A company is manufacturing boxes of three types - plastic, wood and glass. The ratio of the numbers of plastic, wood and glass boxes is 2 : 3 : 5. The difference between the numbers of wood and glass boxes is 144. If the number of plastic boxes is increased by 25%, then what will be the number of plastic boxes now?
SSC CPO 04/10/2023 (1st Shift)
(a) 144 (b) 216 (c) 360 (d) 180

Q.5. Find the fourth proportional of 12^{35} , 3^{12} and 4^{12} .
SSC MTS 14/09/2023 (3rd Shift)
(a) 12^{23} (b) 3^{24} (c) 12^{-11} (d) 4^{24}

Q.6. Three numbers are in the ratio 1 : 2 : 3. By adding 5 to each of the numbers then the ratio changes to 2 : 3 : 4. The smallest number is:
SSC MTS 14/09/2023 (1st Shift)
(a) 5 (b) 15 (c) 12 (d) 20

Q.7. When $x = 1$ then y is directly proportional to $x + 3$ and $y = 8$. When $z = 6$ then x is inversely proportional to z and $x = 2$. Find the relation between y and z .
SSC MTS 12/09/2023 (2nd Shift)
(a) $(y - 6) / z = 24$ (b) $(y - 6) / z = 12$
(c) $z(y - 6) = 12$ (d) $z(y - 6) = 24$

Q.8. The average of the two numbers is 48. If 3 is added to a smaller number, then the ratio between them is 11 : 20. Find the larger number. (Round up to two decimal places).
SSC MTS 05/09/2023 (3rd Shift)
(a) 40.87 (b) 3.87 (c) 63.87 (d) 70.87

Q.9. Arrange the ratios in ascending order $5 : 3\frac{1}{3}$, $6\frac{1}{4} : 3\frac{1}{8}$ and $3.5 : 4.5$.
SSC MTS 05/09/2023 (1st Shift)
(a) $3.5 : 4.5 < 5 : 3\frac{1}{3} < 6\frac{1}{4} : 3\frac{1}{8}$
(b) $6\frac{1}{4} : 3\frac{1}{8} < 5 : 3\frac{1}{3} < 3.5 : 4.5$
(c) $5 : 3\frac{1}{3} < 3.5 : 4.5 < 6\frac{1}{4} : 3\frac{1}{8}$
(d) $5 : 3\frac{1}{3} < 6\frac{1}{4} : 3\frac{1}{8} < 3.5 : 4.5$

Q.10. The incomes of P, Q and R are in the ratio of 9 : 15 : 10 and their savings are in the ratio is 10 : 15 : 12. If R spends 60% of his income, what is the ratio of the expenditures of P, Q and R?
SSC MTS 05/09/2023 (1st Shift)
(a) 18 : 17 : 30 (b) 17 : 30 : 18
(c) 17 : 18 : 30 (d) 18 : 30 : 17

Q.11. Find the fourth proportional to $(a + b)^2 \times (a^3 - b^3)$, $(a^3 + b^3)$, $(a^2 - b^2)$.
SSC MTS 01/09/2023 (2nd Shift)
(a) $\frac{(a^2 - ab + b^2)}{(a + b)}$ (b) $\frac{(a^2 + ab + b^2)}{(a - b)}$
(c) $\frac{(a^2 - ab + b^2)}{(a^2 + ab + b^2)}$ (d) $\frac{(a^2 + ab + b^2)}{(a^2 - b^2)}$

Q.12. Two numbers A and B are such that the sum of 6% of A and 4% of B is $\frac{2}{5}$ of the sum of 7% of A and 11% of B, then the ratio A : B is:
SSC MTS 01/09/2023 (1st Shift)
(a) 3 : 7 (b) 5 : 8 (c) 1 : 8 (d) 1 : 7

Q.13. A certain sum was to be divided between A and B in the ratio 8 : 5. However, by mistake, it was divided in the ratio 5 : 8. Thus, A gets ₹300 less than his original share. Find the sum.
SSC CHSL 17/08/2023 (4th Shift)
(a) ₹2,600 (b) ₹1,500 (c) ₹1,300 (d) ₹600

Q.14. When P is subtracted from each of the numbers 8, 6, 2 and 9, the numbers so obtained in this order are in proportion. What is the mean proportional between

(3P - 6) and (9P - 4) ?
SSC CHSL 10/08/2023 (4th Shift)
(a) 26 (b) 28 (c) 29 (d) 24

Q.15. A man invests ₹44,000 in some shares in the ratio 1 : 4 : 5 which pay dividends of 10%, 15% and 25% on his investment for one year, respectively. His total dividend income is:
SSC CHSL 07/08/2023 (1st Shift)
(a) ₹ 7,850 (b) ₹ 8,580
(c) ₹ 7,970 (d) ₹ 8,510

Q.16. A sum of money is divided among A, B, C in the ratio 3 : 2 : 5, respectively. If the share of C is ₹300 more than B. then how much more money will A get than B ?
SSC CHSL 03/08/2023 (1st Shift)
(a) ₹150 (b) ₹75 (c) ₹100 (d) ₹125

Q.17. The mean proportional of $(a + b)(a - b)^3$, $(a + b)^3(a - b)$ is:
SSC CGL 25/07/2023 (4th shift)
(a) $(a + b)^2(a - b)^2$ (b) $(a + b)^2(a - b)$
(c) $(a + b)(a - b)^2$ (d) $a^2 - b^2$

Q.18. The seats for Mathematics, English and Chemistry in a school are in the ratio of 6 : 4 : 7. If these seats are increased by 20%, 40% and 60%, respectively, the ratio of the increased seats is:
SSC CGL 19/07/2023 (2nd shift)
(a) 9 : 7 : 8 (b) 9 : 7 : 14
(c) 1 : 2 : 3 (d) 7 : 9 : 14

Q.19. In a party hall, there are people in blue and red dresses. The ratio of the number of men in blue to the number of women in red is 3 : 7. The ratio of men in red to the number of women in blue is 2 : 1. If the ratio of the number of people in blue to the number of people in red is 35 : 76, then what is the ratio of the number of men to the number of women ?
SSC MTS 04/05/2023 (Morning)

(a) 47 : 51 (b) 40 : 43
(c) 62 : 57 (d) 52 : 59

Q.20. P's income is 250 percent of Q's income and P's expenditure is 180 percent of Q's expenditure. If P's income is 400 percent of Q's expenditure, then what is the respective ratio of P's savings and Q's savings?
SSC CHSL 17/03/2023 (3rd Shift)
(a) 8 : 3 (b) 11 : 3 (c) 11 : 5 (d) 5 : 2

Q.21. The number of units manufactured by company A was 12500 units in 2019 and 10625 units in 2020. While in company B, the production fell from 34000 units in 2019 to 30600 units in 2020. If X and Y are the percentage decrease in the number of units manufactured by company A and B

respectively from 2019 to 2020, then what will be the ratio of X and Y?

SSC CGL 13/12/2022 (1st Shift)

(a) 5 : 3 (b) 3 : 4 (c) 3 : 2 (d) 8 : 5

Q.22. Find the mean proportion of $\frac{a^3 + b^3}{a - b}$ and $\frac{a^2 - b^2}{a^2 - ab + b^2}$.

SSC CGL 06/12/2022 (3rd Shift)

(a) 1 (b) $a + b$ (c) $\frac{a + b}{a - b}$ (d) $\sqrt{a + b}$

Q.23. In an office with 40 officers, the average salaries of class-A, class-B and class-C officers are ₹600, ₹750 and ₹1,000 a day, respectively. The number of class-A, class-B and class-C officers in the office are in the ratio 5 : 4 : 1, respectively. Find the monthly average salary (in ₹) of an officers, all the 40 of them taken together. [Assume the number of days in a month to be 30.]

SSC MTS 08/07/2022 (Morning)

(a) 25,800 (b) 21,000
(c) 24,030 (d) 24,600

Q.24. The ratio of income of A to that of B is 4 : 3. The expenditure of A is $\frac{5}{12}$ of his income and the expenditure of B is $\frac{7}{15}$ of his income. The difference between their expenditure is ₹4,500. Find the income of B.

SSC CHSL 25/05/2022 (Evening)

(a) ₹49,625 (b) ₹48,625
(c) ₹51,625 (d) ₹50,625

Q.25. A sum of ₹46,800 is divided among A, B, C and D in such a way that the ratio of the combined share of A and D to the combined share of B and C is 8 : 5. The ratio of the share of B to that of C is 5 : 4. A receives ₹18,400. If x is the difference between the shares of A and B and y is the difference between the shares of C and D, then what is the value of (x - y) (in ₹) ?

SSC CGL Tier II (29/01/2022)

(a) 6000 (b) 6500 (c) 5000 (d) 7000

Q.26. If a number of pens and pencils are bought in the ratio of 5 : 3, then Ashok has to pay ₹44. If the ratio is changed as 3 : 5, then he has to pay ₹36. Find the ratio of the price of one pen to one pencil.

SSC MTS 11/10/2021 (Evening)

(a) 5 : 3 (b) 7 : 3 (c) 8 : 5 (d) 7 : 4

Q.27. Atul purchased Bread costing Rs.20 and gave a 100 rupee note to the shopkeeper. The shopkeeper gave the balance money in coins of denomination 2, 5, and 10. If these coins are in the ratio

5 : 4 : 1, then how many 5 coins did the shopkeeper give?

SSC CGL 03/08/2021 (Evening)

(a) 5 (b) 6 (c) 8 (d) 4

Q.28. In a school, $\frac{3}{8}$ of the number of students are girls and the rest are boys, One - third of the number of boys are below 10 years and $\frac{2}{3}$ of the number of girls is also below 10 years. If the number of students of age 10 or more years is 260, then the number of boys in the school is:

SSC CGL Tier II (15/11/2020)

(a) 280 (b) 300 (c) 234 (d) 312

Q.29. By adding 3 and 5 in numerator and denominator of a fraction it becomes $\frac{2}{3}$. If 1 and 3 are subtracted and added from numerator and denominator respectively, it becomes $\frac{2}{5}$.

Find the fraction.

SSC CHSL 19/10/2020 (Morning shift)

(a) $\frac{5}{7}$ (b) $\frac{6}{7}$ (c) $\frac{7}{6}$ (d) $\frac{7}{5}$

Q.30. A, B and C spend 80%, 85% and 75% of their incomes, respectively. If their savings are in the ratio 8 : 9 : 20 and the difference between the incomes of A and C is Rs. 18,000, then the income of B is :

SSC CGL Tier II (13/09/2019)

(a) Rs. 24,000 (b) Rs. 27,000
(c) Rs. 30,000 (d) Rs. 36,000

Q.31. The ratio of income of A and B last year was 4 : 3, respectively. The ratios of their individual incomes of the last year and the present year are 3 : 4 and 5 : 6, respectively. If their total income for the present year is Rs. 8.04 lakh, then the income of B last year was :

SSC CGL Tier II (12/09/2019)

(a) Rs. 2.7 lakh (b) Rs. 3.6 lakh
(c) Rs. 2.4 lakh (d) Rs. 2.8 lakh

Q.32. The ratio of the income of A to that of B is 5 : 7. A and B save Rs. 4000 and Rs. 5000 respectively. If the expenditure of A is equal to $66\frac{2}{3}\%$ of the expenditure

of B, then the total income of A and B is :

SSC CGL Tier II (11/09/2019)

(a) Rs. 25,200 (b) Rs. 24,000
(c) Rs. 26,400 (d) Rs. 28,800

Q.33. The ratio between the divisor and remainder is 3 : 2 and the ratio between the divisor and the quotient is 7 : 12. If the remainder is 14, then find the remainder when the dividend is divided

by 9.

SSC MTS 21/08/2019 (Morning)

(a) 3 (b) 6 (c) 4 (d) 5

Q.34. Ravi's age is $\frac{3}{5}$ of Shyam's age.

After x years the ratio of the ages of Ravi and Shyam becomes 5 : 7. If initially the sum of their ages is 32, then what is the value of x ?

SSC MTS 08/08/2019 (Morning)

(a) 2 (b) 4 (c) 8 (d) 7

Q.35. What is the ratio between the fourth proportional of 3, 4, 9 and the mean proportional between 2 and 98?

SSC CHSL 11/07/2019 (Afternoon)

(a) 7 : 8 (b) 7 : 6 (c) 8 : 7 (d) 6 : 7

Q.36. Rs. 8000 is distributed among A, B and C such that they receive notes of Rs. 500, Rs. 200 and Rs. 100 respectively. The amounts received by them are in the ratio 15 : 2 : 3. What was the ratio of the numbers of notes of Rs. 500, Rs 200 and Rs 100 ?

SSC CHSL 10/07/2019 (Morning)

(a) 3:1:3 (b) 3:3:1 (c) 4:1:2 (d) 3:2:2

Q.37. The ratio of incomes of A and B is 2 : 3 and that of their expenditure is 1 : 2. If 90% of B's expenditure is equal to the income of A, then what is the ratio of the savings of A and B?

SSC CHSL 02/07/2019 (Morning)

(a) 1 : 1 (b) 9 : 8 (c) 8 : 7 (d) 3 : 2

Q.38. A sum of Rs. 4360 was to be divided among A, B, C and D in the ratio of 3 : 4 : 5 : 8, but it was divided in the ratio of $\frac{1}{3} : \frac{1}{4} : \frac{1}{5} : \frac{1}{8}$ by mistake.

As a result:

SSC CHSL 01/07/2019 (Evening)

(a) A received Rs. 956 more
(b) B received Rs. 318 more
(c) D received Rs. 1144 less
(d) C received Rs. 132 less

Q.39. If $a : b = 2 : 5$, $c : b = 3 : 4$, then $a : b : c$ is equal to :

SSC CGL 12/06/2019 (Evening)

(a) 6 : 15 : 20 (b) 8 : 20 : 15
(c) 2 : 5 : 4 (d) 2 : 5 : 3

Q.40. A sum of Rs. x is divided among A, B and C such that the ratio of shares of A and B is 7 : 12 and that of B and C is 8 : 5. If the difference in the shares of A and C is Rs. 214, then the value of x is :

SSC CGL 07/06/2019 (Evening)

(a) 11556 (b) 11128 (c) 11770 (d) 11342

Q.41. The prices of two articles are in the ratio 4 : 5. If the price of the first article is increased by x% and that of the other is decreased by 30%, then the new prices

of A and B will be in the ratio 10 : 7. The value of x is :

SSC CGL 07/06/2019 (Afternoon)

(a) 24.5 (b) 22.5 (c) 25 (d) 20

Q.42. The ratio of the ages of A and B, four years ago, was 4 : 5. Eight years from now, the ratio of the ages of A and B will be 11 : 13. What is the sum of their present ages?

SSC CGL 04/06/2019 (Morning)

(a) 80 years (b) 96 years
(c) 72 years (d) 76 years

Practice Questions

SSC CHSL 2023 Tier - 2

Q.43. A sum of money is divided among A, B, and C in the ratio 4 : 3 : 6. If the share of C is ₹600 more than B, then how much amount will A get more than B ?

SSC CHSL Tier II 02/11/2023

(a) ₹200 (b) ₹150 (c) ₹250 (d) ₹100

Q.44. The ratio of the expenditure to the savings of Richa is 2 : 1. If her income and expenditure increase by 20% and 15%, respectively, then find the percentage increase in her savings.

SSC CHSL Tier II 10/01/2024

(a) 35% (b) 30% (c) 20% (d) 25%

Q.45. The ratio of prices of Coffee and Tea is 3 : 4 and the ratio of their quantities consumed by a family is 4 : 7. Find the ratio of expenditures on Coffee and Tea.

SSC CHSL Tier II 10/01/2024

(a) 3 : 7 (b) 7 : 3 (c) 7 : 4 (d) 4 : 7

Q.46. ₹7,786 is divided among A, B and C in such a manner that the ratio of the amount of A to that of B is 3 : 4 and the ratio of the amount of B to that of C is 12 : 13. The amount of money received by B is:

SSC CHSL Tier II 10/01/2024

(a) ₹7,138 (b) ₹2,977
(c) ₹2,748 (d) ₹5,038

SSC CPO 2023 Tier - 1

Q.47. The areas of the two triangles are in the ratio 4 : 3 and their heights are in the ratio 6 : 5. Find the ratio of their bases.

SSC CPO 03/10/2023 (1st Shift)

(a) 5 : 6 (b) 10 : 9 (c) 9 : 10 (d) 6 : 5

Q.48. If $\frac{x}{4y} = \frac{3}{4}$, then the value of

$\frac{2x + 3y}{x - 2y}$ is

SSC CPO 05/10/2023 (2nd Shift)

(a) 8 (b) 6 (c) 9 (d) 7

SSC MTS 2023 Tier - 1

Q.49. P and Q are inversely proportional to each other and P is 10, when Q is 6. What will be the value of Q when P is 15?

SSC MTS 01/09/2023 (2nd Shift)

(a) 9 (b) 4 (c) 10 (d) 5

Q.50. The mean proportion of $(3 + 2\sqrt{2})^2$ and $(3 - 2\sqrt{2})^2$ is:

SSC MTS 04/09/2023 (1st Shift)

(a) $3 + 2\sqrt{2}$ (b) 5 (c) $2\sqrt{2}$ (d) 1

Q.51. Two numbers are in the ratio of 6 : 7. If the difference between these numbers is 16, then find the numbers.

SSC MTS 04/09/2023 (2nd Shift)

(a) 96, 112 (b) 80, 96
(c) 120, 136 (d) 104, 120

Q.52. Profit of ₹29,120 is divided among A, B and C so that A's share is $2\frac{1}{2}$ times

B's and B's share is $1\frac{1}{5}$ times C. Find B's share.

SSC MTS 05/09/2023 (2nd Shift)

(a) ₹6,940 (b) ₹7,820 (c) ₹6,720 (d) ₹7,140

Q.53. The mean proportion of a and $\frac{a}{b^2}$ is _____.

SSC MTS 05/09/2023 (2nd Shift)

(a) ab (b) $\frac{b}{a}$ (c) $\frac{a}{b}$ (d) $\frac{a}{2b}$

Q.54. Rupees 66,000 is to be divided amongst Asha, Bhupinder and Gargi in such a way that ratio of their shares is to be: Asha : Bhupinder = 9 : 5 and Bhupinder : Gargi = 2 : 1. How much should Asha get ?

SSC MTS 05/09/2023 (3rd Shift)

(a) ₹38,000 (b) ₹36,000
(c) ₹32,000 (d) ₹34,000

Q.55. A certain sum is to be divided between A, B, and C in the ratio 2 : 5 : 7. But by mistake, it was divided in the ratio 5 : 2 : 7. Thus B loses by ₹150. Find the sum.

SSC MTS 06/09/2023 (1st Shift)

(a) ₹700 (b) ₹850 (c) ₹900 (d) ₹625

Q.56. If $p : q = 5 : 2$, then $p^2 - q^2 : p^2 + q^2 = ?$

SSC MTS 06/09/2023 (2nd Shift)

(a) 29 : 21 (b) 4 : 3 (c) 3 : 4 (d) 21 : 29

Q.57. If $A : B = \frac{1}{3} : \frac{1}{5}$, $B : C = \frac{1}{7} : \frac{1}{8}$

and $C : D = \frac{1}{10} : \frac{1}{21}$, then $A : B : C : D$ is:

SSC MTS 06/09/2023 (3rd Shift)

(a) 40 : 24 : 21 : 10 (b) 48 : 23 : 25 : 8

(c) 8 : 3 : 7 : 5 (d) 14 : 24 : 28 : 10

Q.58. The marks scored by a student are directly proportional to the time invested in studies. If the student gets 60 marks when he studies 4 hours a day, then how many hours he should use in his studies daily to get 75 marks ?

SSC MTS 08/09/2023 (1st Shift)

(a) 4.5 (b) 5 (c) 5.5 (d) 6

Q.59. Three numbers are in the ratio 5 : 2 : 7. The sum of their squares is 3822. The biggest number is :

SSC MTS 08/09/2023 (2nd Shift)

(a) 35 (b) 38 (c) 78 (d) 49

Q.60. Among four numbers the sum of the smallest and largest is 72 more than the sum of middle two numbers. If the numbers are in the ratio 6 : 7 : 15 : 19, then the smallest number is :

SSC MTS 08/09/2023 (2nd Shift)

(a) 72 (b) 144 (c) 168 (d) 216

Q.61. If the income of T is 8 times his savings, then his expenditure is _____ times his savings.

SSC MTS 11/09/2023 (3rd Shift)

(a) 6 (b) 7 (c) 8 (d) 5

Q.62. The ratio of number of students in schools A and B is 5 : 8. If the number of students in schools A and B is increased by 20% and 15%, respectively, find the new ratio of students in schools A and B, respectively.

SSC MTS 12/09/2023 (2nd Shift)

(a) 15 : 23 (b) 15 : 13
(c) 13 : 15 (d) 12 : 13

Q.63. The ratio of the number of boys to the number of girls in a school of 1650 students is 6 : 5. If 124 new boys are admitted and few new girls are admitted, the ratio changes to 4 : 3. What is the ratio of the newly admitted girls to the newly admitted boys ?

SSC MTS 13/09/2023 (1st Shift)

(a) 3 : 4 (b) 62 : 9 (c) 4 : 3 (d) 9 : 62

Q.64. In a class of boys and girls, the number of girls is twice the number of boys. If 20 more boys join the class, the ratio of girls and boys is 1 : 1. How many total students were there in the class initially?

SSC MTS 14/09/2023 (1st Shift)

(a) 60 (b) 75 (c) 45 (d) 90

Q.65. If $A : B : C = 3 : 4 : 5$, then $\sqrt{A^2 + B^2} : \sqrt{C^2 - B^2}$ is:

SSC MTS 14/09/2023 (3rd Shift)

(a) 3 : 2 (b) 4 : 5 (c) 5 : 3 (d) 3 : 5

SSC CHSL 2023 Tier - 1

Q.66. Find the third proportional of 25 and 45.

SSC CHSL 02/08/2023 (1st Shift)

(a) 85 (b) 65 (c) 81 (d) 76

Q.67. Find the fourth proportion to 0.24, 0.6 and 20.

SSC CHSL 03/08/2023 (2nd Shift)

(a) 50 (b) 24 (c) 54 (d) 48

Q.68. The cost of 3 cups and 5 plates is ₹1,080 and the cost of 2 cups and 4 plates is ₹840. Find the ratio of the cost of a cup to the cost of a plate.

SSC CHSL 03/08/2023 (3rd Shift)

(a) 1 : 3 (b) 2 : 3 (c) 1 : 1 (d) 1 : 2

Q.69. The compound ratio of 3 : 4 and 6 : 7 is 36 : x. Find x.

SSC CHSL 03/08/2023 (4th Shift)

(a) 64 (b) 75 (c) 56 (d) 48

Q.70. Find 2 numbers such that their mean proportional is 25 and their third proportional is 25.

SSC CHSL 07/08/2023 (1st Shift)

(a) 5 and 5 (b) 5 and 25
(c) 25 and 125 (d) 25 and 25

Q.71. The present age of a father is twice that of his son. 8.5 years hence the ratio would be 7 : 4. Find the son's present age.

SSC CHSL 07/08/2023 (2nd Shift)

(a) 25.5 years (b) 16.5 years
(c) 20.5 years (d) 24.6 years

Q.72. The sum of the amount that Sunaina and Ranjan have is ₹23 and the difference of their amount is ₹7. The ratio of the amount that Sunaina and Ranjan have is:—

SSC CHSL 07/08/2023 (3rd Shift)

(a) 5 : 3 (b) 7 : 15 (c) 15 : 8 (d) 2 : 17

Q.73. Two numbers are in the ratio of 1 : 2. If 7 is added to both, their ratio changes to 3 : 5. The greater number is:

SSC CHSL 07/08/2023 (3rd Shift)

(a) 30 (b) 18 (c) 14 (d) 28

Q.74. The difference between 24 and the mean proportional of 7 and 63 is:

SSC CHSL 07/08/2023 (4th Shift)

(a) 3 (b) 2 (c) 5 (d) 4

Q.75. What is the mean proportional between 3 and 27 ?

SSC CHSL 08/08/2023 (1st Shift)

(a) 30 (b) 81 (c) 15 (d) 9

Q.76. If $a : b = 5 : 7$, then $(6a^2 - 2b^2) : (b^2 - a^2)$ will be :

SSC CHSL 08/08/2023 (4th Shift)

(a) 12 : 5 (b) 21 : 5 (c) 13 : 6 (d) 17 : 8

Q.77. Find the fourth proportion of the numbers $\frac{1}{3}$ rd of 15, $\frac{4}{5}$ th of 25, $\frac{3}{7}$ th of 35:

SSC CHSL 09/08/2023 (1st Shift)

(a) 65 (b) 55 (c) 60 (d) 50

Q.78. The ratio of Ram's salary to Shyam's salary is 3 : 4. The ratio of Shyam's salary to Ghanshyam's salary is 5 : 6. What will be the ratio of Ram's salary to Ghanshyam's salary ?

SSC CHSL 09/08/2023 (3rd Shift)

(a) 3 : 5 (b) 5 : 8 (c) 4 : 5 (d) 3 : 8

Q.79. What is the fourth proportional of $3\sqrt{5}$, $5\sqrt{8}$ and $3\sqrt{10}$?

SSC CHSL 09/08/2023 (4th Shift)

(a) $10\sqrt{5}$ (b) $40\sqrt{2}$ (c) 30 (d) 20

Q.80. The mean proportion of number 0.049 and 0.9 is:

SSC CHSL 10/08/2023 (1st Shift)

(a) 0.021 (b) 0.21 (c) 0.63 (d) 21

Q.81. The ratio of the father's age to his son's age is 5 : 3. The product of the numbers representing their ages is 960. The ratio of their ages after 6 years will be:

SSC CHSL 11/08/2023 (1st Shift)

(a) 23 : 15 (b) 23 : 17
(c) 21 : 15 (d) 21 : 17

Q.82. Find the third proportion of P and 12, when 8, P, 16 and 18 are in proportion.

SSC CHSL 11/08/2023 (2nd Shift)

(a) 16 (b) 8 (c) 18 (d) 12

Q.83. The total number of workers in a company is 960. If the number of female workers in the company is 420, then the respective ratio of the total number of male workers to the total number of female workers in the company is:

SSC CHSL 11/08/2023 (3rd Shift)

(a) 7 : 9 (b) 16 : 9 (c) 16 : 7 (d) 9 : 7

Q.84. Given that x varies directly as y and $x = 45$ when $y = 9$. What is the value of x when $y = 50$?

SSC CHSL 17/08/2023 (1st Shift)

(a) 225 (b) 190 (c) 175 (d) 250

Q.85. If $a : b = 3 : 7$, then $(2a + 3b) : (3a + 2b)$ is:

SSC CHSL 17/08/2023 (2nd Shift)

(a) 27 : 23 (b) 23 : 29 (c) 21 : 29 (d) 21 : 23

Q.86. Divide ₹ 1488 among X, Y and Z. If the shares are in the ratio $X : Y = 4 : 5$, $Y : Z = 7 : 6$, then find out the share of Z.

SSC CHSL 17/08/2023 (3rd Shift)

(a) ₹ 460 (b) ₹ 450 (c) ₹ 470 (d) ₹ 480

SSC CGL 2023 Tier - 1

Q.87. The ratio of the number of boys to that of the girls in a school is 11 : 15. If there are 200 more girls than boys in the school, what is the number of boys in that school ?

SSC CGL 14/07/2023 (4th shift)

(a) 550 (b) 750 (c) 506 (d) 605

Q.88. The ratio between the monthly income and the expenditure of Vaidic is 8 : 5. If his income increases by 20% and his expenditure increases by 30%, then find the percentage increase or decrease in his monthly savings.

SSC CGL 14/07/2023 (4th shift)

(a) $5\frac{2}{3}\%$ increase (b) $5\frac{2}{3}\%$ decrease

(c) $3\frac{1}{3}\%$ increase (d) $3\frac{1}{3}\%$ decrease

Q.89. The ages of Gyanendra and Arbind are in the ratio 6 : 5. If the sum of their ages is 55 years, then what will be the ratio of their ages after seven years from now ?

SSC CGL 14/07/2023 (4th shift)

(a) 5 : 6 (b) 37 : 32 (c) 32 : 37 (d) 6 : 5

Q.90. A bag contains ₹550 in the form of 50p, 25p and 20p coins in the ratio 2 : 3 : 5. The difference between the amounts that are contributed by the 50 p and the 20 p coins is:

SSC CGL 17/07/2023 (3rd shift)

(a) ₹30 (b) ₹20 (c) ₹10 (d) ₹0

Q.91. The third proportional to $(x^2 - y^2)$ and $(x - y)$ is:

SSC CGL 18/07/2023 (1st shift)

(a) $(x - y)$ (b) $\frac{x - y}{x + y}$ (c) $\frac{x + y}{x - y}$ (d) $(x + y)$

Q.92. In a factory, 48% of the number of male workers is equal to two-third of the number of female workers. What is the ratio of the number of males to that of the females in the factory?

SSC CGL 18/07/2023 (2nd shift)

(a) 25 : 12 (b) 12 : 25 (c) 18 : 25 (d) 25 : 18

Q.93. Find the fourth proportional of 144, 192 and 216.

SSC CGL 18/07/2023 (4th shift)

(a) 361 (b) 288 (c) 324 (d) 289

Q.94. Two numbers are, respectively, 17% and 50% more than a third number. The ratio of the two numbers is:

SSC CGL 19/07/2023 (4th shift)

(a) 27 : 25 (b) 39 : 50

(c) 19 : 11 (d) 29 : 25

Q.95. If $a : b :: b : c$, and $b = 96$, then which of the following can be a possible pair of values of a and c ?

SSC CGL 20/07/2023 (2nd shift)

- (a) $a = 24$; $c = 374$ (b) $a = 32$; $c = 288$
 (c) $a = 48$; $c = 168$ (d) $a = 16$; $c = 586$

Q.96. Three numbers are in the ratio $\frac{4}{5}$:

$\frac{5}{6} : \frac{9}{10}$. The difference between the smallest and the greatest numbers is 12. Find the number which is neither the smallest nor the greatest.

SSC CGL 24/07/2023 (2nd shift)

- (a) 96 (b) 108 (c) 100 (d) 104

Q.97. A pole is 9030 cm long. It is colored saffron, white and green in the ratio of 6 : 7 : 8. Find the length of the green part of the pole.

SSC CGL 24/07/2023 (4th shift)

- (a) 3440cm (b) 3010cm
 (c) 3480cm (d) 2580cm

Q.98. If a 7-storey building has a 28 m long shadow, the number of storeys of the building whose shadow is 48 m long is:

SSC CGL 26/07/2023 (3rd shift)

- (a) 14 (b) 24 (c) 16 (d) 12

Q.99. What smallest positive number should be subtracted from 2 and 3 so that 4 is the third proportion to them?

SSC CGL 26/07/2023 (4th shift)

- (a) 1 (b) 2 (c) 4 (d) 3

Q.100. If the third proportional of $3x^2$ and $4xy$ is 48, then find the positive value of y .

SSC CGL 27/07/2023 (2nd shift)

- (a) 3 (b) 5 (c) 2 (d) 4

SSC Selection Post (Phase - XI)

Q.101. The ratio of cheese to sauce for a single pizza is 1 cup to $\frac{1}{2}$ cup. If Rama used 15 cups of sauce to make a number of pizzas, then how many cups of cheese did she use on those pizzas?

Graduate Level 27/06/2023 (Shift - 4)

- (a) 20 cups (b) 10 cups
 (c) 40 cups (d) 30 cups

Q.102. Out of the total income, a family saves 15% and the rest of the amount is spent on three types of expenditure - health, education and food in the ratio of 1 : 2 : 2. The income is Rs.300 less than thrice the expenditure incurred on education. What is the difference between the savings and the expenditure incurred on health?

Graduate Level 27/06/2023 (Shift - 4)

- (a) Rs.300 (b) Rs.350
 (c) Rs.2,250 (d) Rs.500

Q.103. Let M denote the mean proportional between 16 and 9 and let N

denote the third proportional of 9 and 15. Find the value of $5M + 3N$.

Graduate Level 28/06/2023 (Shift - 3)

- (a) 300 (b) 125 (c) 135 (d) 225

Q.104. A company is manufacturing three different products - Glass, Plate and Bowl. The number of Plates is double the number of Glasses. The number of Glasses manufactured is twice the number of Bowls. What is the difference between the number of Plates and the number of Bowls if the total units manufactured by the company is 2380?

Graduate Level 28/06/2023 (Shift - 3)

- (a) 1360 (b) 680 (c) 1120 (d) 1020

Q.105. Two numbers are in the ratio of 7 : 6. If the difference between their squares is 637, then the numbers are Higher Secondary 30/06/2023 (Shift - 2)

- (a) 49 and 42 (b) 49 and 12
 (c) 43 and 36 (d) 14 and 12

Q.106. The students in three batches of a dance class are in ratio 2 : 3 : 5. If 20 students increase in each batch the ratio changes to 4 : 5 : 7. Find the total number of students in the three batches before the increase.

Matriculation Level 27/06/2023 (Shift - 1)

- (a) 120 (b) 100 (c) 150 (d) 80

Q.107. The mean proportional between $(3 + \sqrt{5})$ and $(15 - \sqrt{125})$ is _____.

Matriculation Level 27/06/2023 (Shift - 2)

- (a) $3\sqrt{5}$ (b) $5\sqrt{5}$ (c) $9\sqrt{5}$ (d) $2\sqrt{5}$

Q.108. In an examination, the ratio of practical to theory marks is 3 : 7. If 80% of the marks are scored in the practical and 60% of marks are scored in the theory, what is the ratio of marks scored in the theory and the practical?

Matriculation Level 28/06/2023 (Shift - 1)

- (a) 6 : 7 (b) 5 : 7 (c) 7 : 4 (d) 3 : 7

Q.109. If $a : b : c = 4 : 7 : 9$ and $b : c : d = 28 : 36 : 21$, then $a : b : c : d$ is:

Matriculation Level 28/06/2023 (Shift - 4)

- (a) 12 : 21 : 27 : 17 (b) 16 : 28 : 36 : 21
 (c) 8 : 14 : 18 : 21 (d) 4 : 7 : 9 : 11

Q.110. The ratio of the number of boys to that of girls in a village is 5 : 3. If 40% of the boys and 60% of the girls appeared in an examination, then the ratio of the number of boys and girls who appeared in the examination to the number who did not appear in the same examination, is :-

Matriculation Level 30/06/2023 (Shift - 4)

- (a) 17 : 21 (b) 19 : 21
 (c) 17 : 27 (d) 19 : 23

SSC CHSL 2022 Tier - 2

Q.111. The radii of two cylinders are in the ratio 2 : 1 and their heights are in the ratio 5 : 2. The ratio of their volumes is:

SSC CHSL Tier II (26/06/2023)

- (a) 20 : 1 (b) 15 : 1 (c) 10 : 1 (d) 5 : 1

SSC MTS 2022 Tier - 1

Q.112. The ratio of two numbers is 5 : 9. If 6 is added to the greater number and 6 is subtracted from the smaller number, then the greater number becomes 6 times the smaller number. What are the numbers?

SSC MTS 02/05/2023 (Afternoon)

- (a) 15 and 27 (b) 10 and 18
 (c) 25 and 45 (d) 20 and 36

Q.113. If 750 is divided into two parts such that the ratio of one to the other is in the ratio 8 : 7, then find the smallest number.

SSC MTS 03/05/2023 (Morning)

- (a) 135 (b) 120 (c) 350 (d) 90

Q.114. If the third proportion of 3 and x is 27 and the third proportion of 2 and y is 8, then find $x : y$

SSC MTS 03/05/2023 (Afternoon)

- (a) 7 : 9 (b) 4 : 5 (c) 3 : 7 (d) 9 : 4

Q.115. If $\frac{J}{4} = \frac{K}{5} = \frac{L}{3}$, then what is the value of $(L + J)^2 : (J + K)^2 : (K + L)^2$?

SSC MTS 04/05/2023 (Afternoon)

- (a) 25 : 16 : 9 (b) 64 : 25 : 81
 (c) 49 : 81 : 62 (d) 49 : 81 : 64

Q.116. Salaries of Ankur and Ankit are in the ratio 3 : 2. If the salaries of each is decreased by Rs 2500, the new ratio becomes 8 : 5. What is the sum of their salaries?

SSC MTS 08/05/2023 (Evening)

- (a) Rs. 7500 (b) Rs. 12500
 (c) Rs. 37500 (d) Rs. 28500

Q.117. These positive numbers are in the proportion of 2 : 3 : 5. If the sum of their squares is 342, then find the largest number.

SSC MTS 09/05/2023 (Evening)

- (a) 35 (b) 55 (c) 15 (d) 25

Q.118. If A is the mean proportion of 24 and 6. B is the mean proportion of 81 and 9. Find the value of $3A + B$.

SSC MTS 09/05/2023 (Evening)

- (a) 33 (b) 32 (c) 63 (d) 55

Q.119. In three positive numbers, the ratio of first and second is 3 : 4 and the ratio of second and third is 3 : 5. The

product of first and third is 4500. What is the sum of the three numbers?

SSC MTS 11/05/2023 (Morning)

(a) 282 (b) 235 (c) 198 (d) 205

Q.120. Rs. 6000 was to be divided among P, Q and R in the ratio of 5 : 7 : 8. By mistake it was divided among P, Q and R in the ratio of 9 : 7 : 4. What amount R has lost in the division?

SSC MTS 12/05/2023 (Afternoon)

(a) Rs. 900 (b) Rs. 600
(c) Rs. 300 (d) Rs. 1200

Q.121. When a sum of money was equally distributed among 50 children, each child received Rs. 80. If the same amount is equally distributed among another group of children such that each child get Rs. 50, then what is the number of children in the second group?

SSC MTS 15/05/2023 (Morning)

(a) 50 (b) 80 (c) 60 (d) 40

Q.122. The total number of color blocks is 1260. If the ratio of red block to blue block is 2 : 5 and the ratio of blue block to yellow block is 15 : 7, then how many yellow blocks are there?

SSC MTS 15/05/2023 (Morning)

(a) 315 (b) 675 (c) 270 (d) 745

Q.123. If $(x + y) : (y + z) : (z + x) = 3 : 5 : 7$ and $(x + y + z) = 45$, then what is the value of z ?

SSC MTS 16/05/2023 (Evening)

(a) 18 (b) 30 (c) 27 (d) 20

Q.124. A sum of Rs. 120 is divided among p, q and r in such a way that p gets Rs. 9 more than q and q gets Rs. 6 more than r. What is the ratio of shares of p, q and r respectively?

SSC MTS 17/05/2023 (Morning)

(a) 18 : 15 : 13 (b) 15 : 12 : 10
(c) 16 : 13 : 11 (d) 11 : 8 : 10

Q.125. If 20% of a number is $\frac{1}{6}$ th of another number, what is the ratio of first number to the second number?

SSC MTS 17/05/2023 (Evening)

(a) 4 : 7 (b) 5 : 6 (c) 5 : 3 (d) 4 : 5

Q.126. The salaries of Kabir and Vicky are in the ratio 5:2. If the increments of 20% and 12% are allowed respectively in their salaries, then what will be the new ratio of their salaries?

SSC MTS 18/05/2023 (Evening)

(a) 25 : 73 (b) 81 : 28
(c) 75 : 28 (d) 91 : 25

Q.127. If $a : b : c = 5 : 6 : 4$, then what is the value of $(a + b + c) : (3a + b - c)$?

SSC MTS 19/05/2023 (Evening)

(a) 25 : 23 (b) 18 : 23 (c) 15 : 17 (d) 18 : 19

Q.128. If $A : B : C = 2 : 3 : 4$, then what will be the value of $\frac{A}{B} : \frac{B}{C} : \frac{C}{A}$?

SSC MTS 14/06/2023 (Evening)

(a) 9 : 5 : 36 (b) 8 : 7 : 25
(c) 8 : 9 : 24 (d) 12 : 10 : 13

Q.129. The ratio of P, Q and R is 2 : 5 : 7 respectively. If their sum is 840, then what is the difference between P and R?

SSC MTS 20/06/2023 (Afternoon)

(a) 300 (b) 380 (c) 200 (d) 260

SSC CHSL 2022 Tier - 1

Q.130. The expenditure and savings of a person are in a ratio of 4 : 3. His savings increase by $\frac{1}{5}$ of his income and income remains the same. What is the new ratio of his expenditure and savings?

SSC CHSL 10/03/2023 (1st Shift)

(a) 12 : 17 (b) 14 : 17
(c) 13 : 22 (d) 15 : 13

Q.131. Weight of Ayush and Abhishek are in the ratio of 8 : 5. Abhishek's weight increases by 40 percent and the total weight of Ayush and Abhishek both increases by 60 percent. If the total weight becomes 104 kg, then what is the weight of Ayush after the increment?

SSC CHSL 10/03/2023 (3rd Shift)

(a) 60 kg (b) 75 kg (c) 69 kg (d) 72 kg

Q.132. The ratio of income of Jitendra, and Narendra is 4 : 7 and the ratio of their expenditure is 6 : 5 respectively. If Jitendra saves Rs.400 out of Rs.2,800 income, then find the savings of Narendra.

SSC CHSL 13/03/2023 (2nd Shift)

(a) Rs. 2,900 (b) Rs. 2,000
(c) Rs. 3,900 (d) Rs. 4,900

Q.133. The ratio of incomes of P and Q is 1 : 2. Ratio of income of Q and R is 3 : 2. If one-third of P's income is Rs. 4400 less than the half of P's income, then what is Q's income?

SSC CHSL 14/03/2023 (2nd Shift)

(a) Rs. 56600 (b) Rs. 52800
(c) Rs. 46600 (d) Rs. 41200

Q.134. If the difference between the two numbers is 20 percent of their sum, then what is the ratio of larger number and smaller number?

SSC CHSL 14/03/2023 (4th Shift)

(a) 6 : 5 (b) 5 : 4 (c) 4 : 3 (d) 3 : 2

Q.135. The ratio of incomes of C and D is 3 : 4 respectively. The sum of their expenditure is Rs. 12000. Savings of C is

the same as that of the expenditure of D. What is the sum of their savings?

SSC CHSL 16/03/2023 (2nd Shift)

(a) Rs. 25000 (b) Rs. 15000
(c) Rs. 20000 (d) Rs. 16000

Q.136. Rs. 13000 is divided among X, Y and Z such that 2 times of X's share is equal to 3 times of Y's share which is equal to 4 times of Z's share. What is the share of Y?

SSC CHSL 17/03/2023 (2nd Shift)

(a) Rs. 3200 (b) Rs. 4800
(c) Rs. 5600 (d) Rs. 4000

Q.137. The sum of the present ages of a father and his son is 94 years. 8 years ago their respective ages were in the ratio of 2 : 1. After 10 years what will be the ratio of ages of father and son?

SSC CHSL 17/03/2023 (4th Shift)

(a) 35 : 22 (b) 14 : 13 (c) 25 : 18 (d) 16 : 15

Q.138. Rs. 43,120 is divided among P, Q and R such that P's share is $\frac{4}{7}$ of total share of Q and R together and Q's share is $\frac{2}{5}$ of total share of P and R together.

What is the share of R?

SSC CHSL 21/03/2023 (3rd Shift)

(a) Rs.15120 (b) Rs.14220
(c) Rs.16400 (d) Rs.18050

SSC CGL 2022 Tier - 2

Q.139. The ratio of the incomes of two employees is 7 : 4, and the ratio of their expenditures is 3 : 1. If each of them manages to save ₹4,800 per month, find the sum of their monthly incomes (in ₹).

SSC CGL Tier II (02/03/2023)

(a) 21120 (b) 20120 (c) 21150 (d) 18150

Q.140. The ratio of marks obtained by Rajesh, Rakesh and Ramesh in an exam is 2 : 4 : 9. What are the marks obtained by Rakesh and Ramesh, if Rajesh scored 30 marks in the exam?

SSC CGL Tier II (03/03/2023)

(a) Rakesh = 40, Ramesh = 90
(b) Rakesh = 60, Ramesh = 135
(c) Rakesh = 120, Ramesh = 180
(d) Rakesh = 90, Ramesh = 40

SSC CGL 2022 Tier - 1

Q.141. By mistake, instead of dividing ₹702 among Ram, Ramesh, and Naresh in the ratio $\frac{1}{3} : \frac{1}{4} : \frac{1}{6}$, it was divided in the ratio of 3 : 4 : 6. Who gained the most and by how much?

SSC CGL 02/12/2022 (1st Shift)

(a) Ram ₹158 (b) Ramesh ₹158

(c) Naresh ₹168 (d) Ram ₹168

Q.142. Ratio between the present ages of A and B is 2 : 3, respectively. After 5 years, the ratio between their ages will be 3 : 4. What is B's age at present?

SSC CGL 02/12/2022 (4th Shift)

- (a) 20 years (b) 15 years
(c) 10 years (d) 25 years

Q.143. Find the mean proportion between $(6 + \sqrt{8})$ and $(3 - \sqrt{2})$.

SSC CGL 03/12/2022 (2nd Shift)

- (a) $2\sqrt{12}$ (b) $\sqrt{14}$
(c) $(6 - \sqrt{8})$ (d) $\sqrt{15} - 7$

Q.144. Which smallest positive number should be subtracted from each of 9 and 13 so that 18 is the third proportion to them?

SSC CGL 03/12/2022 (3rd Shift)

- (a) 2 (b) 4 (c) 3 (d) 1

Q.145. The ratio of the present ages of Ram and Ramesh is 3 : 5. After 7 years the ratio of their ages will be 4 : 5. Find the present age of Ramesh.

SSC CGL 05/12/2022 (1st Shift)

- (a) 5 years (b) 15 years
(c) 7 years (d) 12 years

Q.146. A varies directly as the positive square root of B, and inversely as the cube of C. If A = 15, when B = 27 and C = 2, then find B when A = 9 and C = 2.

SSC CGL 05/12/2022 (2nd Shift)

- (a) $\frac{281}{42}$ (b) $\frac{243}{25}$ (c) $\frac{264}{37}$ (d) $\frac{275}{51}$

Q.147. The fourth proportion to 12, 24 and 27 is the same as the third proportion to A and 36. What is the value of A?

SSC CGL 05/12/2022 (3rd Shift)

- (a) 22 (b) 24 (c) 26 (d) 20

Q.148. The mean proportional of a and b is c. What is the mean proportional of a^2c and b^2c ?

SSC CGL 06/12/2022 (1st Shift)

- (a) c (b) 3c (c) c^3 (d) c^2

Q.149. The mean proportional of 6 and 54 is _____ more than 15.

SSC CGL 07/12/2022 (1st Shift)

- (a) 3 (b) 6 (c) 4 (d) 5

Q.150. The third proportional of 'a' and $\frac{b^4}{4a}$ is:

SSC CGL 09/12/2022 (1st Shift)

- (a) $\frac{b^8}{16a^2}$ (b) $\frac{b^8}{8a^2}$ (c) $\frac{b^8}{8a^3}$ (d) $\frac{b^8}{16a^3}$

Q.151. Three positive numbers are in the

ratio of 4 : 5 : 7, and the sum of their squares is 15,210. Find the sum of the three numbers.

SSC CGL 09/12/2022 (3rd Shift)

- (a) 148 (b) 156 (c) 126 (d) 208

Q.152. X is directly proportional to the square of Y. When X is 12, then Y is 2. Find the value of X when Y is 3.

SSC CGL 12/12/2022 (4th Shift)

- (a) 30 (b) 9 (c) 27 (d) 18

SSC CPO 2022 Tier - 1

Q.153. Four numbers are in the ratio 1 : 2 : 3 : 5. Two times their sum is 44. Find the ratio of the numbers obtained by subtracting 3 from the cube of each number.

SSC CPO 09/11/2022 (Morning)

- (a) 2 : 31 : 204 : 999
(b) 3 : 13 : 208 : 992
(c) 11 : 67 : 219 : 1003
(d) 5 : 61 : 213 : 997

Q.154. The price of a bat and a ball are in the ratio 9 : 5. The price of the bat is Rs.380 more than the price of the ball. Find the price of the ball.

SSC CPO 10/11/2022 (Afternoon)

- (a) Rs.470 (b) Rs.485
(c) Rs.475 (d) Rs.480

Q.155. The salaries of A, B and C are in ratio of: $\frac{3}{2}, \frac{6}{5}, \frac{4}{3}$. The salary of A and B together is Rs.40,500. By what percentage is the salary of A more than that of C?

SSC CPO 10/11/2022 (Evening)

- (a) 12% (b) 13% (c) 11.5% (d) 12.5%

Q.156. The ratio of a father's age to his Mother's age is 3 : 2. The product of the numbers representing their age is 486. The ratio of their ages after 5 years will be:

SSC CPO 10/11/2022 (Evening)

- (a) 32:24 (b) 32:21 (c) 32:22 (d) 32:23

SSC CGL 2021 Tier - 2

Q.157. Salaries of B, C, D and E are in the ratio of 2 : 3 : 4 : 5 respectively. Their salaries are increased by 20 percent, 30 percent, 40 percent and 50 percent respectively. If the increased salary of D is ₹560, then what is the sum of the original salaries of B, C, D and E?

SSC CGL Tier II (08/08/2022)

- (a) ₹1400 (b) ₹1560 (c) ₹1820 (d) ₹1260

Q.158. The cost of a diamond is directly proportional to the square of its weight. The cost of a 14 gm diamond is ₹2560. This diamond got broken down into two

pieces in the ratio of 5 : 9. How much loss percent is incurred due to this breakage? (correct to two decimal places)

SSC CGL Tier II (08/08/2022)

- (a) 53.47 percent (b) 49.71 percent
(c) 45.92 percent (d) 55.41 percent

Q.159. A sum of ₹ 1250 has to be distributed among A, B, C and D. Total share of B and D is equal to $(\frac{14}{11})$ of total share of A and C. Share of D is half of share of A. Share of C is 1.2 of share of A. What are the shares of A, B, C and D respectively?

SSC CGL Tier II (08/08/2022)

- (a) ₹ 250, ₹ 575, ₹ 300, ₹ 125
(b) ₹ 350, ₹ 525, ₹ 300, ₹ 125
(c) ₹ 250, ₹ 525, ₹ 300, ₹ 125
(d) ₹ 250, ₹ 575, ₹ 300, ₹ 175

SSC Selection Post (Phase - X)

Q.160. If the mean proportional between 5 and 125 is x and the mean proportional between 4^4 and 6^2 is y, find the value of $2x : y$.

Graduate Level 04/08/2022 (Shift - 2)

- (a) 25 : 48 (b) 25 : 128
(c) 25 : 64 (d) 25 : 96

Q.161. The mean proportional between 75 and x is five times the mean proportional between 15 and 26. The value of x is:

Higher Secondary 05/08/2022 (Shift - 2)

- (a) 100 (b) 120 (c) 110 (d) 130

SSC MTS 2021 Tier - 1

Q.162. The ratio of the number of hens to the number of goats in farm A is 5 : 4 and that on farm B is 5 : 7. What could be the ratio of the total number of hens to the total number of goats in farms A and B taken together if total no. of animals in farm A and B are equal?

SSC MTS 05/07/2022 (Morning)

- (a) 35 : 37 (b) 1 : 2 (c) 9 : 11 (d) 7 : 11

Q.163. A person has some coins of ₹10, ₹5, and ₹2 denominations. The ratio of the products of the numbers of ₹10 and ₹5 coins, the numbers of ₹5 and ₹2 coins, and the numbers of ₹2 and ₹10 coins is 3 : 4 : 2 respectively. What could be the minimum amount of money this person has?

SSC MTS 06/07/2022 (Morning)

- (a) ₹52 (b) ₹88 (c) ₹68 (d) ₹74

Q.164. What number must be subtracted from each of the numbers 55, 23, 43 and

19, so that the new values are in proportion?

SSC MTS 06/07/2022 (Morning)

(a) 9 (b) 11 (c) 7 (d) 5

Q.165. In a housing society, the ratio of two - wheelers and 4 - wheelers is 11 : 9, Among the four-wheelers, one-fourth are petrol vehicles and the remaining 81 are diesel vehicles. How many two -wheelers are there in this housing society?

SSC MTS 06/07/2022 (Evening)

(a) 88 (b) 132 (c) 121 (d) 99

Q.166. There are two numbers 'x' and 'y'. 40% of 'x' is equal to 60% of 'y'. If the average of 'x' and 'y' is 225, then what is the difference between 'x' and 'y'?

SSC MTS 07/07/2022 (Morning)

(a) 85 (b) 80 (c) 75 (d) 90

Q.167. The ratio between the volumes of Sphere 1 and Sphere 2 is 27 : 64. Find the ratio of the surface areas of Sphere 1 and Sphere 2.

SSC MTS 07/07/2022 (Afternoon)

(a) 9 : 16 (b) 16 : 9 (c) 3 : 4 (d) 1 : 3

Q.168. In a municipal election three candidates, Santosh, Ravi, and Vipin, contested and the valid votes they got were in the ratio of 11 : 4 : 9. If the winner defeated his nearest rival by 8350 votes, then how many valid votes were polled in that election?

SSC MTS 07/07/2022 (Evening)

(a) 1,02,000 (b) 2,10,000
(c) 1,90,000 (d) 1,00,200

Q.169. A sum of money is distributed among A, B and C in the ratio 8 : 4 : 3. If A divides his part between D and E in the ratio 9 : 7, then what is the ratio of the share of C to that of D?

SSC MTS 08/07/2022 (Afternoon)

(a) 4 : 5 (b) 3 : 4 (c) 2 : 3 (d) 6 : 7

Q.170. A number when subtracted from 3 times the second number, gives a result equal to $\frac{3}{4}$ of the first number. What is the ratio of the first number to the second number?

SSC MTS 11/07/2022 (Morning)

(a) 11 : 6 (b) 12 : 7 (c) 9 : 5 (d) 10 : 7

Q.171. The difference of two numbers is 24 while the sum of those two numbers is 120. Which of the options below can be the ratio of those two numbers?

SSC MTS 11/07/2022 (Evening)

(a) 3 : 5 (b) 3 : 2 (c) 4 : 5 (d) 5 : 8

Q.172. In a bag there are coins of ₹5, ₹10 and ₹20 denominations. The total number of coins in the bag is 240. If the

number of coins of ₹5, ₹10, ₹20 denominations are in the ratio of 2 : 3 : 5, then what is the total amount of money in the bag?

SSC MTS 12/07/2022 (Afternoon)

(a) ₹3,540 (b) ₹4,600
(c) ₹4,620 (d) ₹3,360

Q.173. A certain sum is divided between four people A, B, C and D in the ratio 3 : 4 : 7 : 8 respectively. If the amount received by D is ₹408 more than the amount received by B, what is the total amount received by C and D?

SSC MTS 13/07/2022 (Evening)

(a) ₹1,630 (b) ₹1,510
(c) ₹1,610 (d) ₹1,530

Q.174. Three numbers a, b, and c are said to be in continued proportion if $a : b :: b : c$. What number must be added to each of the numbers 1, 7 and 25 so that the resulting sums, in that order, may be in continued proportion?

SSC MTS 14/07/2022 (Evening)

(a) 2 (b) 4 (c) 6 (d) 5

Q.175. A household of three humans has cats and dogs in the ratio 1:2. If the total number of legs in the household is 42, what is the total number of cats in the household?

SSC MTS 14/07/2022 (Evening)

(a) 12 (b) 6 (c) 9 (d) 3

Q.176. When 8 is added to the numerator of a fraction and 12 is added to its denominator, the fraction becomes $\frac{1}{2}$.

When 2 is subtracted from its numerator and denominator, the fraction becomes $\frac{1}{8}$. Find the original fraction.

SSC MTS 15/07/2022 (Morning)

(a) $\frac{9}{19}$ (b) $\frac{3}{10}$ (c) $\frac{4}{7}$ (d) $\frac{5}{11}$

Q.177. If $a : b = b : c$, then what is the value of $a^6 : b^6$?

SSC MTS 15/07/2022 (Evening)

(a) $a^3 : b^3$ (b) $b^3 : ac$ (c) $a^3 : c^3$ (d) $a^2 : b^2$

Q.178. In a bag there are Rs 5 and Rs 10 coins such that the product of their numbers is 135 and the total amount in the bag is Rs 165, what is the amount from all Rs 10 coins?

SSC MTS 18/07/2022 (Evening)

(a) 90 (b) 85 (c) 80 (d) 95

Q.179. There are a total of 60 students in a class and the ratio of the number of boys to that of the girls is 4 : 1, respectively. If 22 of the boys from this class are vegetarians, then find the number of non-vegetarian boys from the

same class.

SSC MTS 19/07/2022 (Evening)

(a) 26 (b) 18 (c) 38 (d) 28

Q.180. The ratio of the quantities imported of steel and iron in 2018 was 3 : 2 and in 2020 it was 1 : 1. The ratio of the quantities imported of steel in 2018 and 2020 was 4 : 3. What was the ratio of the quantities imported of iron in 2018 and 2020?

SSC MTS 20/07/2022 (Morning)

(a) 11 : 12 (b) 7 : 5 (c) 8 : 9 (d) 7 : 9

Q.181. On a tree, there are some parrots and some pigeons in the ratio of 7:9, respectively. After an hour, 8 parrots fly away, and 6 pigeons and 10 sparrows come and sit on the tree. The ratio of the parrots and the pigeons on the tree now is 1 : 4. What is the ratio of the parrots and the sparrows that are now on the tree?

SSC MTS 20/07/2022 (Afternoon)

(a) 1 : 1 (b) 1 : 2 (c) 3 : 5 (d) 4 : 5

Q.182. 4 years ago, the age of Garvit was three times of Yuvansh, but one year ago the age of Garvit was two times that of Yuvansh. What is the difference between the ages of Garvit and Yuvansh?

SSC MTS 21/07/2022 (Morning)

(a) 13 years (b) 11 years
(c) 7 years (d) 6 years

Q.183. Three numbers are in the ratio 2 : 3 : 5. The sum of these numbers is 80. What is the product of these three numbers?

SSC MTS 21/07/2022 (Afternoon)

(a) 13200 (b) 15360 (c) 14400 (d) 15630

Q.184. The ratio of the number of boys to the number of girls in a class is 2 : 5. If 10 boys leave the class and 20 new girls join the class, then the new ratio of boys and girls in the class becomes 4 : 25. What is the number of boys in the class now?

SSC MTS 22/07/2022 (Morning)

(a) 12 (b) 8 (c) 9 (d) 22

Q.185. A sum of ₹3400 is divided among P and Q in such a way that $\frac{3}{4}$ of P's

share is equal to $\frac{2}{3}$ of Q's share. What is the positive difference between the shares of P and Q?

SSC MTS 22/07/2022 (Afternoon)

(a) ₹500 (b) ₹400 (c) ₹200 (d) ₹300

Q.186. Rs 21000 is to be distributed between Raj and Karan such that Raj gets Rs 9000 less than Karan. What is the ratio of the amount received by Raj to

that received by Karan?

SSC MTS 22/07/2022 (Evening)

(a) 2 : 5 (b) 3 : 5 (c) 5 : 3 (d) 5 : 2

Q.187. Four years ago, Shyam's age was $\frac{3}{4}$ times that of Ram. Four years hence,

Shyam's age will be $\frac{5}{6}$ times that of Ram

.What is the present age of Shyam?

SSC MTS 26/07/2022 (Morning)

(a) 24 years (b) 20 years
(c) 16 years (d) 15 years

SSC CHSL 2021 Tier - 1

Q.188. If the ratio of the volumes of two cones is 11 : 16 and the ratio of the radii of their bases is 3 : 4, then the ratio of their heights will be:

SSC CHSL 26/05/2022 (Afternoon)

(a) 11 : 9 (b) 4 : 11 (c) 3 : 16 (d) 9 : 11

Q.189. Soham's initial expenditure and savings were in the ratio 5 : 1. His income increases by 25%, while his savings increase by 20%. If Soham's current expenditure is ₹4,347, what was his initial income?

SSC CHSL 31/05/2022 (Morning)

(a) ₹4,130 (b) ₹4,140
(c) ₹4,150 (d) ₹4,160

Q.190. Two numbers, A and B, are such that the sum of 10% of A and 8% of B is $\frac{3}{5}$ th of the sum of 12% of A and 16% of B.

The ratio of A : B is :

SSC CHSL 31/05/2022 (Afternoon)

(a) 2 : 3 (b) 4 : 7 (c) 2 : 7 (d) 4 : 3

Q.191. The monthly income of two persons is in the ratio 4:5 and their expenditures are in the ratio 7:9. If each saves ₹50 per month, then their monthly incomes are:

SSC CHSL 01/06/2022 (Evening)

(a) ₹400; ₹500 (b) ₹700; ₹900
(c) ₹900; ₹500 (d) ₹400; ₹700

Q.192. Two numbers X and Y are such that the sum of 18% of X and 8% of Y is one-third of the sum of 22% of X and 36% of Y. Find the ratio of X and Y.

SSC CHSL 02/06/2022 (Afternoon)

(a) 2 : 5 (b) 3 : 11 (c) 3 : 8 (d) 8 : 3

Q.193. If 60% of a number is equal to three-fourth of another number, what is the ratio of the first number to the second number?

SSC CHSL 07/06/2022 (Evening)

(a) 3 : 7 (b) 2 : 5 (c) 7 : 3 (d) 5 : 4

Q.194. If y varies directly with x, and y = 61 when x = 42, find y when x = 54

(consider integral value).

SSC CHSL 08/06/2022 (Afternoon)

(a) 86 (b) 78 (c) 82 (d) 72

Q.195. P, Q and R are batsmen. The ratio of the runs scored by them in a certain match was P : Q = 16 : 17 and Q : R = 15 : 16. At the end of the match, they scored a total of 956 runs. The number of runs scored by R is (nearest to an integer):

SSC CHSL 09/06/2022 (Afternoon)

(a) 440 (b) 335 (c) 339 (d) 430

Q.196. A varies jointly with B and C. A = 6 when B = 3 and C = 2. Find A when B = 5 and C = 7.

SSC CHSL 10/06/2022 (Evening)

(a) 17.5 (b) 35 (c) 105 (d) 70

SSC CGL 2021 Tier - 1

Q.197. The ratio of the monthly incomes of A and B is 11 : 13 and the ratio of their expenditure is 9 : 11. If both of them manage to save ₹ 4,000 per month, then find the difference in their incomes (in ₹).

SSC CGL 11/04/2022 (Morning)

(a) 2,500 (b) 3,200 (c) 4,000 (d) 3,000

Q.198. At present, A is younger than B by 8 years. If 4 years ago, their ages were in the ratio of 1 : 2, then what is the present age of B (in years)?

SSC CGL 12/04/2022 (Morning)

(a) 11 (b) 20 (c) 12 (d) 18

SSC CGL 2020 Tier - 2

Q.199. If a, b and c are positive numbers such that $(a^2 + b^2) : (b^2 + c^2) : (c^2 + a^2) = 34 : 61 : 45$, then b - a : c - b : c - a = ____.

SSC CGL Tier II (29/01/2022)

(a) 1 : 2 : 3 (b) 3 : 1 : 2
(c) 2 : 1 : 3 (d) 3 : 2 : 1

Q.200. The income of A is $\frac{2}{3}$ of B's income and the expenditure of A is $\frac{3}{4}$ of B's expenditure.

If $\frac{1}{3}$ of the income of B is equal to the expenditure of A, then the ratio of the savings of A to those of B is:

SSC CGL Tier II (29/01/2022)

(a) 3 : 4 (b) 3 : 5 (c) 5 : 3 (d) 4 : 3

Q.201. Alloy A contains metals x and y only in the ratio 5 : 2, while alloy B contains them in the ratio 3 : 4. Alloy C is prepared by mixing alloys A and B in the ratio 4 : 5. The percentage of x in alloy C is:

SSC CGL Tier II (29/01/2022)

(a) $55\frac{4}{9}$ (b) $55\frac{2}{9}$ (c) $55\frac{1}{9}$ (d) $55\frac{5}{9}$

Q.202. One cup has juice and water in the ratio 5 : 2, while another cup of the same capacity has them in the ratio 7 : 4, respectively. If the contents of both the cups (when full) are poured in a vessel, then what will be the final ratio of water to juice in the vessel?

SSC CGL Tier II (29/01/2022)

(a) 25:52 (b) 52:25 (c) 25:26 (d) 26:25

Q.203. The ratio of the incomes of A and B in the last year was 4 : 3. The ratios of their individual incomes in the last year and the present year are 3 : 4 and 5 : 6, respectively. If their total income in the present year is ₹24.12 lakhs, then the sum of the income (in ₹ lakhs) of A in the last year and that of B in the present year is:

SSC CGL Tier II (29/01/2022)

(a) 20.52 (b) 10.98 (c) 22.17 (d) 21.28

Q.204. Let x, y, z be fractions such that $x < y < z$. If z is divided by x, the result is $\frac{5}{2}$,

which exceeds y by $\frac{7}{4}$. If $x + y + z = 1\frac{11}{12}$

then the ratio of (z - x) : (y - x) is:

SSC CGL Tier II (03/02/2022)

(a) 6 : 5 (b) 9 : 5 (c) 5 : 6 (d) 5 : 9

Q.205. In an examination, the number of students who passed and the number of students who failed were in the ratio 25 : 4. If one more student had appeared and passed and the number of failed students was 3 less than earlier, the ratio of passed students to failed students would have become 22 : 3. What is the difference between the number of students who, initially, passed the examination and the number of students who failed the examination?

SSC CGL Tier II (03/02/2022)

(a) 132 (b) 126 (c) 174 (d) 150

SSC MTS 2020 Tier - 1

Q.206. Suman divided a certain sum between her three daughters in the ratio 2 : 3 : 4. Had she divided the sum in the ratio $\frac{1}{2} : \frac{1}{3} : \frac{1}{4}$, the daughter who got the least share earlier, would have got ₹3,500 more. The sum was:

SSC MTS 05/10/2021(Afternoon)

(a) ₹14,000 (b) ₹14,500
(c) ₹13,750 (d) ₹14,625

Q.207. In a library, the ratio of the number of mathematics books to that of physics books, is the same as the ratio of the number of physics books to that of chemistry books. If there are 144 mathematics books and 100 chemistry

books, then the ratio of the total number of mathematics and physics books to the total number of chemistry and physics books is :-

SSC MTS 05/10/2021 (Evening)

(a) 3 : 2 (b) 6 : 5 (c) 5 : 4 (d) 4 : 3

Q.208. 70 males and 100 females are working in a firm. During one mass recruitment drive, an equal number of males and females are recruited and their ratio becomes 3 : 4. The total number of people working in the firm after recruitment is:

SSC MTS 06/10/2021 (Morning)

(a) 240 (b) 200 (c) 210 (d) 220

Q.209. Between two consecutive years, my income is in the ratio 2 : 5 and expenses are in the ratio 4 : 7. If my income in the second year is ₹75,000 and my expenses in the first year are ₹20,000, then my total savings in the two years together are:

SSC MTS 06/10/2021 (Morning)

(a) ₹55,000 (b) ₹60,000
(c) ₹50,000 (d) ₹40,000

Q.210. A sum of ₹3,780 is divided between A, B and C such that if their shares are decreased by ₹130, ₹150 and ₹200, respectively, then they are in the ratio of 5:2:4. What is the original share of C?

SSC MTS 06/10/2021 (Evening)

(a) ₹1,350 (b) ₹1,330 (c) ₹1,400 (d) ₹1,430

Q.211. A, B and C are three boxes containing marbles in the ratio 3:5:7, and the total number of marbles is 75. If 3 marbles are transferred from B to A, and 5 marbles are transferred from C to B, then the new ratio of the marbles is:

SSC MTS 07/10/2021 (Morning)

(a) 6 : 8 : 11 (b) 5 : 6 : 7
(c) 7 : 9 : 10 (d) 6 : 9 : 10

Q.212. When x is subtracted from each of 21, 22, 60 and 64, the numbers obtained in this order are in proportion. What is mean proportional between $(x + 4)$ and

$\left(\frac{x}{2} - 1\right)$?

SSC MTS 07/10/2021 (Evening)

(a) 8 (b) 5 (c) 6 (d) 12

Q.213. If $(a + b) : (b + c) : (c + a) = 7 : 4 : 5$, and $a + b + c = 16$, then

$(a^2 + b^2 + c^2) : (ab + bc + ca)$ is equal to :

SSC MTS 08/10/2021 (Evening)

(a) 24 : 17 (b) 26 : 19
(c) 27 : 20 (d) 23 : 15

Q.214. Seats in a college for a B.Sc. course for Physics, Chemistry and

Biology, are in the proportion 7 : 5 : 8. There is a proposal to increase these seats by 50%, 40% and 25%, respectively. What will be the proportion of increased seats?

SSC MTS 11/10/2021 (Morning)

(a) 12 : 9 : 33 (b) 21 : 14 : 20
(c) 57 : 45 : 33 (d) 7 : 4 : 4

Q.215. Renuka got $1\frac{1}{2}$ times as many marks in Mathematics as in English. Her total marks in Mathematics, English and Science are 190. If the ratio of the marks in Mathematics and Science is 2 : 3, find her Science marks.

SSC MTS 11/10/2021 (Evening)

(a) 92 (b) 90 (c) 85 (d) 88

Q.216. A man ordered 10 physics books and some chemistry books. The price of a chemistry book is twice the price of a Physics book. While preparing the bill the clerk interchanged the number of physics and chemistry books by mistake, which decreased the bill by $12\frac{1}{2}\%$. The ratio of

the number of physics books to the number of chemistry books in the original order is:

SSC MTS 12/10/2021 (Evening)

(a) 3 : 5 (b) 3 : 4 (c) 4 : 5 (d) 2 : 3

Q.217. The ratio of the monthly income and expenditure of Dinesh is 17 : 14, and his monthly savings are ₹12,000. If his monthly income is increased by ₹10,000 and expenditure is decreased by ₹2,000, then the new ratio of his income and expenditure is:

SSC MTS 12/10/2021 (Evening)

(a) 13 : 8 (b) 13 : 9 (c) 11 : 9 (d) 11 : 7

Q.218. If x is subtracted from each of 52, 47, 20 and 19, the numbers so obtained in this order are in proportion. What is the mean proportional between $(x + 13)$ and $(x - 8)$?

SSC MTS 13/10/2021 (Morning)

(a) 12 (b) 10 (c) 15 (d) 9

Q.219. A certain sum is divided among A, B, C and D such that B's share is $\frac{1}{3}$ of A's share, C's share is 40% of B's share, and D's Share is 50% of C's share. If the difference between the shares of B and D is ₹1,600, then the sum is:

SSC MTS 13/10/2021 (Morning)

(a) ₹9,600 (b) ₹9,400 (c) ₹9,000 (d) ₹9,200

Q.220. In an election, the votes cast for two candidates were in the ratio 4 : 9. If the successful candidate received 9,84,321 votes, then the total votes received by the unsuccessful candidate?

SSC MTS 14/10/2021 (Morning)

(a) 421797 (b) 437476
(c) 571279 (d) 912577

Q.221. Eight years ago, the ratio of the ages of A and B was 4 : 5 and the ratio of their ages, 12 years hence, will be 13 : 15.

The present age (In years) of B is:

SSC MTS 14/10/2021 (Afternoon)

(a) 52 (b) 46 (c) 48 (d) 56

Q.222. A, B and C are boxes with marbles in the ratio 2 : 3 : 4. The total number of marbles in the boxes is 360. The above ratio can be changed to 3 : 8 : 7 by which of the following?

(i) By transferring 20 marbles from A to B and 20 marbles from C to B.

(ii) By transferring 20 marbles from A to B and 10 marbles from C to B.

(iii) By transferring 40 marbles from A to B and 20 marbles from C to B

(iv) By transferring 30 marbles from A to B and 30 marbles from C to B.

SSC MTS 18/10/2021 (Morning)

(a) (ii) (b) (iv) (c) (i) (d) (iii)

Q.223. A sum of ₹3,900 is divided among P, Q and R such that R receives two-fifths of P's share and Q receives three times of R's share. What is Q's share (in ₹)?

SSC MTS 18/10/2021 (Afternoon)

(a) 1,500 (b) 600 (c) 1,800 (d) 1,200

Q.224. The number of students in three sections of Grade 10 in a school are in the proportion 3 : 5 : 8. If 15, 30 and 15 more students are admitted in the three sections, respectively, the new proportion becomes 4 : 7 : 9. The total number of students before the new admissions is:

SSC MTS 18/10/2021 (Afternoon)

(a) 320 (b) 160 (c) 240 (d) 400

Q.225. The price of a semi-precious stone weighing 48 g, is ₹2,30,400. It is broken into two pieces whose weights are in the ratio 5 : 7. If the price is proportional to the square of the weight, then the loss (in ₹) incurred, is:

SSC MTS 18/10/2021 (Afternoon)

(a) 1,12,000 (b) 40,000
(c) 1,18,400 (d) 78,400

Q.226. x and y are two numbers having opposite signs such that, $x^2 : y^2 = 49 : 64$. What is the value of $(5x - 6y) : (6x - 7y)$?

SSC MTS 18/10/2021 (Evening)

(a) 44 : 65 (b) 13 : 14
(c) 94 : 117 (d) 83 : 98

Q.227. In an examination, the ratio of the number of candidates who passed to those who failed was 5 : 2. If the number of failed candidates had been 14 more,

then the ratio of the number of passed candidates to those who failed would have been 4 : 3. The number of candidates who appeared in the examination was:

SSC MTS 20/10/2021 (Afternoon)

(a) 126 (b) 98 (c) 70 (d) 100

Q.228. The ratio of the number of students in Science to that in Humanities, in Grade 12 of 360 students, is 2 : 3. When some new students are admitted, the number of Science students increases by 18 and the ratio of the Science students to Humanities students changes to 3 : 5. What is the number of new Humanities students admitted?

SSC MTS 20/10/2021 (Evening)

(a) 54 (b) 45 (c) 63 (d) 72

Q.229. The ratio of the present ages of a mother and daughter is 3 : 1. Five years ago, the ratio of their ages was 5 : 1. Find the age of the mother 41 years hence.

SSC MTS 22/10/2021 (Afternoon)

(a) 68 years (b) 65 years

(c) 71 years (d) 70 years

Q.230. The ratio of the present incomes of X and Y is 5 : 8 and the ratios of their individual incomes, in the last year and in the present year, are 2 : 3 and 3 : 4, respectively. If their total income for the last year was ₹56,000, then what is the present income of Y?

SSC MTS 26/10/2021 (Afternoon)

(a) ₹48,000 (b) ₹36,000

(c) ₹40,000 (d) ₹30,000

Q.231. A shopkeeper placed a supply order for jeans, shirts and t-shirts at the rates of ₹500, ₹350 and ₹200 per piece respectively. He buys articles of each type in the ratio 1 : 2 : 3 for ₹46,800. For how many t-shirts did he place the order?

SSC MTS 27/10/2021 (Morning)

(a) 104 (b) 26 (c) 78 (d) 52

Q.232. The present ages of a mother and her son are in the ratio of 6 : 1. When the son becomes as old as his mother now, then the ratio of his father's age to his mother's age is 13 : 11. When the son becomes as old as his father now, then the sum of the ages of the father and son will be 115 years. The present age of the father is:

SSC MTS 27/10/2021 (Evening)

(a) 40 years (b) 45 years

(c) 42 years (d) 44 years

Q.233. The ratio of the incomes of A and B is 1 : 2 and that of their expenditure is 2 : 3. If 80% of B's expenditure is equal to the income of A, then what is the ratio of

the savings of B to the savings of A ?

SSC MTS 02/11/2021 (Afternoon)

(a) 2 : 9 (b) 9 : 2 (c) 7 : 3 (d) 3 : 7

Q.234. In an examination, the ratio of the number of candidates who passed, to those who failed was 5 : 2. If the number of failed candidates had been 14 more, then the ratio of the number of passed candidates to those who failed would have been 4 : 3. How many candidates passed the examination?

SSC MTS 02/11/2021 (Afternoon)

(a) 126 (b) 70 (c) 100 (d) 98

Q.235. The father of a girl is three years older than her mother. Six years ago, the average age of the girl and her parents was 32 years, and the age of the father was twice the present age of his daughter. What is the present age (in years) of the mother ?

SSC MTS 02/11/2021 (Evening)

(a) 48 (b) 42 (c) 45 (d) 49

Q.236. The ratio of two numbers is 4 : 5.

If one is subtracted from the first number, and two is added to the second number, then the ratio becomes 3 : 4. What will be the ratio when eight and four are, respectively, added to the first and the second number?

SSC MTS 02/11/2021 (Evening)

(a) 2 : 3 (b) 8 : 9 (c) 6 : 7 (d) 1 : 5

Q.237. A batter missed her century by four runs in an innings, in which the scoring shots in the form of sixes, fours, threes, two's and singles, were in the proportion 1 : 3 : 2 : 7 : 10. How many runs did she score in singles?

SSC MTS 02/11/2021 (Evening)

(a) 20 (b) 16 (c) 10 (d) 12

SSC CGL 2020 Tier - 1

Q.238. A, B and C divide a certain sum of money among themselves. The average of the amounts with them is 4520. Share of A is $10\frac{2}{3}\%$ more than share of B and

$33\frac{1}{3}\%$ less than share of C. What is the share of B (in rupee)?

SSC CGL 13/08/2021 (Evening)

(a) 3500 (b) 5976 (c) 3600 (d) 3984

Q.239. The ratio of present ages of A and B is 7 : 8. After 6 years from now, the ratio of their ages will be 8 : 9. If C's present age is 10 years more than the present age of A, then the present age (in years) of C is:

SSC CGL 17/08/2021 (Evening)

(a) 56 (b) 52 (c) 59 (d) 45

Q.240. Alloy A contains metal x and y in the ratio 5 : 2 and alloy B contains these metals in the ratio 3 : 4. Alloy C is prepared by mixing A and B in the ratio 4 : 5. The percentage of y in alloy C is:

SSC CGL 18/08/2021 (Evening)

(a) $44\frac{4}{9}\%$ (b) $33\frac{4}{9}\%$ (c) $66\frac{4}{9}\%$ (d) $55\frac{5}{9}\%$

Q.241. Fourth proportion to 12, 18, 6 is equal to the third proportion to 4, k. What is the value of k?

SSC CGL 23/08/2021 (Morning)

(a) 6 (b) $4\sqrt{3}$ (c) 6.5 (d) 4

Q.242. Monthly salaries of Anil and Kumud are in the ratio 19 : 17, If Anil and Kumud get salary hike of Rs. 2000 and Rs. 1000 respectively, then the ratio in their salaries becomes 8 : 7. What is the present salary of Kumud (in Rs) ?

SSC CGL 24/08/2021 (Afternoon)

(a) 18000 (b) 38000 (c) 34000 (d) 35000

SSC CHSL 2020 Tier - 1

Q.243. If 9685 is divided into three parts in such a way that one-fourth of the first part, one-third of the second part and one-sixth of the third part are equal, then what is the first part?

SSC CHSL 15/04/2021 (Afternoon)

(a) 4470 (b) 2253 (c) 2980 (d) 2235

Q.244. A bag contains coins of denominations Rs.1, Rs.2 and Rs.5 in the ratio of 4 : 5 : 8. If the total value of these coins is Rs.432, then what is the number of Rs.2 coins?

SSC CHSL 05/08/2021 (Morning)

(a) 40 (b) 30 (c) 50 (d) 60

Q.245. The ratio of the incomes of A and B is 7 : 5. A and B save Rs. 4,000 and Rs. 3,500, respectively. If the expenditure of B is half the expenditure of A, then the total income of A and B (in Rs.) will be:

SSC CHSL 05/08/2021 (Afternoon)

(a) 10,000 (b) 13,500 (c) 12,000 (d) 15,000

Q.246. In a proportion, the product of the first and fourth terms is 70 and that of the second and third items is $3.5y$. The value of y is.

SSC CHSL 09/08/2021 (Morning)

(a) 17 (b) 20 (c) 15 (d) 22

Q.247. A sum of Rs. x is divided among A, B and C such that the ratio of the shares of A and C is 8 : 7 and that of B and C is 3 : 2. If the difference between the shares of A and B is Rs. 240, then what is the value of x?

SSC CHSL 12/08/2021 (Morning)

(a) 2544 (b) 2580 (c) 2490 (d) 2448

SSC CPO 2020 Tier - 1

Q.248. Seven years ago, the ratio of the ages of A and B was 4 : 5. Eight years hence, the ratio of the ages of A and B will be 9 : 10. What is the sum of their present ages in years?

SSC CPO 23/11/2020 (Morning)
(a) 41 (b) 82 (c) 32 (d) 56

Q.249. In a school, $\frac{5}{12}$ of the number of students are girls and the rest are boys, $\frac{4}{7}$ of the number of boys are below 14 years of age, and $\frac{2}{5}$ of the number of girls are 14 years or above 14 years of age. If the number of students below 14 years of age is 1120, then the total number of students in the school is:
SSC CPO 25/11/2020 (Evening)
(a) 1820 (b) 1900 (c) 1920 (d) 1290

SSC CGL 2019 Tier - 2

Q.250. The sum of the present ages of father and a son is 52 years. Four years hence, the son's age will be $\frac{1}{4}$ that of the father. What will be the ratio of the ages of the son and father, 10 years from now?

SSC CGL Tier II (15/11/2020)
(a) 2 : 7 (b) 2 : 5 (c) 3 : 8 (d) 1 : 3

Q.251. The numerator of a fraction is 3 more than the denominator. When 5 is added to the numerator and 2 is subtracted from the denominator, the fraction becomes $\frac{8}{3}$. When the original fraction is divided by $5\frac{1}{2}$, the fraction so obtained is:

SSC CGL Tier II (15/11/2020)
(a) $\frac{2}{3}$ (b) $\frac{1}{4}$ (c) $\frac{1}{2}$ (d) $\frac{3}{4}$

Q.252. A person divided a certain sum between his three sons in the ratio 3 : 4 : 5. Had he divided the sum in the ratio $\frac{1}{3}$, $\frac{1}{4}$, $\frac{1}{5}$, the son, who got the least share earlier, would have got ₹1,188 more. The sum (in ₹) was :

SSC CGL Tier II (15/11/2020)
(a) 6,840 (b) 6,768 (c) 7,008 (d) 5,640

Q.253. The ratio of the monthly income of X and Y is 5 : 4 and that of their monthly expenditure is 9 : 7. If the income of Y is equal to the expenditure of X, then what is the ratio of the savings of X and Y ?

SSC CGL Tier II (15/11/2020)
(a) 6 : 7 (b) 8 : 9 (c) 7 : 6 (d) 9 : 8

Q.254. If $a : b : c = \frac{1}{4} : \frac{1}{3} : \frac{1}{2}$, then

$$\frac{a}{b} : \frac{b}{c} : \frac{c}{a} = ?$$

SSC CGL Tier II (16/11/2020)
(a) 12 : 9 : 8 (b) 8 : 9 : 24
(c) 9 : 8 : 24 (d) 9 : 12 : 8

SSC CHSL 2019 Tier - 1

Q.255. The sum of three numbers is 79. If the ratio of the first number to the second number is 4 : 7 and that of the second number to the third number is 4 : 5, then the second number is:
SSC CHSL 12/10/2020 (Evening)
(a) 15 (b) 28 (c) 35 (d) 12

Q.256. If $2145 : x :: 3003 : 42$, then the value of y so that $x : 2508 :: y : 11704$, is
SSC CHSL 13/10/2020 (Morning)
(a) 212 (b) 156 (c) 140 (d) 96

Q.257. The ratio of man's age to his father's age is 4:5, and the ratio of his age to his son's age is 6:1. Four years ago these ratios were 11:14 and 11:1, respectively. The ratio of the age of the grandfather to that of the grandson 12 years from now will be:

SSC CHSL 13/10/2020 (Afternoon)
(a) 12 : 5 (b) 14 : 3 (c) 18 : 5 (d) 18 : 7

Q.258. Six years ago, the average of the ages of Ravi, Mohan and Govind was 32 years. If Shyam joins them now, the average of the ages of all four of them is 36 years. The present age of Shyam is:
SSC CHSL 14/10/2020 (Morning)
(a) 35 years (b) 32 years
(c) 40 years (d) 30 years

Q.259. The perimeter of a rectangular field is 32 meters and its sides are in the ratio 5 : 3. Then the sides of the field are:
SSC CHSL 26/10/2020 (Afternoon)
(a) 10 m and 6 m (b) 9 m and 7 m
(c) 12 m and 10 m (d) 5 m and 3 m

Q.260. The length and breadth of a rectangle are in the ratio 5 : 3. If the length is 8 m more than the breadth, what is the area of the rectangle?
SSC CHSL 17/03/2020 (Afternoon)

(a) $240 m^2$ (b) $380 m^2$
(c) $360 m^2$ (d) $400 m^2$

SSC CGL 2019 Tier - 1

Q.261. The ratio of the number of boys to the number of girls in a school of 640 students is 5 : 3. If 30 more girls are admitted in the school, then how many

more boys should be admitted so that the ratio of boys to that of the girls, becomes 14 : 9

SSC CGL 03/03/2020 (Morning)
(a) 15 (b) 30 (c) 25 (d) 20

Q.262. The ratio of boys and girls in a group is 7 : 6. If 4 more boys join the group and 3 girls leave the group, then the ratio of boys to girls becomes 4:3. What is the total number of boys and girls initially in the group?

SSC CGL 04/03/2020 (Afternoon)
(a) 117 (b) 78 (c) 91 (d) 104

Q.263. When x is added to each of 10, 16, 22 and 32, the numbers obtained in this order are in proportion? What is the mean proportional between the numbers $(x + 1)$ and $(3x + 1)$?

SSC CGL 05/03/2020 (Afternoon)
(a) 15 (b) 12 (c) 10 (d) 9

Q.264. In a wallet, there are notes of the denominations of ₹ 10 and ₹ 50. The total notes is 12. The number of ₹ 10 and ₹ 50 notes are in the ratio of 1 : 2. Total money in the wallet is:

SSC CGL 06/03/2020 (Afternoon)
(a) ₹280 (b) ₹360 (c) ₹440 (d) ₹110

SSC CPO 2019 Tier - 1

Q.265. A sum of Rs. 12000 is divided between A, B, C and D such that the ratio of share of A and B is 8 : 9, that of B and C is 2 : 3 and that of C:D is 9 : 13. What is the difference between the shares of B and D ?

SSC CPO 13/12/2019 (Morning)
(a) Rs. 1320 (b) Rs. 2400
(c) Rs. 2520 (d) Rs. 2760

SSC CGL 2018 Tier - 2

Q.266. If $(a + b) : (b + c) : (c + a) = 7 : 6 : 5$ and $a + b + c = 27$, then what will be the value of $\frac{1}{a} : \frac{1}{b} : \frac{1}{c}$?

SSC CGL Tier II (11/09/2019)
(a) 3 : 6 : 4 (d) 3 : 2 : 4
(c) 4 : 3 : 6 (d) 3 : 4 : 2

Q.267. Two numbers are in the ratio 3 : 5. If 13 is subtracted from each, the new numbers are in the ratio 10 : 21. If 5 is added to each of the original numbers, then the ratio becomes :

SSC CGL Tier II (13/09/2019)
(a) 19 : 30 (b) 23 : 33 (c) 4 : 5 (d) 24 : 35

SSC CGL 2018 Tier - 1

Q.268. If x is subtracted from each of 23, 39, 32 and 56, the numbers so obtained

in this order are in proportion. What is the mean proportional between $(x + 4)$ and $(3x + 1)$?

SSC CGL 04/06/2019 (Afternoon)

(a) 15 (b) 10 (c) 12 (d) 14

Q.269. If $a : b = 3 : 2$, then $(5a + 2b) : (3a + 4b)$ is equal to:

SSC CGL 10/06/2019 (Afternoon)

(a) 16 : 15 (b) 8 : 7 (c) 19 : 17 (d) 17 : 14

Answer Key :-

1.(d)	2.(a)	3.(d)	4.(d)
5.(a)	6.(a)	7.(d)	8.(c)
9.(a)	10.(b)	11.(c)	12.(c)
13.(c)	14.(b)	15.(b)	16.(c)
17.(a)	18.(b)	19.(d)	20.(b)
21.(c)	22.(b)	23.(b)	24.(d)
25.(a)	26.(b)	27.(c)	28.(b)
29.(a)	30.(b)	31.(a)	32.(b)
33.(d)	34.(c)	35.(d)	36.(a)
37.(c)	38.(c)	39.(b)	40.(d)
41.(c)	42.(a)	43.(a)	44.(b)
45.(a)	46.(c)	47.(b)	48.(c)
49.(b)	50.(d)	51.(a)	52.(c)
53.(c)	54.(b)	55.(a)	56.(d)
57.(a)	58.(b)	59.(d)	60.(b)
61.(b)	62.(a)	63.(d)	64.(a)
65.(c)	66.(c)	67.(a)	68.(a)
69.(c)	70.(d)	71.(a)	72.(c)
73.(d)	74.(a)	75.(d)	76.(c)
77.(c)	78.(b)	79.(d)	80.(b)
81.(a)	82.(a)	83.(d)	84.(d)
85.(a)	86.(d)	87.(a)	88.(c)
89.(b)	90.(d)	91.(b)	92.(d)
93.(b)	94.(b)	95.(b)	96.(c)
97.(a)	98.(d)	99.(a)	100.(a)
101.(d)	102.(a)	103.(c)	104.(d)
105.(a)	106.(b)	107.(d)	108.(c)
109.(b)	110.(b)	111.(c)	112.(b)
113.(c)	114.(d)	115.(d)	116.(c)
117.(c)	118.(c)	119.(d)	120.(d)
121.(b)	122.(a)	123.(c)	124.(c)
125.(b)	126.(c)	127.(c)	128.(c)
129.(a)	130.(c)	131.(c)	132.(a)
133.(b)	134.(d)	135.(d)	136.(d)
137.(a)	138.(a)	139.(a)	140.(b)
141.(c)	142.(b)	143.(b)	144.(d)
145.(c)	146.(b)	147.(b)	148.(c)

149.(a)	150.(d)	151.(d)	152.(c)
153.(d)	154.(c)	155.(d)	156.(d)
157.(a)	158.(c)	159.(a)	160.(a)
161.(d)	162.(a)	163.(c)	164.(c)
165.(b)	166.(d)	167.(a)	168.(d)
169.(c)	170.(b)	171.(b)	172.(d)
173.(d)	174.(a)	175.(d)	176.(b)
177.(c)	178.(a)	179.(a)	180.(c)
181.(c)	182.(d)	183.(b)	184.(a)
185.(c)	186.(a)	187.(c)	188.(a)
189.(b)	190.(b)	191.(a)	192.(c)
193.(d)	194.(b)	195.(c)	196.(b)
197.(c)	198.(b)	199.(c)	200.(b)
201.(d)	202.(a)	203.(a)	204.(a)
205.(b)	206.(d)	207.(b)	208.(c)
209.(c)	210.(c)	211.(d)	212.(c)
213.(b)	214.(b)	215.(b)	216.(d)
217.(b)	218.(b)	219.(d)	220.(b)
221.(c)	222.(c)	223.(c)	224.(c)
225.(a)	226.(d)	227.(b)	228.(a)
229.(c)	230.(a)	231.(c)	232.(a)
233.(b)	234.(b)	235.(c)	236.(b)
237.(a)	238.(c)	239.(b)	240.(a)
241.(a)	242.(c)	243.(c)	244.(a)
245.(c)	246.(b)	247.(d)	248.(a)
249.(c)	250.(d)	251.(b)	252.(b)
253.(d)	254.(c)	255.(b)	256.(c)
257.(c)	258.(d)	259.(a)	260.(a)
261.(d)	262.(d)	263.(a)	264.(c)
265.(c)	266.(c)	267.(a)	268.(c)
269.(c)			

Solutions:-

Sol.1.(d)

Ratio of their concentration = 10 : 20 : 30

Ratio of their quantity = 2 : 3 : x

∴ Concentration of mixed solution =

$$\frac{10 \times 2 + 20 \times 3 + 30 \times x}{5 + x} = 23$$

$$20 + 60 + 30x = 115 + 23x$$

$$7x = 35, x = 5$$

Sol.2.(a) Total students = 720

$$\text{No. of Boys} = \frac{3}{8} \times 720 = 270$$

$$\text{No. of Girls} = \frac{5}{8} \times 720 = 450$$

According to question

$$\frac{270 + x}{450 + 18} = \frac{2}{3}$$

$$= 810 + 3x = 2 \times 468 \Rightarrow 3x = 936 - 810$$

$$x = \frac{126}{3} = 42$$

Sol.3.(d) According to the question,

$$\text{Ram} = \frac{12}{5} = 2.4, \text{Shyam} = \frac{18}{9} = 2$$

$$\text{Preetham} = \frac{4}{8} = 0.5, \text{Vinay} = \frac{10}{7} = 1.42$$

So Ram has the greater ratio of hens to cows.

Sol.4.(d) Let the number of plastic, wood and glass = $2x$, $3x$ and $5x$

According to question,

$$= 5x - 3x = 144 \Rightarrow x = 72$$

$$\text{Number of plastic woods} = 2x = 144$$

Now,

$$\text{After increasing } 25\% = 144 \times \frac{5}{4} = 180$$

Sol.5.(a) Fourth proportional

$$= \frac{3^{12} \times 4^{12}}{12^{35}} \Rightarrow \frac{3^{12} \times 4^{12}}{3^{35} \times 4^{35}}$$

$$= 3^{-23} \times 4^{-23} = 12^{-23}$$

Sol.6.(a) According to the question,

$$\frac{x+5}{2x+5} = \frac{2}{3} \Rightarrow x = 5$$

Smallest number (x) = 5

Sol.7.(d) Given, $y \propto (x + 3)$

$$y = k_1(x + 3) \dots (I)$$

When $y = 8$, and $x = 1$

$$\Rightarrow 8 = k_1 \times 4 \Rightarrow k_1 = 2$$

$$x \propto \frac{1}{z}$$

$$x = k_2 \times \frac{1}{z} \dots (II)$$

$x = 2$ and $z = 6$.

$$\Rightarrow 2 = k_2 \times \frac{1}{6} \Rightarrow k_2 = 12$$

at $k_2 = 12$ equation (II) become

$$x = \frac{12}{z}, \text{ put this value in eq. (I)}$$

$$y = 2\left(\frac{12}{z} + 3\right) \Rightarrow y = 2\left(\frac{12 + 3z}{z}\right)$$

$$\Rightarrow y \times z = 24 + 6z \Rightarrow z(y - 6) = 24$$

Sol.8.(c) According to the question,

$x + y = 96$ {let $x < y$ }

$$\text{So, } \frac{x+3}{96-x} = \frac{11}{20} \Rightarrow 20x + 60$$

$$= 96 \times 11 - 11x$$

$$31x = 996 \Rightarrow x = \frac{996}{31} = 32.12$$

Largest number (y)

$$= 96 - x = 96 - 32.12 \approx 63.87$$

Sol.9.(a) Given :-

$$5 : 3\frac{1}{3}, 6\frac{1}{4} : 3\frac{1}{8} \text{ and } 3.5 : 4.5$$

$$5 : 3\frac{1}{3} = 5 : \frac{10}{3} \Rightarrow 3 : 2 = 1.5$$

$$6\frac{1}{4} : 3\frac{1}{8} = \frac{25}{4} : \frac{25}{8} \Rightarrow 2 : 1 = 2$$

$$3.5 : 4.5 = 35 : 45 \Rightarrow 7 : 9 = 0.78$$

Required ascending order = $0.78 < 1.5 < 2$

$$\text{OR } 3.5 : 4.5 < 5 : 3\frac{1}{3} < 6\frac{1}{4} : 3\frac{1}{8}$$

Sol.10.(b)

Let income of P, Q, and R is $9x$, $15x$ and $10x$
and savings of P, Q, and R is $10y$, $15y$, $12y$
According to the question,

$$10x - 12y = 10x \times \frac{60}{100}$$

$$10x - 12y = 6x \Rightarrow \frac{x}{y} = \frac{3}{1}$$

Ratio of expenditure of P, Q, and R

$$\begin{aligned} &= (9x - 10y) : (15x - 15y) : (10x - 12y) \\ &= (27 - 10) : (45 - 15) : (30 - 12) \\ &= 17 : 30 : 18 \end{aligned}$$

Sol.11.(c) Fourth proportional

$$\begin{aligned} &= \frac{(a^3 + b^3) \times (a^2 - b^2)}{(a + b)^2 \times (a^3 - b^3)} \\ &= \frac{(a + b)(a^2 - ab + b^2) \times (a + b)(a - b)}{(a + b)^2(a - b)(a^2 + ab + b^2)} \\ &= \frac{(a^2 - ab + b^2)}{(a^2 + ab + b^2)} \end{aligned}$$

Sol.12.(c) According to the question,
($A \times 6\% + B \times 4\%$)

$$= (A \times 7\% + B \times 11\%) \times \frac{2}{5}$$

$$A \times 30\% + B \times 20\% = A \times 14\% + B \times 22\%$$

$$A \times 16\% = B \times 2\% \Rightarrow A : B = 1 : 8$$

Sol.13.(c)

$$\begin{aligned} \text{Ratio} &\rightarrow A : B \\ \text{Initial} &\rightarrow 8x : 5x = 13 \\ \text{Final} &\rightarrow 5x : 8x = 13 \end{aligned}$$

According to the question,

$$\text{Difference in A's share } 8x - 5x = 300$$

$$x = \frac{300}{3} = 100$$

$$\text{Total sum} = (8x + 5x) = 13x = ₹1300$$

Sol.14.(b)

$$\begin{aligned} \text{Required number} &= \frac{ad - bc}{(a + d) - (b + c)} \\ &= \frac{8 \times 9 - 6 \times 2}{(8 + 9) - (6 + 2)} \\ &= \frac{72 - 12}{9} = \frac{20}{3} \end{aligned}$$

$$\text{Now, } (3P - 6) = 14 \text{ and } (9P - 4) = 56$$

$$\text{Mean proportional} = \sqrt{14 \times 56} = 28$$

Sol.15.(b) Net % dividends

$$= \frac{1 \times 10\% + 4 \times 15\% + 5 \times 25\%}{1 + 4 + 5} = 19.5\%$$

$$\begin{aligned} \text{Total dividend income} &= 44,000 \times 19.5\% \\ &= ₹8580 \end{aligned}$$

Sol.16.(c) $A : B : C = 3 : 2 : 5$

Difference between share of C and B

$$= 3 \text{ units} \Rightarrow 3 \text{ units} = ₹300$$

$$\text{Share of A more than B} = 1 \text{ units} = ₹100$$

Sol.17.(a) Mean proportional

$$= \sqrt{(a + b)(a - b)^3 \times (a - b)(a + b)^3}$$

$$\begin{aligned} &= \sqrt{(a - b)^4 \times (a + b)^4} \\ &= (a - b)^2 \times (a + b)^2 \end{aligned}$$

Sol.18.(b)

Ratio - Maths : English : Chemistry

$$\text{Initial} - 6 : 4 : 7$$

$$\text{Final} - 7.2 : 5.6 : 11.2$$

$$\begin{aligned} \text{Required ratio} &= 7.2 : 5.6 : 11.2 \\ &= 9 : 7 : 14 \end{aligned}$$

Sol.19.(d)

$$\text{Blue}_{(\text{men})} : \text{Red}_{(\text{women})} = 3x : 7x;$$

$$\text{Red}_{(\text{men})} : \text{Blue}_{(\text{women})} = 2y : y;$$

$$\frac{3x + y}{7x + 2y} = \frac{35}{76}$$

$$\Rightarrow 228x + 76y = 245x + 70y \Rightarrow \frac{x}{y} = \frac{6}{17}$$

$$\text{Number of men} = 3x + 2y = 52$$

$$\text{Number of women} = 7x + y = 59$$

$$\text{Required ratio} = 52 : 59$$

Sol.20.(b)

$$\text{Ratio} \rightarrow P : Q$$

$$\text{Income} \rightarrow 40 : 16$$

$$\text{Expenditure} \rightarrow 18 : 10$$

$$\text{Savings} \rightarrow 11 : 3$$

Therefore, the ratio of the savings of P
And Q = $11 : 3$

Sol.21.(c) % decrease (X) in the number
of units manufactured by company A =

$$\frac{12500 - 10625}{12500} \times 100$$

$$= \frac{1,875}{12500} \times 100 = 15\%$$

% decrease (Y) in the number of units
manufactured by company

$$B = \frac{34,000 - 30,600}{34,000} \times 100$$

$$= \frac{3,400}{34,000} \times 100 = 10\%$$

$$\text{Required ratio} = 15 : 10 = 3 : 2$$

Sol.22.(b)

Mean proportion of 'a' and 'b' is $\sqrt{ab}$

Mean proportion

$$= \sqrt{\left(\frac{a^3 + b^3}{a - b}\right) \left(\frac{a^2 - b^2}{a^2 - ab + b^2}\right)}$$

$$= \sqrt{\left(\frac{(a + b)(a^2 - ab + b^2)}{a - b}\right) \left(\frac{(a + b)(a - b)}{(a^2 - ab + b^2)}\right)}$$

Mean proportion

$$= \sqrt{\left(\frac{(a + b)}{1}\right) \left(\frac{(a + b)}{1}\right)} = (a + b)$$

Sol.23.(b)

$$\text{Ratio} \rightarrow A : B : C$$

$$\text{No of officers} \rightarrow 5 : 4 : 1$$

$$\text{Average} \rightarrow (600) : (600 + 150) : (600 + 400)$$

So, the average of 40 officers $\rightarrow$

$$= 600 + \frac{(150 \times 4) + (400 \times 1)}{(5 + 4 + 1)}$$

$$= 600 + \frac{1000}{10} = 700$$

$$\begin{aligned} \text{So, the monthly salary of an officer} \\ &= 700 \times 30 = ₹21,000 \end{aligned}$$

Sol.24.(d)

Ratio $\rightarrow$ Income = exp. + Savings

$$A \Rightarrow 12 = 5 + 7 \Rightarrow 4 \times 5$$

$$B \Rightarrow 15 = 7 + 8 \Rightarrow 3 \times 4$$

Difference b/w their expenditure $\rightarrow$

$$(4 \times 5 \times 5) - (3 \times 4 \times 7) \text{ units} = 4500 \text{ Rs.}$$

$$16 \text{ units} = 4500 \text{ Rs.}$$

Income of B ($3 \times 4 \times 15$) units

$$= \frac{4500}{16} \times 180 = 50625 \text{ Rs.}$$

Sol.25.(a)

$$\text{Share of A} = \frac{18400}{46800} = \frac{46}{117} \text{ of the sum.}$$

Let total sum be $117a$, share of A = $46a$

$$\text{Share of A + D} = \frac{8}{13} \times 117a = 72a$$

$$\text{Share of D} = 72a - 46a = 26a$$

$$\text{Share of B + C} = \frac{5}{13} \times 117a = 45a$$

Given, share of B : share of C = $5 : 4$

So, share of B = $25a$

and share of C = $20a$

x = difference of share of A and B

$$= 46a - 25a = 21a.$$

y = Difference of share of C and D

$$= 26a - 20a = 6a$$

$$x - y = 21a - 6a = 15a$$

$$117a = 46,800, a = 400$$

$$\text{So, } 15a = 15 \times 400 = 6000$$

Sol.26.(b) Let price of one pen = x
and price of one pencil = y

$$\text{So, } 5x + 3y = 44 \dots\dots(1)$$

$$3x + 5y = 36 \dots\dots(2)$$

On solving (1) and (2), we get

$$x = 7 \text{ and } y = 3$$

$$\text{So required ratio} = 7 : 3$$

Sol.27.(c)

$$\text{Denomination} \rightarrow ₹2 : ₹5 : ₹10$$

$$\text{Number of coins} \rightarrow 5 : 4 : 1$$

$$\text{Amount Ratio} \rightarrow 10 : 20 : 10$$

$$\text{Total Amount} = 40 \text{ unit} = 80\text{Rs}$$

$$1 \text{ unit} = 2\text{Rs.}$$

$$\text{Number of 5 rupees coin} = 4 \times 2 = 8$$

Sol.28.(b) Number of boys = $\frac{5}{8}$

$$\text{Boys Below 10 years} = \frac{5}{8} \times \frac{1}{3} = \frac{5}{24}$$

$$\text{Girls Below 10 years} = \frac{3}{8} \times \frac{2}{3} = \frac{6}{24}$$

$$\text{Students Below 10 years} = \frac{11}{24}$$

$$\text{Students Above 10 years} = \frac{13}{24}$$

$$\text{Total students} = 260 \times \frac{24}{13} = 480$$

$$\text{Number of boys} = 480 \times \frac{5}{8} = 300$$

Sol.29.(a) Go through option (a)

$$\frac{5+3}{7+5} = \frac{8}{12} = \frac{2}{3} \text{ (which satisfy)}$$

$$\frac{5-1}{7+3} = \frac{4}{10} = \frac{2}{5} \text{ (which satisfy)}$$

Sol.30.(b)

Ratio $\Rightarrow$ Income = expenditure + saving

$$A (80\%) \Rightarrow 5 = 4 + 1 \times 8$$

$$B (85\%) \Rightarrow 20 = 17 + 3 \times 3$$

$$C (75\%) \Rightarrow 4 = 3 + 1 \times 20$$

According to question, Difference of income of A and C (80 - 40) units $\Rightarrow$ (40 units) = 18000 Rs.

Then, Income of B (60 units) = 27000 Rs

Sol.31. (a)

Ratio $\rightarrow A : B$

Last year Income $\rightarrow 4 : 3$

Now,

Last Year : Present Year

$$A \rightarrow (3 : 4) \times 4 \times 5$$

$$B \rightarrow (5 : 6) \times 3 \times 3$$

Last Year : Present Year

$$A \rightarrow 60 : 80$$

$$B \rightarrow 45 : 54$$

According to the question,

$$(80 + 54) \text{ units} = 8.04 \text{ lakh}$$

$$1 \text{ unit} = 0.06 \text{ lakh}$$

$$\text{Last year income of B (45 units)} = 45 \times$$

$$0.06 = 2.7 \text{ lakh}$$

Sol.32.(b)

$$\text{Fractional Value} \rightarrow 66\frac{2}{3}\% = \frac{2}{3}$$

$$\text{Ratio} \rightarrow A : B$$

$$\text{Income} \rightarrow 5 : 7$$

$$\text{Expenditure} \rightarrow 2 : 3$$

$$\text{Savings} \rightarrow 4000 : 5000$$

Now,

$$(5 \times 3) - (7 \times 2) \text{ units}$$

$$= (3 \times 4000) - (2 \times 5000)$$

$$\Rightarrow 1 \text{ unit} = 2000 \text{ Rs.}$$

$$\text{Total income of A and B} \rightarrow (5 + 7) \text{ units}$$

$$12 \text{ units} = 12 \times 2000 = 24000 \text{ Rs.}$$

Sol.33. (d)

Ratio $\rightarrow$ Divisor : Quotient : Remainder

$$\begin{array}{ccc} 7 & : & 12 \\ 3 & & : 2 \end{array}$$

$$21 : 36 : 14$$

According to the question

$$14 \text{ unit} = 14 \Rightarrow 1 \text{ unit} = 1$$

$$21 \text{ unit (Divisor)} = 21$$

$$\text{Required remainder} = \frac{21 \times 36 + 14}{9}$$

$$= 5 \text{ (Rem.)}$$

Sol.34.(c)

Let the age of shyam = 5 units and

age of Ravi = 3 units

According to the question

$$(5 + 3) \text{ unit} = 32 \text{ years} \Rightarrow 1 \text{ unit} = 4 \text{ years}$$

$$\text{Age of Shyam} = 5 \times 4 = 20 \text{ years and}$$

$$\text{Age of Ravi} = 3 \times 4 = 12 \text{ years}$$

$$\text{Now, } \frac{20+x}{12+x} = \frac{7}{5}$$

$$100 + 5x = 84 + 7x \Rightarrow x = 8 \text{ years}$$

Short Trick :-

$$\text{Ratio} \Rightarrow \text{Ravi} : \text{Shyam}$$

$$\text{Present} \Rightarrow 3 : 5$$

$$\begin{array}{cc} +2 & +2 \\ \text{After} \Rightarrow 5 & : 7 \end{array}$$

According to question,

$$\text{Sum of present age (3 + 5) units} = 32 \text{ yrs.}$$

$$\text{Then (2 units)} = 8 \text{ yrs.}$$

Sol.35.(d) Fourth proportional of

$$3, 4, 9 = \frac{9 \times 4}{3} = 12$$

Mean proportional between 2 and 98

$$= \sqrt{2 \times 98} = 14$$

$$\text{Required ratio} = 12 : 14 = 6 : 7$$

Sol.36.(a)

$$\text{notes worth} \rightarrow 5 : 2 : 1$$

$$\text{Amount} \rightarrow 15 : 2 : 3$$

$$\text{No. of notes} = \frac{\text{Total amount}}{\text{note's worth}}$$

$$\text{Ratio of number of notes} = \frac{15}{5} : \frac{2}{2} : \frac{3}{1} = 3 : 1 : 3$$

Sol.37.(c) Let, the incomes of A and B are 2I and 3I respectively.

The expenditure of A and B are 1E and 2E respectively.

According to the question

$$2I = 2E \times \frac{90}{100} \Rightarrow \frac{I}{E} = \frac{9}{10}$$

$$\text{Saving of A} = 2(9) - 1(10) = 8 \text{ unit}$$

$$\text{Saving of B} = 3(9) - 2(10) = 7 \text{ unit}$$

$$\text{Required ratio} = 8 : 7$$

Short Trick :-

$$\text{Ratio} \rightarrow A : B$$

$$\text{Income} \rightarrow 9 \times (2 : 3)$$

$$\text{expenditure} \rightarrow 1 : 2 \times 10$$

$$\text{Saving of A} \rightarrow (9 \times 2) - (1 \times 10) = 8$$

$$\text{Saving of B} \rightarrow (9 \times 3) - (2 \times 10) = 7$$

$$\text{Saving} \rightarrow A : B = 8 : 7$$

Sol.38. (c) Actual ratio = 3 : 4 : 5 : 8

$$\Rightarrow (3 + 4 + 5 + 8) \text{ unit} = 4360$$

$$\Rightarrow 20 \text{ unit} = 4360 \Rightarrow 1 \text{ unit} = 218$$

$$A's \text{ share (3 unit)} = 3 \times 218 = 654$$

$$B's \text{ share (4 unit)} = 4 \times 218 = 872$$

$$C's \text{ share (5 unit)} = 5 \times 218 = 1090$$

$$D's \text{ share (8 unit)} = 8 \times 218 = 1744$$

$$\text{Mistaken ratio} = \frac{1}{3} : \frac{1}{4} : \frac{1}{5} : \frac{1}{8}$$

$$= 40 : 30 : 24 : 15$$

$$= (40 + 30 + 24 + 15) \text{ unit} = 4360$$

$$= 1 \text{ unit} = 40$$

$$A's \text{ share (40 unit)} = 40 \times 40 = 1600$$

$$B's \text{ share (30 unit)} = 30 \times 40 = 1200$$

$$C's \text{ share (24 unit)} = 24 \times 40 = 960$$

$$D's \text{ share (15 unit)} = 15 \times 40 = 600$$

$$\text{Clearly D get } 1744 - 600 = \text{Rs. } 1144 \text{ less}$$

Sol.39. (b)

$$\text{Ratio} \rightarrow A : B : C$$

$$2 : 5 \times 4$$

$$4 : 3 \times 5$$

$$A : B : C = 8 : 20 : 15$$

Sol.40.(d)

$$\text{Ratio} \rightarrow A : B : C$$

$$7 : 12$$

$$8 : 5$$

$$14 : 24 : 15$$

According to the question

$$(15 - 14) = 1 \text{ unit} = 214$$

$$\text{Sum (x) 53 unit} = (14 + 24 + 15) \times 214 = 11342$$

Sol.41.(c) Let the price of the articles be 400 and 500.

According to the question,

$$\frac{400 \times \frac{100+x}{100}}{500 \times \frac{100-30}{100}} = \frac{10}{7}$$

$$\Rightarrow \frac{4(100+x)}{5 \times 70} = \frac{10}{7}$$

$$\Rightarrow 400 + 4x = 500 \Rightarrow x = 25\%$$

Sol.42.(a) Let the ages of A and B 4 years ago be 4x and 5x respectively.

According to the question

$$\frac{4x+12}{5x+12} = \frac{11}{13} \Rightarrow 52x + 156 = 55x + 132$$

$$3x = 24 \Rightarrow x = 8$$

$$\text{Sum of their present age} = 4x + 4 + 5x + 4 = 9(8) + 8 = 80$$

Short Trick :-

$$\text{Ratio of age} \rightarrow A : B$$

$$\text{Before 4 yrs.} \rightarrow 4 : 5 \times 2$$

$$\text{After 8 yrs.} \rightarrow 11 : 13 \times 1$$

$$\text{Difference} \rightarrow (11 \times 1) - (4 \times 2) = 3 \text{ units}$$

$$3 \text{ units} = 12 \text{ yrs.}, 1 \text{ unit} = 4 \text{ years}$$

$$\text{Sum of present age of A and B} \rightarrow$$

$$\{(4 + 5) \times 2\} \text{ units} + 8 \text{ yrs.}$$

$$18 \text{ units} + 8 \text{ yrs.} = 18 \times 4 + 8$$

$$= 72 \text{ yrs.} + 8 \text{ yrs.} = 80 \text{ yrs.}$$

Sol.43.(a)

$$\text{Ratio of share of A, B and C} = 4 : 3 : 6$$

According to question,

$$\text{Difference between B and C} = 3 \text{ units}$$

$$\text{Here 3 units} = ₹ 600$$

$$\therefore \text{ difference between A and B} =$$

$$1 \text{ unit} = \frac{600}{3} \times 1 = ₹ 200$$

Sol.44.(b)

$$\text{Ratio of expenditure and saving} = 2 : 1,$$

$$\text{So income} = 3 \text{ unit}$$

Now,

Income $\rightarrow 3 \times 20\% = 60\%$
 Expenditure $\rightarrow 2 \times 15\% = 30\%$
 Saving $\rightarrow 1 \times ? = 30\%$

$$\therefore ? = \frac{30}{1} = 30\%$$

% increase in her saving = 30%

Sol.45.(a)

Expenditure = price $\times$ consumption

Ratio of price = 3 : 4

Ratio of quantity = 4 : 7

$\therefore$ Ratio of expenditure = 12 : 28 or 3 : 7

Sol.46.(c)

A : B = 3 : 4

B : C = 12 : 13

A : B : C = 9 : 12 : 13

Here total amount = 34 unit = ₹ 7786

$\therefore$ B's amount = 12 unit

$$= \frac{7786}{34} \times 12 = ₹ 2748$$

Sol.47.(b)

Area of triangle = $\frac{1}{2} \times \text{base} \times \text{height}$

Area $\rightarrow 4 : 3$

Height $\rightarrow 6 : 5$

Then ratio of bases $\rightarrow \frac{4}{6} : \frac{3}{5} = 10 : 9$

Sol.48.(c) According to the question,

$$\frac{X}{Y} = \frac{3}{1} \text{ (put the value in equation)}$$

$$\frac{2X + 3Y}{X - 2Y} = \frac{2 \times 3 + 3 \times 1}{3 - 2 \times 1} = 9$$

$$\text{Sol.49.(b)} \quad P \propto \frac{1}{Q} \Rightarrow P = k \times \frac{1}{Q}$$

$$10 = k \times \frac{1}{6} \Rightarrow k = 60$$

$$\text{now, } 15 = 60 \times \frac{1}{Q} \Rightarrow Q = 4$$

Sol.50.(d) Mean proportion

$$= \sqrt{(3 + 2\sqrt{2})^2 \times (3 - 2\sqrt{2})^2}$$

$$= (3 + 2\sqrt{2}) \times (3 - 2\sqrt{2}) \Rightarrow 9 - 8 = 1$$

Sol.51.(a)

Lets two numbers are 6x and 7x

According to the question,

$$7x - 6x = 16 \Rightarrow x = 16$$

Numbers are 6x = 6 $\times$ 16 = 96 and

$$7x = 7 \times 16 = 112$$

Sol.52.(c) According to the question,

Ratio - A : B : C

Profit - 15x : 6x : 5x

$$\text{B's share} = 29120 \times \frac{6x}{15x + 6x + 5x}$$

$$= ₹ 6720$$

Sol.53.(c) According to the question,

Mean proportion of a and $\frac{a}{b^2}$

$$= \sqrt{a \times \frac{a}{b^2}} = \frac{a}{b}$$

Sol.54.(b)

Asha : Bhupinder : Gargi

9 : 5 : 5

2 : 2 : 1

18 : 10 : 5

Hence, shares of Asha

$$= \frac{66000}{(18 + 10 + 5)} \times 18 = ₹ 36,000$$

Sol.55.(a) According to the question,

$$5x - 2x = ₹ 150 \Rightarrow x = ₹ 50$$

$$\text{Hence, total sum} = (2x + 5x + 7x) = 14x$$

$$= ₹ 700$$

Sol.56.(d) p : q = 5 : 2,

$$\text{then } p^2 - q^2 : p^2 + q^2 = 25 - 4 : 25 + 4$$

$$= 21 : 29$$

$$\text{Sol.57.(a)} \quad A : B = \frac{1}{3} : \frac{1}{5} = 5 : 3$$

$$B : C = \frac{1}{7} : \frac{1}{8} = 8 : 7$$

$$C : D = \frac{1}{10} : \frac{1}{21} = 21 : 10$$

A : B : C : D

5 : 3 : 3 : 3

8 : 8 : 7 : 7

21 : 21 : 21 : 10

40 : 24 : 21 : 10

Sol.58.(b)

Marks obtained by the student after

$$\text{studying for one hour} = \frac{60}{4} = 15 \text{ Marks}$$

So the time taken by him to study to get

$$75 \text{ marks} = \frac{75}{15} = 5 \text{ hours}$$

Sol.59.(d) According to the question

Let the numbers be 5x, 2x, 7x

According to the question,

$$(5x)^2 + (2x)^2 + (7x)^2 = 3822$$

$$= 25x^2 + 4x^2 + 49x^2 = 3822$$

$$= 78x^2 = 3822 \Rightarrow x = 7$$

$$\text{So the largest no.} = 7x = 7 \times 7 = 49$$

Sol.60.(b) According to the question,

$$(19 + 6) - (7 + 15) \text{ units} = 72$$

3 units = 72

Smallest number (6 units)

$$= \frac{72}{3} \times 6 = 144$$

Sol.61.(b) Let the savings of T = 1 unit

Then income of T = 8 units

Expenditure of T = (8 - 1) = 7 units

Hence, expenditure of T is 7 times of his savings

Sol.62.(a) Required ratio

$$= 5 \times \frac{120}{100} : 8 \times \frac{115}{100} = 6 : 9.2 = 15 : 23$$

Sol.63.(d) Total students = 1650

Ratio $\rightarrow$ Boys : Girls

6 : 5

$$\text{Number of boys} = 1650 \times \frac{6}{11} = 900$$

$$\text{Number of girls} = 1650 \times \frac{5}{11} = 750$$

After Admission -

$$\text{Total boys} = 900 + 124 = 1024$$

New Ratio $\rightarrow$ Boys : Girls

4 : 3

$$4 \text{ unit} = 1024, 3 \text{ unit} = 768$$

$$\text{Newly admitted girls} = 768 - 750 = 18$$

$$\text{So, required ratio} = 18 : 124 = 9 : 62$$

Sol.64.(a) According to the question,

$$\frac{x + 20}{2x} = \frac{1}{1} \Rightarrow x = 20$$

$$\text{Number of student initially} = 3x = 60$$

Sol.65.(c)

$$\sqrt{A^2 + B^2} : \sqrt{C^2 - B^2}$$

$$= \sqrt{9 + 16} : \sqrt{25 - 16} \Rightarrow \sqrt{25} : \sqrt{9} = 5 : 3$$

Sol.66.(c)

$$\text{Third proportional} = \frac{45 \times 45}{25} = 81$$

Sol.67.(a)

$$\text{Fourth proportional} = \frac{0.6 \times 20}{0.24} = 50$$

Sol.68.(a)

$$3C + 5P = 1080 ₹ \dots \text{e.q. (1)}$$

$$2C + 4P = 840 ₹ \dots \text{e.q. (2)}$$

$$C + P = 240 ₹$$

$$\Rightarrow 2C + 2P = 480 ₹ \dots \text{e.q. (3)}$$

On solving e.q. (3) and e.q. (2), we get

$$2P = 360 \Rightarrow P = 180 ₹ \text{ and } C = 60 ₹$$

$$\text{Required ratio} = 1 : 3$$

Sol.69.(c) Compound ratio :-

$$\frac{3 \times 6}{4 \times 7} = \frac{36}{x} \Rightarrow x = 56$$

Sol.70.(d) Mean proportional (25) = $\sqrt{ab}$

$$\Rightarrow a = \frac{625}{b}$$

$$\text{Third proportional (25)} = \frac{b^2}{a}$$

$$\Rightarrow b^2 = 25a = 25 \times \frac{625}{b}$$

$$\Rightarrow b = 25 \text{ and } a = 25$$

$$\text{Sol.71.(a)} \quad \frac{2x + 8.5}{x + 8.5} = \frac{7}{4}$$

$$x = 8.5 \times 3 = 25.5 \text{ years}$$

Sol.72.(c) Required ratio

$$= \left(\frac{23 + 7}{2}\right) : \left(\frac{23 - 7}{2}\right) = 15 : 8$$

$$\text{Sol.73.(d)} \quad \frac{x+7}{2x+7} = \frac{3}{5}$$

$$x = 7 \times 2 = 14$$

$$\text{Greatest number } (2x) = 28$$

Sol.74.(a)

$$\text{Mean proportional} = \sqrt{7 \times 63} = 21$$

$$\text{Required difference} = 24 - 21 = 3$$

Sol.75.(d)

$$\text{Mean proportional} = \sqrt{3 \times 27} = 9$$

$$\text{Sol.76.(c)} \quad \frac{(6a^2 - 2b^2)}{(b^2 - a^2)} = \frac{(6 \times 25 - 2 \times 49)}{(49 - 25)}$$

$$= \frac{(150 - 98)}{(24)} = \frac{(52)}{(24)} = \frac{(13)}{(6)} = 13 : 6$$

Sol.77.(c)

$$\frac{1}{3} \text{rd of } 15 = 5, \frac{4}{5} \text{th of } 25 = 20$$

$$\text{and } \frac{3}{7} \text{th of } 35 = 15$$

$$\text{Fourth proportional} = \frac{20 \times 15}{5} = 60$$

Sol.78.(b) Ratio of salaries :-

$$\text{Ram : Shyam} = 3 : 4 \text{ or } 15 : 20$$

$$\text{Shyam : Ghanshyam} = 5 : 6 \text{ or } 20 : 24$$

$$\text{Ram : Ghanshyam} = 15 : 24 \text{ or } 5 : 8$$

Sol.79.(d)

$$\text{Fourth proportional} = \frac{3\sqrt{10} \times 5\sqrt{8}}{3\sqrt{5}}$$

$$= \sqrt{2} \times 5\sqrt{8} = 5 \times 4 = 20$$

Sol.80.(b)

$$\text{Mean proportional} = \sqrt{0.049 \times 0.9} = 0.21$$

Sol.81.(a) Let their age be $5x$ and $3x$, then

$$5x \times 3x = 960 \Rightarrow x = 8$$

$$\text{Their present age} = 40 \text{ years and } 24 \text{ years}$$

$$\text{Required ratio} = 40 + 6 : 24 + 6 = 23 : 15$$

Sol.82.(a) Given numbers are in

$$\text{proportional, so } \frac{8}{P} = \frac{16}{18} \Rightarrow P = 9$$

$$\text{Third proportional} = \frac{12 \times 12}{9} = 16$$

Sol.83.(d) Number of male worker

$$= 960 - 420 = 540$$

$$\text{Required ratio} = 540 : 420 = 9 : 7$$

Sol.84.(d) Given:- $x = 45$, $y = 9$, $x = Ky$

$$45 = K \times 9, K = 5$$

$$\text{Now, } x = ?, y = 50$$

$$x = 5 \times 50 = 250$$

Sol.85.(a) According to the question,

$$\frac{a}{b} = \frac{3}{7}$$

$$\text{So, } (2a + 3b) : (3a + 2b)$$

$$= (2 \times 3 + 3 \times 7) : (3 \times 3 + 2 \times 7)$$

$$= 27 : 23$$

Sol.86.(d)

$$\text{Ratio - X : Y : Z}$$

$$4 : 5 : 5$$

$$7 : 7 : 6$$

$$\text{Final - } 28 : 35 : 30$$

$$\text{Share of Z} = \frac{1488}{(28 + 35 + 30)} \times 30 = ₹480$$

Sol.87.(a) Ratio :- Boys : Girls = 11 : 15
difference between the number of boys and girls (4 units) = 200

$$\text{No. of boys (11 units)} = \frac{200}{4} \times 11 = 550 \text{ boys}$$

Sol.88.(c)

$$\text{Ratio - Income} = \text{expenditure} + \text{savings}$$

$$\text{Initial - } 80 = 50 + 30$$

$$\text{Final - } 96 = 65 + 31$$

$$\text{Change \% in savings} = \frac{1}{30} \times 100$$

$$= 3\frac{1}{3} \% \text{ increased}$$

Sol.89.(b)

$$\text{Ratio of ages :- Gyanendra : Arbind}$$

$$6 : 5$$

$$11 \text{ units} = 55 \text{ years}$$

$$\text{Gyanendra ages after 7 years}$$

$$= \left(\frac{55}{11} \times 6\right) + 7 = 37 \text{ yrs.}$$

$$\text{Arbind ages after 7 years}$$

$$= \left(\frac{55}{11} \times 5\right) + 7 = 32 \text{ yrs.}$$

$$\text{So, required ratio} = 37 : 32$$

Sol.90.(d)

$$\text{Ratio - } 0.5 ₹ : 0.25 ₹ : 0.20 ₹$$

$$\text{Coins - } 2 : 3 : 5$$

$$\text{Amount - } 1 : 0.75 : 1$$

$$\text{Required difference} = (1 - 1) = ₹ 0$$

$$\text{Sol.91.(b)} \text{ Third proportional} = \frac{b^2}{a}$$

According to question,

$$\frac{(x-y)^2}{(x^2-y^2)} = \frac{(x-y)(x-y)}{(x+y)(x-y)} = \frac{(x-y)}{(x+y)}$$

Sol.92.(d) 48% of male workers

$$= \frac{2}{3} \times \text{female worker}$$

$$\frac{\text{Male workers}}{\text{Female workers}} = \frac{2}{3} \times \frac{100}{48}$$

$$= \frac{100}{72} = \frac{25}{18}$$

So, required ratio :-

$$\text{Male worker : female worker} = 25 : 18$$

$$\text{Sol.93.(b)} \text{ Fourth proportional} = \frac{bc}{a}$$

According to question,

$$\frac{bc}{a} = \frac{192 \times 216}{144} = 288$$

Sol.94.(b) Let the third number be 100.

According to question,

$$\text{The two numbers} = 117 \text{ and } 150$$

$$\text{Required ratio} = 117 : 150 = 39 : 50$$

Sol.95.(b)

$$\text{According to question, } \frac{96}{a} = \frac{c}{96}$$

Going through all the options, while putting option (b)

$$a = 32 \text{ and } c = 288$$

$$\frac{96}{32} = \frac{288}{96} \Rightarrow 3 = 3 \text{ [satisfy]}$$

Sol.96.(c)

$$\text{Ratio of no.} = \frac{4}{5} : \frac{5}{6} : \frac{9}{10} = 24 : 25 : 27$$

$$\text{Difference between greatest and smallest} = (27 - 24) \text{ units} \Rightarrow 3 \text{ units} = 12$$

$$\text{Middle number (25 units)} = \frac{12}{3} \times 25 = 100$$

Sol.97.(a) Let the length of colored saffron, white and green be $6x$, $7x$ and $8x$

According to question,

$$6x + 7x + 8x = 9030 \Rightarrow x = 430$$

$$\text{Length of green part} = 8x = 8 \times 430 = 3440 \text{ cm}$$

Sol.98.(d) Let the number of storey be x .

According to question,

$$\frac{28}{7} = \frac{48}{x} \Rightarrow x = 12$$

Sol.99.(a) Let the number subtracted be x .

$$\frac{(2-x)}{(3-x)} = \frac{(3-x)}{4}$$

$$\Rightarrow 8 - 4x = 9 + x^2 - 6x$$

$$\Rightarrow x^2 - 2x + 1 = 0 \Rightarrow (x-1)(x-1) = 0$$

$$\Rightarrow (\text{smallest possible number}) x = 1$$

Sol.100.(a) Third proportional to $3x^2$ and

$$4xy = \frac{4xy \times 4xy}{3x^2} = \frac{16}{3} y^2$$

$$\frac{16}{3} y^2 = 48 \Rightarrow y^2 = \frac{48}{16} \times 3 = 9 \Rightarrow y = 3$$

Sol.101.(d)

$$\text{Ratio} \rightarrow \text{Cheese : sauce} = 2 : 1$$

According to the question,

$$\text{Sauce used to make pizza (1 unit)}$$

$$= 15 \text{ cups}$$

$$\text{Cheese to be used (2 unit)} = 30 \text{ cups}$$

Sol.102.(a) Let total Income = 100 units

$$\text{Then, Savings} = 15 \text{ units}$$

$$\text{Expenditure on health, education and food be } 17 \text{ units, } 34 \text{ units and } 34 \text{ units respectively.}$$

According to the question,

$$3 \times (34 \text{ units}) - 100 \text{ units} = 300 \text{ Rs.}$$

$$2 \text{ units} = 300 \text{ Rs.}$$

$$\text{Required difference}$$

$$= (17 \text{ units} - 15 \text{ units}) = 2 \text{ units} = 300 \text{ Rs.}$$

$$\text{Sol.103.(c)} M = \sqrt{16 \times 9} = 12$$

$$N = \frac{15^2}{9} = 25$$

$$5M + 3N = 5 \times 12 + 3 \times 25 = 60 + 75 = 135$$

Sol.104.(d)Ratio of manufacturing $\rightarrow$

Plate : Glasses = 2 : 1

and Glasses : Bowl = 2 : 1

Plate : Glasses : Bowl = 4 : 2 : 1

According to the question ,

Total manufacturing (7 units) = 2380

Required difference (3 units) =

$$\frac{2380}{7} \times 3 = 1020 \text{ units}$$

Sol.105.(a) Let the numbers be $7x$ and $6x$

According to question,

$$(7x)^2 - (6x)^2 = 637$$

$$= 49x^2 - 36x^2 = 637$$

$$= x^2 = 49 \Rightarrow x = 7$$

Therefore, the numbers are 49 and 42.

Sol.106.(b) Ratio of Students $\rightarrow$

Initially = 2 : 3 : 5

Finally = 4 : 5 : 7

Increase in ratio (2 units) = 20 students

Initially total students (10 units) = 100 students

Sol.107.(d) Mean proportional

$$= \sqrt{(3 + \sqrt{5})(15 - \sqrt{125})}$$

$$= \sqrt{(3 + \sqrt{5})5(3 - \sqrt{5})}$$

$$= \sqrt{5(9 - 5)} = 2\sqrt{5}$$

Sol.108.(c) Let the marks of practical and theory be 300 and 700.

According to question,

Required ratio

$$= (700 \times \frac{60}{100}) : (300 \times \frac{80}{100})$$

$$= 420 : 240 = 7 : 4$$

Sol.109.(b) $a : b : c = 4 : 7 : 9 \times 4$

$$\Rightarrow a : b : c = 16 : 28 : 36$$

$$\text{and } b : c : d = 28 : 36 : 21$$

$$\text{or } a : b : c : d = 16 : 28 : 36 : 21$$

Sol.110.(b) Let the number of boys and girls in the village be 500 and 300.

According to question,

Total number of boys and girls who appeared in the examination $\rightarrow$

$$(500 \times \frac{40}{100}) + (300 \times \frac{60}{100}) = 380$$

Total number of boys and girls who did not appear in the examination

$$\rightarrow 800 - 380 = 420$$

$$\text{Required ratio} = 380 : 420 = 19 : 21$$

Sol.111.(c)Volume of cylinder = $\pi r^2 h$ Ratio $\rightarrow$ Initial : Final

$$(\text{radius})^2 \rightarrow (2)^2 : (1)^2$$

$$\text{height} \rightarrow 5 : 2$$

$$\text{Volume} \rightarrow 20 : 2 = 10 : 1$$

Sol.112.(b) Let the numbers be $= 5x, 9x$

According to the question,

$$9x + 6 = 6(5x - 6)$$

$$9x + 6 = 30x - 36 \Rightarrow 42 = 21x \Rightarrow x = 2$$

So the required numbers = $5 \times 2 = 10$,

$$9 \times 2 = 18$$

Sol.113.(c)

$$\text{Smallest number} = 750 \times \frac{7}{8+7} = 350$$

Sol.114.(d) 3rd proportion of $a, b = \frac{b^2}{a}$

$$\text{So the value of } x \Rightarrow 27 = \frac{x^2}{3} \Rightarrow x = 9$$

$$\text{and the value of } y \Rightarrow 8 = \frac{y^2}{2} \Rightarrow y = 4$$

$$x = 9, y = 4$$

Required ratio = 9 : 4

$$\text{Sol.115.(d)} \quad \frac{J}{4} = \frac{K}{5} = \frac{L}{3}$$

$$J = 4, K = 5, L = 3$$

According to the question,

$$(L + J)^2 : (J + K)^2 : (K + L)^2$$

$$(3 + 4)^2 : (4 + 5)^2 : (5 + 3)^2$$

$$49 : 81 : 64$$

Sol.116.(c) According to the question,

$$\frac{3x - 2500}{2x - 2500} = \frac{8}{5}$$

$$15x - 12500 = 16x - 20000$$

$$x = 7500$$

$$\text{Sum of their salary } (5x) = 5 \times 7500 = ₹ 37500$$

Sol.117.(c) Let numbers be $2x, 3x$ and $5x$

According to the question,

$$(2x)^2 + (3x)^2 + (5x)^2 = 342$$

$$38x^2 = 342 \Rightarrow x^2 = 9 \Rightarrow x = 3$$

$$\text{Largest number } (5x) = 15$$

Sol.118.(c) Mean proportion of 24 and 6

$$\Rightarrow (A) = \sqrt{24 \times 6} = 12$$

Mean proportion of 81 and 9

$$\Rightarrow (B) = \sqrt{81 \times 9} = 27$$

$$\text{Then, } (3A + B) = 3 \times 12 + 27 = 63$$

Sol.119.(d) 1st no. : 2nd no. : 3rd no.

$$3 : 4 : 4$$

$$3 : 3 : 5$$

$$9 : 12 : 20$$

According to the question,

$$(9x \times 20x) \text{ unit} = ₹ 4500$$

$$x^2 = \frac{450}{18} \Rightarrow x = \sqrt{\frac{225}{9}} = 5$$

$$\text{Sum of three numbers} = (41x) = 41 \times 5 = 205$$

Sol.120.(d) Ratio $\rightarrow P : Q : R$

$$\text{Original} \rightarrow 5 : 7 : 8$$

$$\text{Faulty} \rightarrow 9 : 7 : 4$$

Here, Difference between Original and Faulty amount of $R = 4$ unit

$$(9 + 7 + 4) \text{ unit} = 6000$$

$$4 \text{ unit} = \frac{6000 \times 4}{20} = \text{Rs. } 1200$$

Sol.121.(b) Total amount = 50×80

$$\text{Number of children} = \frac{50 \times 80}{50} = 80$$

Sol.122.(a)

Ratio - red : blue : yellow

$$2 : 5$$

$$15 : 7$$

$$\text{Final} - 6 : 15 : 7$$

Number of yellow blocks

$$= 1260 \times \frac{7}{6+15+7} = 315$$

Sol.123.(c)

$$(x + y) : (y + z) : (z + x) = 3 : 5 : 7$$

$$\text{Let, } x + y = 3a, y + z = 5a, z + x = 7a$$

$$2(x + y + z) = 15a \text{ -----e.q. (1)}$$

$$45 = \frac{15a}{2} \Rightarrow a = 6$$

$$(x + y) = 3a = 18$$

$$\text{Now, } (x + y + z) = 45$$

$$18 + z = 45 \Rightarrow z = 27$$

Sol.124.(c) Let share of r be x .Then, share of q and $p = (x+6)$ and $(x+15)$

According to question,

$$x + (x + 6) + (x + 15) = 120$$

$$3x = 99 \Rightarrow x = 33$$

$$\text{Required ratio} = 48 : 39 : 33 = 16 : 13 : 11$$

Sol.125.(b) According to the question,

$$x \times \frac{1}{5} = y \times \frac{1}{6}$$

$$\text{Ratio of the numbers} \Rightarrow x : y = 5 : 6$$

Sol.126.(c)

$$\text{New salary of Kabir} = 5 \times \frac{120}{100} = 6 \text{ unit}$$

$$\text{New salary of Vickey} = 2 \times \frac{112}{100}$$

$$= 2.24 \text{ unit}$$

$$\text{Required ratio} = 600 : 224 = 75 : 28$$

Sol.127.(c) $a : b : c = 5 : 6 : 4$ then,

$$(a + b + c) : (3a + b - c) = (5 + 6 + 4) : (3 \times 5 + 6 - 4) = 15 : 17$$

Sol.128.(c)

$$A : B : C = 2 : 3 : 4$$

$$\frac{A}{B} : \frac{B}{C} : \frac{C}{A} = \frac{2}{3} : \frac{3}{4} : \frac{4}{2}$$

$$= 8 : 9 : 24$$

Sol.129.(a) According to the question,

$$(2 + 5 + 7) \text{ unit} = 840 \Rightarrow 1 \text{ unit} = 60$$

$$\text{Difference between P and R } (7 - 2) \text{ unit} = 5 \times 60 = 300$$

Sol.130.(c) According to the question,
Ratio $\rightarrow$ Income : expenditure : savings

Initial $\rightarrow 7 : 4 : 3$

Change $\rightarrow -1.4 \quad +1.4$

Final $\rightarrow 7 : 2.6 : 4.4$

His savings increased $\frac{1}{5}$ th of their total

income and income remains same

Therefore, the new ratio of his expenditure and savings = $2.6 : 4.4 = 13 : 22$

Sol.131.(c) Let, weight of Ayush and Abhishek be $8x$ and $5x$ respectively.
Increased weight of Abhishek =

$$5x \times \frac{140}{100} = 7x$$

Overall increased weight

$$\rightarrow (8x + 5x) \times \frac{160}{100} = 20.8x$$

According to the question,

$$20.8x = 104 \text{ kg.} \Rightarrow x = 5 \text{ kg.}$$

Increased weight of Ayush

$$\rightarrow 20.8x - 7x = 13.8x$$

$$13.8x = 13.8 \times 5 = 69 \text{ kg.}$$

Sol.132.(a) Let the income of Jitendra and Narendra = $4x$ and $7x$

And expenditure of Jitendra and Narendra = $6y$ and $5y$

According to the question,

$$4x = 2800 \rightarrow x = 700$$

$$\text{Savings of Jitendra} \Rightarrow 4x - 6y = 400$$

$$\Rightarrow y = 400$$

$$\text{Savings of Narendra} \Rightarrow 7x - 5y$$

$$= 4900 - 2000 = 2900 \text{ Rs.}$$

Sol.133.(b) Let income of P = x

According to the question;

$$\frac{1}{3}x + 4400 = \frac{1}{2}x \Rightarrow \frac{1}{6}x = 4400$$

$$x = \text{Rs. } 26400$$

$$\frac{P}{Q} = \frac{1}{2} \Rightarrow Q = 26400 \times 2 = \text{Rs. } 52800$$

Sol.134.(d) Let the larger number be x and the smaller number be y .

According to question,

$$x - y = \frac{20}{100} \times (x + y) \Rightarrow \frac{x - y}{x + y} = \frac{1}{5}$$

$$x - y = 1 \quad \dots\dots\dots \text{e.q. (1)}$$

$$x + y = 5 \quad \dots\dots\dots \text{e.q. (2)}$$

On solving e.q. (1) and (2); we get

$$x = 3 \text{ and } y = 2$$

required ratio, $x : y = 3 : 2$

Sol.135.(d) Total expenditure = 12000₹

According to the question,

Expenditure of D = Saving of C = x

Then, Expenditure of C = $(12000 - x)$

Ratio $\rightarrow$ C : D

Income $\rightarrow$ 3 : 4

Expenditure $\rightarrow (12000 - x) : x$

Saving $\rightarrow x : (y)$

In 'C' case $\Rightarrow$ Income - Expenditure = Savings

$$3 \text{ units} - (12000 - x) = x$$

$$\Rightarrow 1 \text{ unit} = 4000 \text{ ₹}$$

$$\text{Total Income (7 units)} = 28000 \text{ ₹}$$

Now,

$$\text{total saving} = 28000 - 12000 = 16000 \text{ ₹}$$

Short Trick :-

Ratio $\rightarrow$ C : D

Income $\rightarrow$ 3 : 4

Expenditure $\rightarrow$ 2 : 1

Saving $\rightarrow$ 1 : 3

$$\text{Total expenditure (3 units)} = 12000 \text{ ₹}$$

$$\text{Total Savings (4 units)} = 16000 \text{ ₹}$$

Sol.136.(d) According to the question,

$$\text{Share of } 2x = 3y = 4z \rightarrow x : y : z = 12 : 8 : 6$$

According to the question,

$$\text{Total amount } (12x + 8x + 6x) = 13000$$

$$\Rightarrow x = 500$$

$$\text{Then, share of } y (8x) = 4000 \text{ Rs.}$$

Sol.137.(a) Let current age of father be x and son be y

$$x + y = 94 \quad \dots\dots\dots \text{e.q. (1)}$$

$$(8 - x) = 2(8 - y)$$

$$2y - x = 8 \quad \dots\dots\dots \text{e.q. (2)}$$

On adding e.q. (1) and e.q. (2); we get

$$y = 34 \text{ and } x = 60$$

$$\text{Required ratio} = 70 : 44 = 35 : 22$$

Sol.138.(a) Ratio $\rightarrow$

$$P : Q + R = 4 : 7 \times 7$$

$$Q : P + R = 2 : 5 \times 11$$

Total amount of P, Q, R = 77 units

$$\text{And R's unit} = 77 - (7 \times 4 + 11 \times 2) = 27 \text{ units}$$

$$77 \text{ units} = ₹43120$$

$$(R) 27 \text{ units} = \frac{43120}{77} \times 27 \rightarrow ₹15120$$

Sol.139.(a)

Ratio $\rightarrow$ A : B

Income $\rightarrow$ 7 : 4 $\times 2$

Expenditure $\rightarrow$ 3 : 1 $\times 3$

$$\text{Savings of A} = \{(7 \times 2) - (3 \times 3)\}$$

$$= 5 \text{ units} \Rightarrow 5 \text{ units} = 4800 \text{ ₹}$$

Total Income of A and B

$$= (7 + 4) \times 2 = 22 \text{ units}$$

$$22 \text{ units} \rightarrow \frac{4800}{5} \times 22 = 21120 \text{ ₹}$$

Sol.140.(b) Ratio of their marks $\rightarrow 2 : 4 : 9$

According to the question,

Marks obtained by Rajesh (2 units) = 30

Marks obtained by Rakesh (4 units) = 60

Marks obtained by Ramesh (9 units) = 135

Sol.141.(c)

Ram : Ramesh : Naresh

$$\frac{1}{3} : \frac{1}{4} : \frac{1}{6} \Rightarrow 4 : 3 : 2$$

Ratio $\rightarrow$ Ram : Ramesh : Naresh

Initial $\rightarrow 4 : 3 : 2 = 9 \text{ units} \times 13$

$$\text{Final} \rightarrow 3 : 4 : 6 = 13 \text{ units} \times 9$$

Make units equal as money divided is same

Ram : Ramesh : Naresh

$$52 : 39 : 26 = 117 \text{ units}$$

$$27 : 36 : 54 = 117 \text{ units}$$

$$-25u \quad -3u \quad +28u$$

According to question,

$$117 \text{ units} = \text{Rs. } 702$$

$$\text{Then, } 28 \text{ units} = \text{Rs } 168$$

Clearly, Naresh got Rs.168 More

Sol.142.(b) Let, the present age of A and B is $2x$ and $3x$

Now, according to the question,

After 5 years

$$\frac{2x + 5}{3x + 5} = \frac{3}{4} \Rightarrow 8x + 20 = 9x + 15$$

$$x = 5 \text{ yrs.}$$

$$\text{B present age} = 3x = 3 \times 5 = 15 \text{ yrs.}$$

Short Trick:-

$$\begin{array}{ccc} \text{Present age} & \Rightarrow & A : B \\ & & 2 : 3 \\ & +5 & +5 \\ +5 \text{ years} & \Rightarrow & 3 : 4 \\ & & 1 \text{ unit} = 5 \text{ years} \end{array}$$

Then, B's present age = 3 units

$$= 3 \times 5 \text{ yrs} = 15 \text{ yrs.}$$

Sol.143.(b) Mean proportional between a and b is $\sqrt{ab}$.

Mean proportional of $(6 + \sqrt{8})$ and

$$(3 - \sqrt{2}) = \sqrt{(6 + \sqrt{8})(3 - \sqrt{2})}$$

$$= \sqrt{(18 - 6\sqrt{2} + 3\sqrt{8} - \sqrt{16})}$$

$$= \sqrt{(18 - 6\sqrt{2} + 6\sqrt{2} - 4)} \Rightarrow \sqrt{14}$$

Sol.144.(d) If c is the third proportion of

the number a and b , then $c = \frac{b^2}{a}$

Let, x be the no. which is subtracted from 9 and 13 so that 18 is the third proportional to them.

According to question,

$$\frac{(13 - x)(13 - x)}{(9 - x)} = 18$$

$$169 - 26x + x^2 = 162 - 18x$$

$$7 - 8x + x^2 = 0$$

$$x^2 - 7x - x + 7 = 0$$

$$\Rightarrow (x - 7)(x - 1) = 0 \Rightarrow x = 7 \text{ and } x = 1$$

Since, smallest positive number to be subtracted. Hence, $x = 1$

Sol.145.(c) Let the present ages of ram and ramesh are $3x$ and $5x$ respectively

Their ages after 7 years = $3x + 7$ and $5x + 7$, According to question,

$$\frac{3x + 7}{5x + 7} = \frac{4}{5} \Rightarrow 5x = 7 \Rightarrow x = \frac{7}{5}$$

Now, present age of ramesh = $5x$

$$= 5 \times \frac{7}{5} = 7 \text{ years}$$

Sol.146.(b) $A \propto \sqrt{B} \propto \frac{1}{C^3}$

$$A = \frac{k \times \sqrt{B}}{C^3} \Rightarrow 15 = \frac{k \times \sqrt{27}}{2^3}$$

$$k = \frac{120}{3\sqrt{3}} \dots\dots\dots \text{e.q. (1)}$$

Again ,

$$A = \frac{k \times \sqrt{B}}{C^3} \Rightarrow 9 = \frac{120 \times \sqrt{B}}{3\sqrt{3} \times 2^3}$$

$$\sqrt{B} = \frac{9\sqrt{3} \times 8}{40} \Rightarrow B = \frac{243}{25}$$

Sol.147.(b) Fourth proportional of 12, 24 and 27 = $\frac{24 \times 27}{12} = 54$

Third proportional of A and 36

$$= \frac{36 \times 36}{A} = 54$$

$$\text{Then, } A = \frac{36 \times 36}{54} = 24$$

Sol.148.(c)

Mean proportional of a and b is 'c'

$$= \sqrt{ab} \Rightarrow c^2 = ab$$

Now, Mean proportional of a^2c and b^2c

$$= \sqrt{a^2c \times b^2c} = \sqrt{a^2c^2b^2} = abc = c^3$$

Sol.149.(a) Mean proportion of 6 and 54

$$\Rightarrow \sqrt{6 \times 54} = 18$$

so, 18 is 3 more than 15.

Sol.150.(d) Third proportional of a and

$$\frac{b^4}{4a} = \frac{\left(\frac{b^4}{4a}\right)^2}{a} = \frac{b^8}{16a^2 \times a} = \frac{b^8}{16a^3}$$

Sol.151.(d)

Let the three no's be 4x, 5x and 7x

According to question,

$$(4x)^2 + (5x)^2 + (7x)^2 = 15,210$$

$$16x^2 + 25x^2 + 49x^2 = 15,210$$

$$90x^2 = 15,210 \Rightarrow x = \sqrt{\frac{15,210}{90}} = 13$$

So, the sum of three no's = 16x

$$= 16 \times 13 = 208$$

Sol.152.(c)

As $X \propto Y^2 \Rightarrow X = kY^2$ where k is constant

$$\Rightarrow 12 = k \times 2^2 \Rightarrow 12 = 4k \Rightarrow k = 3$$

$$\text{When } Y = 3, \text{ then, } X = 3 \times 3^2 = 27$$

Sol.153.(d)

Let the four no's be x, 2x, 3x and 5x

According to the question,

$$2(x + 2x + 3x + 5x) = 44$$

$$\Rightarrow 11x = 22 \Rightarrow x = \frac{22}{11} = 2$$

So, the no's are 2, 4, 6 and 10

$$\text{Required ratio} = 2^3 - 3 : 4^3 - 3 : 6^3 - 3 : 10^3 - 3 = 5 : 61 : 213 : 997$$

Sol.154.(c) Let the price of bat and ball

be 9x and 5x respectively

According to question,

$$9x - 5x = 380 \Rightarrow 4x = 380 \Rightarrow x = \frac{380}{4} = 95$$

$$\text{So, the price of ball} = 5x = 95 \times 5 = ₹475$$

Sol.155.(d) LCM of (2, 5, 3) = 30

Then, the ratio of the salary of A, B and C

$$= \frac{3}{2} \times 30 : \frac{6}{5} \times 30 : \frac{4}{3} \times 30$$

$$= 45 : 36 : 40$$

$$\text{Required percentage} = \frac{45 - 40}{40} \times 100$$

$$= \frac{5}{40} \times 100 = 12.5\%$$

Sol.156.(d) Let the age of father and mother be 3x and 2x respectively

According to the question,

$$(3x) \times (2x) = 486$$

$$\Rightarrow 6x^2 = 486 \Rightarrow x^2 = \frac{486}{6} = 81$$

$$\Rightarrow x = \sqrt{81} = 9 \text{ yrs.}$$

Then, the age of father and mother

$$= 9 \times 3 = 27 \text{ yrs and } 2 \times 9 = 18 \text{ yrs.}$$

So, the ratio of the ages after 5 yrs.

$$= (27 + 5) : (18 + 5) = 32 : 23$$

Sol.157.(a)

Ratio $\rightarrow B : C : D : E$

Initial Salaries $\rightarrow 20 : 30 : 40 : 50$

Final salary $\rightarrow 24 : 39 : 56 : 75$

According to the question,

Increased salary of D = 56 unit

56 unit = ₹560

Sum of original salaries of B, C, D and E

$$= \frac{560}{56} \times 140 = ₹1,400$$

Sol.158.(c) It is given that, $CP \propto W^2$

$$\text{Initial weight of a diamond} = (5 + 9)^2$$

$$= 14^2 = 196 \text{ unit}$$

After break down, weight of a diamond

$$= 5^2 + 9^2 = 25 + 81 = 106 \text{ unit}$$

Required loss% in weight of diamond

$$= \frac{196 - 106}{196} \times 100$$

$$= \frac{90}{196} \times 100 = 45.918 \sim 45.92\%$$

Sol.159.(a) According to the question,

$$D = \frac{A}{2}, C = 1.2A$$

$$\text{Also, } B + D = \frac{14}{11} (A + C)$$

$$B + \frac{A}{2} = \frac{14}{11} (A + 1.2A)$$

$$B + \frac{A}{2} = \frac{14}{11} (2.2A)$$

$$B + 0.5A = 2.8A \Rightarrow B = 2.3A$$

So, ratio of sharing among A : B : C : D

$$= A : 2.3A : 1.2A : 0.5A = 1 : \frac{23}{10} : \frac{12}{10}$$

$$: \frac{5}{10} = 10 : 23 : 12 : 5$$

$$\text{Share of A} = \frac{10}{50} \times 1250 = ₹250$$

$$\text{Share of B} = \frac{23}{50} \times 1250 = ₹575$$

$$\text{Share of C} = \frac{12}{50} \times 1250 = ₹300$$

$$\text{Share of D} = \frac{5}{50} \times 1250 = ₹125$$

Hence, option (a) is the correct answer.

$$\text{Sol.160.(a)} \quad x = \sqrt{125 \times 5} = \sqrt{625} = 25$$

$$y = \sqrt{4^4 \times 6^2} = 4^2 \times 6 = 96$$

$$\text{So, } 2x : y = 25 \times 2 : 96 = 50 : 96 = 25 : 48$$

Sol.161.(d) According to question

$$\Rightarrow \sqrt{75x} = 5 \times \sqrt{15 \times 26}$$

$$\Rightarrow 5\sqrt{3x} = 5 \times \sqrt{15 \times 26}$$

$$\Rightarrow \sqrt{3x} = \sqrt{3} \times \sqrt{5 \times 26} \Rightarrow x = 5 \times 26 = 130$$

Sol.162.(a)

No. of hens : No. of goats

$$\text{Farm A} \quad 5 : 4 = 9) \times 4$$

$$\text{Farm B} \quad 5 : 7 = 12) \times 3$$

$$\text{No. of hens in farm A and B} = 5 \times 4 + 5 \times 3 = 20 + 15 = 35$$

No of goats in farm A and B

$$= 4 \times 4 + 7 \times 3 = 16 + 21 = 37$$

So, the required ratio = 35 : 37

Sol.163.(c) Let the no. of coins of ₹10, ₹5 and ₹2 be a, b, c respectively

According to question,

$$ab : bc : ca = 3 : 4 : 2$$

$$a : b : c = 3 \times 2 : 3 \times 4 : 4 \times 2$$

$$= 6 : 12 : 8 \text{ or } 3 : 6 : 4$$

So, the minimum amount of money that

$$\text{person has} = 3 \times 10 + 6 \times 5 + 4 \times 2$$

$$= 30 + 30 + 8 = ₹68$$

Sol.164.(c)

The difference between (55 and 43) = 12

The difference between (23 and 19) = 4

$$\text{Ratio of their difference} \rightarrow \frac{12}{4}$$

$$\text{Difference} = (12 - 4) = 8 \text{ units}$$

Again ,

the difference between 55 and 23 = 32

Making previous difference equal to 32,

$$\frac{12 \times 4}{4 \times 4} = \frac{48}{16}$$

Now , compare with initial fraction we

$$\text{get, } \frac{55}{23} \rightarrow \frac{48}{16}$$

Clearly, we can see that (55 - 48) = 7 must be subtracted from each of the given no. to get the values in proportion.

Short trick :- required number

$$= \frac{ad - bc}{(a + d) - (b + c)} = \frac{55 \times 19 - 23 \times 43}{(55 + 19) - (23 + 43)}$$

$$= \frac{1045 - 989}{8} = 7$$

Sol.165.(b)

Ratio $\rightarrow$ 2-wheelers : 4-wheelers

$$11 : 9$$

$$\text{No. of petrol vehicle} = 9 \times \frac{1}{4} = 2.25 \text{ unit}$$

$$\text{No. of diesel vehicle} = 9 - 2.25 = 6.75 \text{ unit}$$

$$6.75 \text{ unit corresponds to } 81$$

$$9 \text{ unit corresponds to } \frac{81}{6.75} \times 9 = 108$$

$$11 \text{ unit corresponds to } \frac{108}{9} \times 11 = 132$$

$$\text{Number of 2-wheelers in the housing society} = 11 \text{ unit} = \frac{108}{9} \times 11 = 132$$

Sol.166.(d) The two numbers are x and y
According to question,
40% of $x = 60\%$ of y

$$\frac{x}{y} = \frac{60}{40} = \frac{3}{2}$$

$$\text{The average of } x \text{ and } y = \frac{3+2}{2} = \frac{5}{2}$$

$$= 2.5 \text{ unit which corresponds to } 225$$

$$\text{So, the difference of } x \text{ and } y = 1 \text{ unit}$$

$$1 \text{ unit} = \frac{225}{2.5} = 90$$

Sol.167.(a)

$$\frac{\text{volume of sphere 1}}{\text{volume of sphere 2}} = \frac{\frac{4}{3}\pi r_1^3}{\frac{4}{3}\pi r_2^3}$$

$$\Rightarrow \frac{27}{64} = \frac{r_1^3}{r_2^3} \quad \text{So, } \frac{r_1}{r_2} = \frac{3}{4}$$

$$\text{Surface area of sphere 1} = 4\pi r_1^2$$

$$= 4\pi \times 3^2 = 36\pi$$

$$\text{Surface area of sphere 2} = 4\pi r_2^2$$

$$= 4\pi \times 4^2 = 64\pi$$

$$\text{So, the required ratio} = 36\pi : 64\pi = 9 : 16$$

Sol.168.(d)

$$\text{Ratio} \rightarrow \text{Santosh} : \text{Ravi} : \text{Vipin}$$

$$\text{Valid Votes} \rightarrow 11 : 4 : 9$$

According to question,

$$\text{Winner - nearest rival} = (11 - 9) \text{ unit}$$

$$= 2 \text{ unit}$$

$$2 \text{ unit} = 8350 \text{ votes}$$

$$\text{Total valid votes polled} = \frac{8350}{2} \times 24$$

$$= 1,00,200$$

Sol.169.(c)

$$\begin{array}{ccc} A & : & B & : & C \\ 8 & : & 4 & : & 3 \end{array} (\times 2)$$

$$16 : 8 : 6$$

A divides his parts

$$\begin{array}{ccc} A=(16) & & A=D+E \\ \swarrow & & \searrow \\ D=9 & & E=7 \end{array}$$

$$16\text{unit}=16\text{unit}$$

Therefore,

$$C : D = 6 : 9 = 2 : 3$$

Sol.170.(b) let the first number = a

And second number = b

According to the question ,

$$3b - a = \frac{3a}{4} \Rightarrow 3b = \frac{3a}{4} + a$$

$$\Rightarrow 3b = \frac{7a}{4} \Rightarrow \frac{a}{b} = \frac{12}{7}$$

$$\text{therefore ratio} \Rightarrow a : b = 12 : 7$$

Sol.171.(b) Let the numbers are a and b ,

According to the question,

$$a + b = 120$$

$$a - b = 24$$

$$\frac{a+b}{a-b} = \frac{120}{24} = \frac{5}{1}$$

By Componendo and dividendo rule

$$\frac{a}{b} = \frac{5+1}{5-1} = \frac{6}{4} = \frac{3}{2}$$

$$\text{so, } a : b \Rightarrow 3 : 2$$

Sol.172.(d)

Denomination of coin : ₹5 : ₹10 : ₹20

$$\text{Ratio of coin} \rightarrow : 2x : 3x : 5x$$

$$\text{Value} \rightarrow 10x : 30x : 100x$$

$$10x = 240 \Rightarrow x = 24$$

Now total amount of money in the bag

$$= 10x + 30x + 100x$$

$$= 140x = 140 \times 24 = \text{Rs } 3360$$

Sol.173.(d) A : B : C : D

$$3 : 4 : 7 : 8$$

$$\Rightarrow D - B = 8x - 4x \Rightarrow 4x = 408 \Rightarrow x = 102$$

$$C + D = 7x + 8x = 15x = 15 \times 102 = 1530 \text{ Rs}$$

Sol.174.(a)

Let x be added to each numbers

$$\Rightarrow (1+x) : (7+x) :: (7+x) : (25+x)$$

$$\Rightarrow (1+x)(25+x) = (7+x)(7+x)$$

$$\Rightarrow 25 + 26x = 49 + 14x$$

$$\Rightarrow 12x = 24 \Rightarrow x = 2$$

Sol.175.(d)

Let the cat be x and dogs be $2x$.

$$\Rightarrow (4 \times x) + (4 \times 2x) + 3 \times 2 = 42$$

$$\Rightarrow 12x = 36 \Rightarrow x = 3$$

$$\text{Total number of cats} = 3$$

Sol.176.(b) Let the numerator be x and

$$\text{denominator be } y \Rightarrow \frac{x+8}{y+12} = \frac{1}{2}$$

$$\Rightarrow 2x - y = -4 \dots\dots\dots(i)$$

$$\Rightarrow \frac{x-2}{y-2} = \frac{1}{8}$$

$$8x - y = 14 \dots\dots\dots(ii)$$

On solving eq (i) and (ii), we get

$$x = 3, y = 10$$

$$\text{Original fraction} = \frac{3}{10}$$

$$\text{Sol.177.(c)} \quad \frac{a}{b} = \frac{b}{c} \Rightarrow ac = b^2$$

$$\text{Now, } a^6 : b^6 = a^6 : (b^2)^3$$

$$\Rightarrow a^6 : b^6 = a^6 : (ac)^3$$

$$\Rightarrow a^6 : b^6 = a^6 : a^3 c^3$$

$$\Rightarrow a^6 : b^6 = a^3 : c^3$$

Sol.178.(a)

Let the ₹5 coin be x and ₹10 coins be y
 $xy = 135$ and $5x+10y=165$

$$\text{So, } 5\left(\frac{135}{y}\right) + 10y = 165$$

$$2y^2 - 33y + 135 = 0$$

$y = 9$ or 7.5 (number of coins cannot be in decimal)

$$\text{So, amount} = 10 \times 9 = ₹90$$

Sol.179.(a) Total students = 60

$$B : G$$

$$4 : 1$$

$$\Rightarrow 4+1 = 5 \rightarrow 60 \quad (\text{given})$$

$$\text{So number of boys} = 4 \rightarrow \frac{60}{5} \times 4 = 48$$

$$\text{Number of non-vegetarian} = 48 - 22 = 26$$

Sol.180.(c) According to the question,

$$\text{Ratio of steel (2018 : 2020)} = 4 : 3 = 4 : 3$$

$$\begin{array}{cc} \text{Steel} & \text{iron} \end{array}$$

$$2018 \rightarrow (3 : 2) \times 4$$

$$2020 \rightarrow (1 : 1) \times 3 \times 3$$

Here,

$$\begin{array}{cc} \text{Steel} & \text{iron} \end{array}$$

$$2018 \rightarrow 12 : 8$$

$$2020 \rightarrow 9 : 9$$

$$\text{Therefore ratio of iron} = 2018 : 2020$$

$$8 : 9$$

Sol.181.(c)

Let the parrots be $7x$ and pigeon be $9x$.

According to the question,

$$\frac{7x-8}{9x+6} = \frac{1}{4} \Rightarrow 28x - 32 = 9x + 6$$

$$\Rightarrow 19x = 38 \Rightarrow x = 2$$

$$\text{Initial Numbers of parrots} = 7 \times 2 = 14$$

$$\text{Remaining parrots} = 14 - 8 = 6$$

Therefore ratio of parrots and sparrows

$$\Rightarrow 6 : 10 = 3 : 5$$

Sol.182.(d)

Let the age of Yuvansh be x year.

$$\text{Ratio} \rightarrow \text{Garvit} : \text{Yuvansh}$$

$$4 \text{ years ago} \rightarrow 3x : x$$

$$1 \text{ year ago} \rightarrow 3x+3 : x+3$$

$$\text{Present age} \rightarrow 3x+4 : x+4$$

According to question ,

$$3x+3 = 2 \times (x+3) \Rightarrow x = 3$$

Difference in their age=

$$(3x+4) - (x+4) = 13 - 7 = 6 \text{ years.}$$

Sol.183.(b)

Let the numbers are $2x$, $3x$ and $5x$

$$2x + 3x + 5x = 80 \Rightarrow 10x = 80 \Rightarrow x = 8$$

$$\text{Product of the no. } 16 \times 24 \times 40 = 15360$$

Sol.184.(a) Let the number of boys be $2x$ and girls be $5x$

$$\text{Now, } \frac{2x-10}{5x+20} = \frac{4}{25} \Rightarrow 50x - 20x = 330$$

$$\Rightarrow 30x = 330 \Rightarrow x = 11$$

$$\text{Number of boys now} = 11 \times 2 = 22$$

$$= 22 - 10 = 12$$

$$\text{Sol.185.(c)} \quad \frac{3}{4} \times P = \frac{2}{3} \text{ of } Q$$

$$\Rightarrow \frac{P}{Q} = \frac{8}{9} \Rightarrow P + Q = 8 + 9 = 17 \text{ unit}$$

⇒ positive difference = Q - P = 9 - 8 = 1 unit
 ⇒ 17 unit = 3400 ⇒ 1 unit = Rs 200

Sol.186.(a) Let Karan gets = x

Raj gets = x - 9000

As per question,

$$\Rightarrow x + x - 9000 = 21000 \Rightarrow x = 15,000$$

$$\text{Ratio of Raj : Karan} = 6000 : 15000 = 2 : 5$$

Sol.187.(c) 4 years ago, ratio of age of

Shyam : Ram = 3 : 4

4 years hence, ratio of age of

Shyam : Ram = 5 : 6

According to question,

$$\Rightarrow \frac{3x+8}{4x+8} = \frac{5}{6} \Rightarrow 18x + 48 = 20x + 40$$

$$2x = 8 \Rightarrow x = 4$$

$$\text{Present age of Shyam} = 3x + 4 = 3 \times 4 + 4 = 16 \text{ years}$$

Sol.188.(a) Volume of Cone = $\frac{1}{3}\pi r^2 h$

Ratio → 1st Cone : 2nd Cone

$$(\text{radius})^2 \rightarrow 3^2 : 4^2$$

$$\text{Height} \rightarrow x : y$$

$$\text{Volume} \rightarrow 11 : 16$$

$$\text{Ratio of height} = \frac{11}{9} : \frac{16}{16}$$

$$x : y = 11 \times 16 : 16 \times 9 = 11 : 9$$

Sol.189.(b) Let the expenditure and savings of Soham be 500 and 100 units

$$\text{Income} = \text{Expenditure} + \text{Saving}$$

$$\text{Original} \rightarrow 600 = 500 + 100$$

$$\text{New} \rightarrow 750 = 630 + 120$$

According to question,

$$\text{Current expenditure (630 units)} \rightarrow ₹4347$$

$$\text{So, his Initial income (600 units)} \rightarrow$$

$$\frac{4347}{630} \times 600 = ₹4140$$

Sol.190.(b) According to question,

$$10\% \text{ of } A + 8\% \text{ of } B = \frac{3}{5} \text{ (12\% of } A + 16\% \text{ of } B)$$

$$50\% \text{ of } A + 40\% \text{ of } B = 36\% \text{ of } A + 48\% \text{ of } B$$

$$14\% \times A = 8\% \times B \Rightarrow \frac{A}{B} = \frac{4}{7}$$

Sol.191.(a)

$$A : B$$

$$\text{Income} \rightarrow 4x : 5x$$

$$\text{Expenditure} \rightarrow 7y : 9y$$

$$\text{Saving} \rightarrow 4x - 7y : 5x - 9y$$

According to question,

$$4x - 7y = 5x - 9y \Rightarrow 2y = x$$

$$\text{Now, } 4x - 7y = 50 \Rightarrow 4 \times 2y - 7y = 50$$

$$y = ₹50 \text{ and } x = ₹100$$

$$\text{The income of } A = 4x = 4 \times 100 = ₹400$$

$$\text{The income of } B = 5x = 5 \times 100 = ₹500$$

Short Trick :-

$$\text{Ratio} \rightarrow A : B$$

$$\text{Income} \rightarrow 4 : 5 \times 2$$

$$\text{Expenses} \rightarrow 7 : 9 \times 1$$

$$\text{Savings} \rightarrow 1 : 1$$

According to question,

$$\text{Saving (1 unit)} = 50 \text{ Rs.}$$

$$\text{Income of A} \rightarrow (4 \times 2) \text{ units} = 400 \text{ Rs.}$$

$$\text{Income of B} \rightarrow (5 \times 2) \text{ units} = 500 \text{ Rs.}$$

$$\text{Sol.192.(c)} (18x + 8y)\% = \frac{1}{3}(22x + 36y)\%$$

$$54x + 24y = 22x + 36y$$

$$32x = 12y \Rightarrow \frac{x}{y} = \frac{12}{32} = \frac{3}{8}$$

$$\text{So, the ratio of } x \text{ and } y = 3 : 8$$

Sol.193.(d) Let the number be x and y.

According to question,

$$60\% \times x = \frac{3}{4} \times y$$

$$\frac{3}{5} \times x = \frac{3}{4} \times y \Rightarrow \frac{x}{y} = \frac{5}{4}$$

$$\text{So, the required ratio} = 5 : 4$$

Sol.194.(b) According to question,

$$y = kx \rightarrow k = \frac{y}{x}$$

$$\text{When, } x = 42 \text{ then } y = 61$$

$$\frac{y_1}{x_1} = \frac{y_2}{x_2}$$

$$y_2 \rightarrow \frac{61 \times 54}{42} = 78.42 \approx 78$$

Sol.195.(c) Ratio → P : Q : R

$$16 : 17 \times 15$$

$$15 : 16 \times 17$$

$$P : Q : R = 240 : 255 : 272$$

$$\text{Total run scored by all batsmen} =$$

$$(240 + 255 + 272) \text{ units} = 767 \text{ units}$$

$$767 \text{ units} \rightarrow 956 \text{ Run}$$

$$\text{Run scored by R (272 units)} =$$

$$\frac{956}{767} \times 272 = \frac{260032}{767} \approx 339 \text{ Run}$$

Sol.196.(b) According to the question,

A varies with B and C

$$A = B \times C \Rightarrow 6 = 3 \times 2$$

$$\text{So, } A = 5 \times 7 = 35$$

Sol.197.(c) Ratio → A : B

$$\text{Income} \rightarrow 11 : 13$$

$$\text{Expenses} \rightarrow 9 : 11$$

$$\text{Savings} \rightarrow 2 : 2$$

According to question,

$$\text{Saving (2 units)} = 4000 \text{ Rs.}$$

$$\text{Difference b/w their income (13 - 11) units}$$

$$\rightarrow 2 \text{ units} = 4000 \text{ Rs.}$$

Sol.198.(b) Let the present ages of A and

B be a and b years respectively.

According to the question,

$$\frac{a-4}{b-4} = \frac{1}{2}$$

$$2a - 8 = b - 4 \Rightarrow 2a - b = 4 \dots\dots(1)$$

$$\text{Also, } a = b - 8 \Rightarrow a - b = -8 \dots\dots(2)$$

On solving equations (1) and (2), we get

$$a = 12 \text{ and } b = 20$$

So present age of B = 20 years

Sol.199.(c) Given,

$$(a^2 + b^2) : (b^2 + c^2) : (c^2 + a^2)$$

$$= 34 : 61 : 45,$$

$$\text{Adding all we get, } 2(a^2 + b^2 + c^2) = 140$$

$$(a^2 + b^2 + c^2) = 70.$$

$$c^2 = (a^2 + b^2 + c^2) - (a^2 + b^2)$$

$$= 70 - 34 = 36.$$

$$\text{Similarly, } b^2 = 25 \text{ and } a^2 = 9$$

$$\text{So, } a : b : c = \sqrt{9} : \sqrt{25} : \sqrt{36} = 3 : 5 : 6$$

$$b - a : c - b : c - a$$

$$= (5 - 3) : (6 - 5) : (6 - 3) \Rightarrow 2 : 1 : 3$$

Sol.200.(b) A : B

$$\text{Income} \quad 2x : 3x$$

$$\text{Expenditure} \quad 3y : 4y$$

$$\text{Savings} \quad (2x - 3y) : (3x - 4y)$$

According to the question,

$$\frac{1}{3} \times 3x = 3y, \Rightarrow x : y = 3 : 1$$

$$\text{Savings ratio} = (2x - 3y) : (3x - 4y)$$

$$\Rightarrow (2 \times 3 - 3 \times 1) : (3 \times 3 - 4 \times 1)$$

$$= 3 : 5$$

Short Trick :-

$$\text{Ratio} \rightarrow A : B$$

$$\text{Income} \rightarrow 2 : 3 \times 3$$

$$\text{Expenses} \rightarrow 3 : 4 \times 1$$

$$\text{Savings of A} = (2 \times 3) - (3 \times 1) = 3$$

$$\text{Savings of B} = (3 \times 3) - (4 \times 1) = 5$$

$$\text{Required Ratio} \rightarrow 3 : 5$$

Sol.201.(d)

$$\text{Ratio} \rightarrow x : y = \text{Total}$$

$$\text{Alloy A} \rightarrow 5 : 2 = 7 \times 4$$

$$\text{Alloy B} \rightarrow 3 : 4 = 7 \times 5$$

Quantity of metal 'x' in Alloy C →

$$(5 \times 4) + (3 \times 5) = 35 \text{ units}$$

Total quantity of Alloy C →

$$(7 \times 4) + (7 \times 5) = 63 \text{ units}$$

$$\text{Required percentage} = \frac{35}{63} \times 100 = 55\frac{5}{9}\%$$

Sol.202.(a)

$$\text{Ratio} \rightarrow \text{Juice} : \text{water} = \text{total}$$

$$\text{Cup 1} \rightarrow 5 : 2 = 7 \times 11$$

$$\text{Cup 2} \rightarrow 7 : 4 = 11 \times 7$$

$$\text{Cup 3} \Rightarrow \text{Juice} : \text{Water}$$

$$(5 \times 11) + (7 \times 7) : (2 \times 11) + (4 \times 7)$$

$$104 : 50 = 52 : 25$$

$$\text{Ratio} \rightarrow \text{water} : \text{juice} = 25 : 52$$

Sol.203.(a) Ratio → Last year Income

$$A : B = 4 : 3$$

Now,

$$\text{Last Year} : \text{Present Year}$$

$$A \rightarrow 4 \times 5 \text{ (3 : 4)}$$

$$B \rightarrow 3 \times 3 \text{ (5 : 6)}$$

Last Year : Present Year

A → 60 : 80

B → 45 : 54

According to the question ,

$(80 + 54)$ units = 24.12 lakh

1 unit = 0.18 lakh

Last year income of A + Present year

income of B = $(60 + 54)$ units = 114 units

114 units = $114 \times 0.18 = 20.52$ Lakh

$$\text{Sol.204.(a)} \quad \frac{z}{x} = \frac{5}{2} \Rightarrow z = \frac{5x}{2}$$

According to the question ,

$$\frac{5}{2} = y + \frac{7}{4} \Rightarrow y = \frac{5}{2} - \frac{7}{4} = \frac{3}{4}$$

$$\text{Also, } x + y + z = 1 \frac{11}{12}$$

Putting the values of z and y as obtained in above equation, we get

$$x + \frac{3}{4} + \frac{5x}{2} = \frac{23}{12}$$

$$\frac{4x + 3 + 10x}{4} = \frac{23}{12} \Rightarrow 14x + 3 = \frac{23}{3}$$

$$42x = 23 - 9 \Rightarrow x = \frac{1}{3}$$

$$z = \frac{5 \times \frac{1}{3}}{2} = \frac{5}{6}, \text{ So, } (z - x) : (y - x)$$

$$= \left(\frac{5}{6} - \frac{1}{3}\right) : \left(\frac{3}{4} - \frac{1}{3}\right)$$

$$= \frac{3}{6} : \frac{5}{12} = 6 : 5$$

Sol.205.(b) Let , initial passed students be $25x$ and initial failed be $4x$.

According to the question,

$$\frac{25x + 1 + 3}{4x - 3} = \frac{22}{3}$$

$$75x + 12 = 88x - 66$$

$$13x = 78 \rightarrow x = 6$$

$$\text{Initial passed} - \text{initial failed} \rightarrow 25x - 4x$$

$$21x = 21 \times 6 = 126$$

Sol.206.(d) Let the total sum = ₹ x

Old ratio of share → 2 : 3 : 4

New ratio of share → 6 : 4 : 3

According to question,

$$\frac{6x}{13} - \frac{2x}{9} = 3500 \text{ Rs.}$$

$$\frac{28x}{117} = 3500 \Rightarrow x = 14,625 \text{ Rs.}$$

Sol.207.(b) Let number of Mathematics books = M, number of Physics books = P and number of Chemistry books = C

According to question,

$$\frac{M}{P} = \frac{P}{C} \Rightarrow P^2 = MC = 144 \times 100 = 14400$$

$$\text{So, } P = 120$$

$$\text{So, required ratio} \rightarrow (M + P) : (C + P)$$

$$= (144 + 120) : (100 + 120)$$

$$= 264 : 220 = 6 : 5$$

Sol.208.(c) Let x males and x females are added in the recruitment drive.

According to the question,

$$\frac{70 + x}{100 + x} = \frac{3}{4}$$

$$280 + 4x = 300 + 3x \Rightarrow x = 20$$

$$\text{Total males now} = 70 + 20 = 90,$$

$$\text{Total females now} = 100 + 20 = 120$$

$$\text{Total working population} = 90 + 120 = 210$$

Sol.209.(c)

Ratio → 1st year : 2nd year

Income → 2x : 5x

Expenses → 4y : 7y

According to question,

$$5x = 75000 \Rightarrow x = 15000$$

$$4y = 20000 \Rightarrow y = 5000$$

Savings of both years combined =

$$(2x - 4y) + (5x - 7y) =$$

$$(2 \times 15000 - 4 \times 5000) + (5 \times 15000 - 7 \times 5000) = 10000 + 40000 = 50000 \text{ Rs.}$$

Sol.210.(c) Let the shares of A, B and C be $5x + 130$, $2x + 150$ and $4x + 200$ respectively.

According to the question,

$$5x + 130 + 2x + 150 + 4x + 200 = 3780$$

$$11x + 480 = 3780 \Rightarrow 11x = 3300 \Rightarrow x = 300$$

$$\text{Share of C } (4x + 200) \rightarrow 4 \times 300 + 200$$

$$= 1400 \text{ Rs.}$$

Sol.211.(d)

$$\text{Marbles in Box A} : \frac{3}{15} \times 75 = 15,$$

$$\text{Marbles in Box B} : \frac{5}{15} \times 75 = 25,$$

$$\text{Marbles in Box C} : \frac{7}{15} \times 75 = 35$$

On transferring the marbles as per directions given in question, we get

$$\text{Total marbles in Box A} = 15 + 3 = 18$$

$$\text{Total marbles in Box B} = 25 - 3 + 5 = 27$$

$$\text{Total marbles in Box C} = 35 - 5 = 30$$

$$\text{So, required ratio} = 18 : 27 : 30 = 6 : 9 : 10$$

$$\text{Sol.212.(c)} \quad \frac{21 - x}{22 - x} = \frac{60 - x}{64 - x}$$

$$(21 - x)(64 - x) = (22 - x)(60 - x)$$

$$1344 + x^2 - 21x - 64x = 1320 + x^2 - 60x - 22x$$

$$1344 - 85x = 1320 - 82x$$

$$24 = 3x \Rightarrow x = 8$$

So, Mean proportional

$$= \sqrt{(x + 4) \left(\frac{x}{2} - 1\right)}$$

$$= \sqrt{(8 + 4) \left(\frac{8}{2} - 1\right)}$$

$$= \sqrt{(12)(3)} = \sqrt{36} = 6$$

Sol.213.(b)

$$\text{Let } (a + b) = 7x, (b + c) = 4x, (c + a) = 5x$$

So, according to question

$$7x + 4x + 5x = 2(a + b + c)$$

$$7x + 4x + 5x = 2(16)$$

$$16x = 32 \Rightarrow x = 2$$

$$\text{So, } a + b = 7x = 14,$$

$$b + c = 4x = 8 \text{ and } c + a = 5x = 10$$

$$\text{So, } a = 8, b = 6, c = 2 \text{ Now,}$$

$$(a^2 + b^2 + c^2) : (ab + bc + ca)$$

$$(8^2 + 6^2 + 2^2) : 8 \times 6 + 6 \times 2 + 8 \times 2$$

$$= (64 + 36 + 4) : (48 + 12 + 16)$$

$$= (104) : (76) = 26 : 19$$

Sol.214.(b) Proportion of increased seats

$$= 7 \times \frac{150}{100} : 5 \times \frac{140}{100} : 8 \times \frac{125}{100}$$

$$= \frac{21}{2} : 7 : 10 = 21 : 14 : 20$$

Sol.215.(b)

Ratio → Maths : English : Science

$$3 : 2 : 9$$

$$2 : 3 : 3$$

$$-----$$

$$6 : 4 : 9$$

$$\text{Total marks } (6 + 4 + 9) \text{ units} = 190$$

$$\text{Marks in Science } (9 \text{ units}) = 90$$

Sol.216.(d) Let the number of chemistry books = x and let the price of each physics book and each chemistry book be 1 and 2 respectively.

So, total correct amount

$$= 10 \times 1 + x \times 2 = 10 + 2x$$

Total incorrect amount

$$= 10 \times 2 + x \times 1 = 20 + x$$

According to the question,

$$12.5 = \frac{(10 + 2x) - (20 + x)}{10 + 2x} \times 100$$

$$\frac{1}{8} = \frac{(10 + 2x) - (20 + x)}{10 + 2x}$$

$$\frac{1}{8} = \frac{x - 10}{10 + 2x} \Rightarrow 10 + 2x = 8x - 80$$

$$6x = 90 \Rightarrow x = 15$$

$$\text{So, required ratio} = 10 : x = 10 : 15 = 2 : 3$$

Sol.217.(b)

Income = Expenditure + Savings

$$17 = 14 + 3$$

According to question ,

Saving (3 units) = 12000 Rs.

$$\Rightarrow 1 \text{ unit} = 4000$$

Income (17 units) = 68000 Rs.

Expenditure (14 units) = 56000 Rs.

Now ,

$$\text{New income} = 68000 + 10000 = ₹78,000$$

$$\text{New expenditure} = 56000 - 2000$$

$$= ₹54,000$$

$$\text{Required ratio} \rightarrow 78000 : 54000 = 13 : 9$$

Sol.218.(b) According to the question,

$$\Rightarrow 52 - x : 47 - x :: 20 - x : 19 - x$$

$$\Rightarrow (52 - x)(19 - x) = (47 - x)(20 - x)$$

$$\Rightarrow 988 - 71x + x^2 = 940 - 67x + x^2$$

$$\Rightarrow 48 = 4x \Rightarrow x = 12$$

Mean proportional between $(x + 13)$ and

$$(x - 8) = \sqrt{(12 + 13)(12 - 8)} = \sqrt{100} = 10$$

Sol.219.(d)

Ratio → A : B : C : D

$$3 : 1$$

$$5 : 2$$

$$2 : 1$$

Final $\rightarrow 30 : 10 : 4 : 2$

Difference between B and D = $(10 - 2)$ units

8 units = ₹1600 $\rightarrow 1$ unit = 200 Rs.

Total sum $(30 + 10 + 4 + 2) =$

46 units = $46 \times 200 = 9200$ Rs.

Sol.220.(b)

Votes received by unsuccessful

candidates = $\frac{4 \times 9,84,321}{9} = 4,37,476$

Sol.221. (c) Let the age of A and B 8 years ago be $4x$ and $5x$ respectively

$$\frac{4x + 8 + 12}{5x + 8 + 12} = \frac{13}{15} \Rightarrow \frac{4x + 20}{5x + 20} = \frac{13}{15}$$

$$60x + 300 = 65x + 260$$

$$5x = 40 \Rightarrow x = 8$$

Present age of B = $5x + 8 = 5(8) + 8$

= 48 years

Sol.222.(c) Here we need to check through options, and by doing so we can see that option (c) satisfies the conditions given in question.

Ratio of marbles $\rightarrow A : B : C = 2 : 3 : 4$

Total marbles (9 units) = 360

1 unit = 40

Box A = 80, Box B = 120, Box C = 160

So on applying option (c), we get

$(80 - 20) : (120 + 20 + 20) : (160 - 20)$

$\Rightarrow 60 : 160 : 140 = 3 : 8 : 7$

Sol.223.(c) Let the share of P be 5 units,

Now share of R = $\frac{2}{5} \times 5 = 2$ units.

Share of Q = $3 \times 2 = 6$ units.

So, ratio of share of P : Q : R = 5 : 6 : 2.

Share of Q = $\frac{6}{13} \times 3900 = 1,800$.

Sol.224. (c)

If we observe the given ratio we get

$$\begin{array}{ccc} 3 : 5 : 8 \\ 4 : 7 : 9 \\ \hline 1 : 2 : 1 \\ \swarrow \quad \downarrow \quad \searrow \\ 15 \quad 30 \quad 15 \end{array}$$

Clearly, 1 unit = 15,

Number of students before new

admissions $\rightarrow (3 + 5 + 8)$ units = 16 units

Required number of students

= $16 \times 15 = 240$

Sol.225.(a)

Ratio of weight $\rightarrow 5 : 7 = 12$

Ratio of Price $\rightarrow 5^2 : 7^2 = 12^2$

Now, According to question,

Total Price before stone broken = 144 units

144 units = 230400 Rs. $\Rightarrow 1$ unit = 1600 Rs

Total Price after stone broke into two

parts $(25 + 49)$ units $\rightarrow 74$ units

Loss $(144 - 74) = 70$ units

$\Rightarrow 70$ units = 112000 Rs.

Sol.226.(d) Given, $x^2 : y^2 = 49 : 64$

As x and y have opposite signs, let x be positive and y is negative.

$x = +7$ and $y = -8$

Putting these values in

$(5x - 6y) : (6x - 7y)$ we get

$5 \times 7 - 6 \times (-8) : 6 \times 7 - 7 \times (-8)$

$35 + 48 : 42 + 56 \Rightarrow 83 : 98$

Sol.227.(b)

Passed	Failed
5	2
4	3
-14	+14

$15 - 8 = 7$
 $56 - (-42) = 98$

Total number of students appeared

= 7 unit = 98

Sol.228.(a) Students in Science = $2x$

Students in Humanities = $3x$

$2x + 3x = 360$, $5x = 360$, $\Rightarrow x = 72$

Let students added in Humanities be y .

$$\text{Now, } \frac{2x + 18}{3x + y} = \frac{3}{5} \Rightarrow \frac{144 + 18}{216 + y} = \frac{3}{5}$$

$$162 \times 5 = 648 + 3y$$

$$3y = 810 - 648 \Rightarrow y = \frac{162}{3} = 54$$

Sol.229.(c) Let Present age of mother and daughter be $3x$ and x respectively.

$$\text{So, } \frac{3x - 5}{x - 5} = \frac{5}{1}$$

$$3x - 5 = 5x - 25$$

$$2x = 20, \text{ so } x = 10$$

Mother's age 41 years hence = $3x + 41$

= $3(10) + 41 = 30 + 41 = 71$ years

Short Tricks :-

Ratio Mother : Daughter

Present	3	:	1
-5 years	5	:	1
	-5	:	-5

$(1 \times 3) - (5 \times 1) = -2$ units
 $(-5 \times 5) - (-5 \times 1) = -20$

$$-2 \text{ units} = -20 \rightarrow 1 \text{ unit} = 10$$

Mother's age 41 years hence $\rightarrow$

$(3 \times 10) + 41 = 71$ years

Sol.230.(a)

Ratio $\rightarrow X : Y$

Present income $\rightarrow 5 : 8$

Ratio $\rightarrow$ Present year : Last year

X $\rightarrow 3 : 2$) $\times 4 \times 5$

Y $\rightarrow 4 : 3$) $\times 3 \times 8$

Total income for last year of X and Y = 40 unit and 72 unit

40 unit + 72 unit = 112 unit

112 unit = ₹56,000

Present income of y (96 unit)

$$= \frac{56000}{112} \times 96 = ₹48,000$$

Sol.231.(c)

Ratio $\rightarrow$ Jeans : Shirts : t-shirts

Quantity $\rightarrow 1 : 2 : 3$

Price $\rightarrow 500 : 350 : 200$

Total $\rightarrow 500 : 700 : 600$

So, Total amount paid = 1800 units

1800 units = 46,800

1 unit = 26

Number of t-shirts (3 unit) = $3 \times 26 = 78$

Sol.232.(a) Ratio of Present age $\rightarrow$

Mother : Son = $6x : x$

After $5x$ year $\rightarrow$

Mother : Son = $11x : 6x$

Father : Mother = $13x : 11x$

Present age of father = $13x - 5x = 8x$

Again Ratio of Present age $\rightarrow$

Father : Mother : Son = $8x : 6x : x$

After $7x$ year $\rightarrow$

Father : Mother : Son = $15x : 13x : 8x$

According to question,

Sum of father and son $(15x + 8x) = 23x$

$23x = 115$ years $\rightarrow x = 5$ years

Present age of father = $8x = 8 \times 5$

= 40 years

Sol.233.(b)

Ratio $\rightarrow A : B$

Income $\rightarrow 1 : 2$) $\times 2.4$

Expenditure $\rightarrow 2 : 3$

Savings of B $\rightarrow (2.4 \times 2 - 3) = 1.8$

Savings of A $\rightarrow (2.4 \times 1 - 2) = 0.4$

Required Ratio $\rightarrow 1.8 : 0.4 = 9 : 2$

Sol.234.(b) Let the number of candidates who passed be $5x$ and that of who failed be $2x$.

According to question,

$$\Rightarrow \frac{5x - 14}{2x + 14} = \frac{4}{3}$$

$$\Rightarrow 5x \times 3 - 14 \times 3 = 4 \times 2x + 14 \times 4$$

$$\Rightarrow 15x - 42 = 8x + 56$$

$$\Rightarrow 7x = 56 + 42, \text{ so } x = 14.$$

Number of students who passed = $5x$

= $5 \times 14 = 70$

Short Trick:-

Ratio $\Rightarrow$ Passed : Failed

Before $\Rightarrow 5 : 2$

After $\Rightarrow 4 : 3$

According to question,

1 unit = 14 students

Number of students who passed (5

units) $\rightarrow 5 \times 14 = 70$

Sol.235.(c)

Let, the present age of the girl be x

Age of father six years ago = $2x$

Ratio of Present Age $\Rightarrow$

Girl : Father : Mother

$x : 2x + 6 : 2x + 3$

Their present average age $\rightarrow$

$32 + 6 = 38$ years

Now, $x + (2x + 6) + (2x + 3) = 38 \times 3$

$$5x + 9 = 114 \Rightarrow x = 21$$

Present age of her mother $(2x + 3) =$

$$2x + 3 = 2 \times 21 + 3 = 45 \text{ years}$$

Sol.236.(b) Let the numbers be $4x$ and $5x$ respectively.

According to question, $\frac{4x-1}{5x+2} = \frac{3}{4}$

$$\Rightarrow 16x - 4 = 15x + 6 \Rightarrow x = 10.$$

So the numbers are 40 and 50 respectively.

Now, required ratio $\Rightarrow 40 + 8 : 50 + 4$

$$\Rightarrow 48 : 54 = 8 : 9$$

Short Trick:-

1st no. 2nd no.

$(4 \times 4) - (3 \times 5) = 1 \text{ unit}$
 $(3 \times 2) - (4 \times -1) = 10$

According to question, 1 unit = 10

Then, numbers are 40 and 50

Required Ratio $\rightarrow (40 + 8) : (50 + 4) = 8 : 9$

Sol.237.(a) Ratio $\rightarrow$

sixes : fours : threes : twos : singles

$$1 : 3 : 2 : 7 : 10$$

Value $\rightarrow 1 \times 6 : 3 \times 4 : 2 \times 3 : 7 \times 2 : 10 \times 1$

$$\rightarrow 6 : 12 : 6 : 14 : 10 = 48 \text{ units}$$

$$48 \text{ units} = 96 \text{ runs}$$

Total singles (10 units) = $10 \times 2 = 20 \text{ runs}$

Sol.238.(c) $10\frac{2}{3}\% = \frac{32}{300}$

Ratio $\rightarrow A : B = 332 : 300$

$A : C = 2 : 3$

After Balancing the ratio $\rightarrow$

$$A : B : C = 332 : 300 : 498$$

According to question,

$$(332 + 300 + 498) \text{ units} = 4520 \times 3$$

$$1130 \text{ units} = 13560 \text{ Rs.}$$

$$1 \text{ unit} = 12 \text{ Rs.}$$

Share of B (300 units) = $300 \times 12 = 3600$

Sol.239.(b)

Ratio $\rightarrow A : B$

Present age $\rightarrow 7 : 8$

+6 years $\rightarrow 8 : 9$

1 unit age gap = 6 years

A's present age = $7 \times 6 = 42 \text{ years}$

B's present age = $8 \times 6 = 48 \text{ years}$

C's present age = A's present age + 10
 $= 52 \text{ years}$

Sol.240.(a)

Ratio $\rightarrow X : Y$

$A \rightarrow 5 : 2 \times 4$

$B \rightarrow 3 : 4 \times 5$

$C \rightarrow (20 + 15) : (8 + 20) = 35 : 28$

Required % of Y = $\frac{28}{63} \times 100 = 44\frac{4}{9}\%$

Sol.241.(a)

$$\frac{12}{18} = \frac{6}{x} \Rightarrow x = 9 \Rightarrow \frac{4}{k} = \frac{k}{9} \Rightarrow k = 6$$

Sol.242.(c) Let salary of Anil and Kumud are $19x$ and $17x$.

According to the question

$$= \frac{19x + 2000}{17x + 1000} = \frac{8}{7}$$

$$133x + 14000 = 136x + 8000$$

$$3x = 6000 \Rightarrow x = 2000$$

Present salary = $17 \times 2000 = 34000$

Sol.243.(c)

Let the three parts be a, b and c .

$$\frac{a}{4} = \frac{b}{3} = \frac{c}{6} \rightarrow a : b : c = 4 : 3 : 6$$

First part (a) = $\frac{4}{4+3+6} \times 9685 = 2980$

Sol.244.(a)

Coins Value $\rightarrow 1 : 2 : 5$

Quantity $\rightarrow 4 : 5 : 8$

Total Value $\rightarrow 4 : 10 : 40$

Total value (54 units) = 432 Rs.

1 unit = 8 Rs., 10 unit = 80 Rs

Number of Rs.2 coins (5 units) = $\frac{80}{2} = 40$

Sol.245.(c)

Expenditure ratio of A to B = $2 : 1$

$$\frac{7x - 4000}{5x - 3500} = \frac{2}{1}$$

$$7x - 4000 = 10x - 7000$$

$$3000 = 3x \Rightarrow x = 1000$$

Total Income $(7x + 5x) \text{ unit} = 12x$

$$= 12 \times 1000 = 12000 \text{ Rs.}$$

Sol.246.(b) When a, b, c and d are in

proportion, then $ad = bc$

According to question,

$$ad = 70 \text{ and } bc = 3.5y$$

$$70 = 3.5y \Rightarrow y = 20$$

Sol.247.(d)

Ratio $\rightarrow A : C = 8 : 7 \times 2$

$B : C = 3 : 2 \times 7$

Combining the ratio, we get

$$A : B : C = 16 : 21 : 14$$

Difference between share of A and B

$$(21 - 16) \text{ units} = 5 \text{ units} = 240 \text{ Rs.}$$

Total sum (51 units) = $\frac{240}{5} \times 51 = 2448 \text{ Rs.}$

Sol.248.(a)

Ratio $A : B$

-7 years $\left(\begin{array}{cc} 4 & 5 \\ +5 & +5 \end{array} \right)$

+8 years $\left(\begin{array}{cc} 9 & 10 \end{array} \right)$

According to question,

$$5 \text{ units} = 15 \text{ years} \rightarrow 1 \text{ unit} = 3 \text{ years}$$

Sum of their present age =

$$(4 + 5) \text{ units} + 14 \text{ years}$$

$$= 9 \times 3 + 14 = 41 \text{ years}$$

Sol.249.(c)

Ratio $\rightarrow \text{Boys} : \text{Girls} = 7 : 5$

Girls below 14 years of age = $5 \times \frac{3}{5} = 3$

Boys below 14 years of age = $7 \times \frac{4}{7} = 4$

According to question,

$$(3 + 4) = 7 \text{ units} = 1120$$

$$1 \text{ unit} = 160$$

Total Students $(7 + 5) = 12 \times 160 = 1920$

Sol.250.(d)

Sum of Present age = 52 years

Sum of their age after 4 years = 60 years

Son's age after 4 year = $60 \times \frac{1}{5} = 12 \text{ years}$

Present age of son = 8 years

Present age of father = 44 years

After 10 years son = 18 years

After 10 years father = 54 years

Their age ratios $\rightarrow 18 : 54 = 1 : 3$

Sol.251.(b) Let, the fraction = $\frac{x+3}{x}$

According to the question,

$$\frac{x+8}{x-2} = \frac{8}{3} \rightarrow 5x = 40 \Rightarrow x = 8$$

Then the original Fraction = $\frac{11}{8}$

After dividing Original Fraction by $\frac{11}{2} \rightarrow$

$$\frac{11}{8} \times \frac{2}{11} = \frac{1}{4}$$

Sol.252.(b)

Initial Ratio $\rightarrow 3 : 4 : 5 = 12 \text{ units} \times 47$

Final Ratio $\rightarrow 20 : 15 : 12 = 47 \text{ units} \times 12$

Equating the total units, we get

Initial Ratio $\rightarrow 141 : 188 : 235 = 564 \text{ units}$

Final Ratio $\rightarrow 240 : 180 : 144 = 564 \text{ units}$

Least share difference = $240 - 141 = 99$

99 units = 1188 Rs., 1 unit = 12 Rs.

Total sum (564 units) = $564 \times 12 = 6768 \text{ Rs.}$

Sol.253.(d)

Ratio $\rightarrow X : Y$

Income $\rightarrow 5 : 4 \times 9$

Expenditure $\rightarrow 4 \times 9 : 7$

Savings of X = $(5 \times 9) - (4 \times 9) = 9$

Savings of Y = $(4 \times 9) - (4 \times 7) = 8$

Required Ratio = $9 : 8$

Sol.254.(c) LCM of 4, 3, 2 is 12

Multiply the ratio by 12

$$a : b : c = 3 : 4 : 6$$

$$\frac{a}{b} : \frac{b}{c} : \frac{c}{a} = \frac{3}{4} : \frac{4}{6} : \frac{6}{3}$$

LCM of 4, 6 and 3 is 12.

Multiply the ratio by 12

$$\frac{a}{b} : \frac{b}{c} : \frac{c}{a} = 9 : 8 : 24$$

Sol.255.(b)

Ratio $\rightarrow a : b : c$

$$4 : 7 : 7$$

$$4 : 4 : 5$$

$$16 : 28 : 35$$

Sum of three numbers $(16 + 28 + 35)$ units

→ 79 Units → 79

Second number (28 units) = 28

Sol.256.(c) $2145 : x :: 3003 : 42$

$$\frac{2145}{x} = \frac{3003}{42} \Rightarrow x = 30$$

Now, $x : 2508 :: y : 11704$

$$\frac{30}{2508} = \frac{y}{11704} \Rightarrow y = 140$$

Sol.257.(c) $\frac{4x-4}{5x-4} = \frac{11}{14}$

$$56x - 56 = 55x - 44 \Rightarrow x = 12$$

Present age →

Man = 48 years, Father = 60 years

Man's age (6 unit) = 48 years,

$$\text{Son's Age (1 unit)} = \frac{1}{6} \times 48 = 8 \text{ years}$$

After 12 years, Required ratio →

$$(60 + 12) : (8 + 12) = 18 : 5$$

Sol.258.(d) Present age of Ravi, Mohan

and Govind = $3 \times 32 + 3 \times 6 = 114$

Sum of ages of all four = $4 \times 36 = 144$

The present age of shyam = $144 - 114$
= 30 years

Sol.259.(a)

Perimeter of rectangle = $2(l + b)$

$$32 \text{ m} = 2(5x + 3x) \Rightarrow x = 2 \text{ m}$$

Length $(5x) = 10 \text{ m}$, Breadth $(3x) = 6 \text{ m}$

Sol.260.(a)

Ratio → Length : Breadth = 5 : 3

Difference (2 units) = 8 m

Length (5 units) = 20 m

Breadth (3 units) = 12 m

$$\text{Area of Rectangle} = 20 \times 12 = 240 \text{ m}^2$$

Sol.261.(d)

Total students in school = 640

Ratio → Boys : Girls = 5 : 3

8 units = 640

Boys (5 units) = 400, Girls (3 units) = 240

30 more girls added →

Total girls = $240 + 30 = 270$

Let 'x' boys be added to make the ratio of boys to girls → 14 : 9

According to question,

$$\frac{400 + x}{270} = \frac{14}{9} \Rightarrow x = 20$$

Sol.262.(d)

Let boys be '7x' and girls be '6x'

As per given question:

$$\frac{7x+4}{6x-3} = \frac{4}{3} \Rightarrow x = 8$$

Initially total number of boys and girls →

$$13x = 13 \times 8 = 104$$

Sol.263.(a)

$$x = \frac{ad - bc}{(a + d) - (b + c)}$$

$$x \Rightarrow \frac{10 \times 32 - 16 \times 22}{(10 + 32) - (16 + 22)} = 8$$

Value of $x = 8$

Mean proportional of $(x + 1)$ and $(3x + 1)$

$$\text{is : } \sqrt{(9)(25)} = 15$$

Sol.264.(c)

Denomination → ₹ 10 : ₹ 50

Number of notes → 1 : 2

3 units = 12 ⇒ 1 unit = 4

Total money in wallet →

$$(10 \times 1 \times 4) + (50 \times 2 \times 4) = 440 \text{ Rs.}$$

Sol.265.(c)

Ratio → A : B : C : D

$$8 : 9 : 9 : 9$$

$$2 : 2 : 3 : 3$$

$$9 : 9 : 9 : 13$$

$$16 : 18 : 27 : 39$$

On adding the ratios = 100 units

100 units = 12000 ⇒ 1 unit = 120

difference between the shares of B and D

$$= 21 \times 120 = \text{Rs } 2520$$

Sol.266.(c) Let $(a + b) = 7$ units

$(b + c) = 6$ units, $(c + a) = 5$ units

Adding all three, we get

$$(a + b) + (b + c) + (c + a) = (7 + 6 + 5) \text{ units}$$

$$(a + b + c) = 9 \text{ units}$$

According to the question,

$$9 \text{ units} = 27 \Rightarrow 1 \text{ unit} = 3$$

$$\Rightarrow a = (a + b + c) - (b + c)$$

$$= 9 - 6 = 3 \text{ units}$$

$$3 \text{ unit} = 3 \times (3) = 9$$

Similarly, $b = 12$ and $c = 6$

$$\Rightarrow \frac{1}{a} : \frac{1}{b} : \frac{1}{c} = \frac{1}{9} : \frac{1}{12} : \frac{1}{6} = 4 : 3 : 6$$

Sol.267.(a)

Let the numbers are 3k and 5k

$$\Rightarrow \frac{3k - 13}{5k - 13} = \frac{10}{21}$$

$$63k - 273 = 50k - 130 \Rightarrow k = 11$$

Required ratio =

$$3(11) + 5 : 5(11) + 5 = 19 : 30$$

Sol.268.(c) To find the value of x :-

Method :1 $x = \frac{ad - bc}{(a + d) - (b + c)}$

$$x \Rightarrow \frac{23 \times 56 - 39 \times 32}{(23 + 56) - (39 + 32)} = 5$$

Method :2 $\frac{23 - x}{39 - x} = \frac{32 - x}{56 - x} \dots (i)$

By hit and trial, we can find that $x = 5$

satisfies the eq. (i)

Mean proportion of $(x + 4)$ and $(3x + 1)$

$$\text{i.e. } 9 \text{ and } 16 = \sqrt{9 \times 16} = 12$$

Sol.269.(c) Given, $a : b = 3 : 2$

Let $a = 3k$ and $b = 2k$

$$(5a + 2b) : (3a + 4b)$$

$$= \{5(3k) + 2(2k)\} : \{3(3k) + 4(2k)\}$$

$$= 19k : 17k = 19 : 17$$